PARABOLIC VARIFOLDS v0.1

JOSHUA L. FLYNN AND IGNACIO TEJEDA

Abstract. We develop a theory of codimension-one parabolic varifolds using chronomorphisms as admissible deformations. We prove an Allard-type rectifiability theorem: locally bounded spatial first chronovariation, together with a half-time condition on the weight, implies parabolic rectifiability on the positive lower-density locus. At almost every point of this locus with respect to the weight measure, the density exists and the geometrically normalized mass and varifold blow-ups converge weakly to unique flat limits.

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1. Introduction

We develop a variational theory for parabolic rectifiable sets and prove a rectifiability theorem for parabolic varifolds. The starting point is Mattila's theory of parabolic rectifiability [26, Sections 3–4], which relates approximation by parabolic Lipschitz graphs to the existence of tangent planes and the structure of tangent measures. Our original motivation was to study these sets through a variational framework in the spirit of Allard [1, Introduction, pp. 417–418]. Allard's rectifiability theorem asserts that the positive-lower-density part of a varifold with locally bounded first variation is rectifiable; we recall its precise statement in Theorem 4.6.1. A parabolic counterpart requires both a suitable class of deformations and a notion of first variation that accounts for the different roles of space and time.

We work in $\mathbb{P}^n = \mathbb{R}^n_x \times \mathbb{R}_t$, $n \geq 1$, with the square-sum parabolic metric (5.1.1), and put $q = n + 1$. The model tangent planes are the vertical planes $V = L \times \mathbb{R}_t$, where L is an $(n-1)$-dimensional linear subspace of $\mathbb{R}^n$. These planes have parabolic Hausdorff dimension q by Lemma 5.6.1; their collection is denoted by $\mathbb{P}G$. A parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$ is a positive Radon measure W on $\Omega \times \mathbb{P}G$, with weight $\mu = \|W\|$; see Section 6. Write W_+ for its restriction to the set E_+ of positive lower q-dimensional parabolic density, and $\mu_+ = \|W_+\|$, as in (3.0.4).

The Main Theorem in Section 3 states that W_+ is a rectifiable parabolic varifold whenever W has locally bounded spatial first chronovariation and the half-time condition $\mathrm{HT}(\mu, a)$ holds for μ_+-almost every a. Thus there are a Borel countably $\mathbb{P}$-rectifiable set $K \subset E_+$ and a Borel function $\theta : K \to (0, \infty)$ such that $\mu_+ = \theta \mathscr{H}^q_p \llcorner K$, and the plane distribution of W_+ is concentrated on the rectifiable tangent plane of K at μ_+-almost every point. Here countable $\mathbb{P}$-rectifiability requires, for every positive Lipschitz bound, a countable cover by vertical parabolic Lipschitz graphs modulo an $\mathscr{H}^q_p$-null set (Definition 5.7.2). Theorems 1 and 2 give the two stages of the proof and the accompanying density and convergence conclusions. In particular, finite upper density is a conclusion of Theorem 1(4).

The requirement of arbitrarily small Lipschitz bounds is essential: Mattila constructs a vertical parabolic Lipschitz graph with no approximate parabolic tangent planes [26, Example 8.1, pp. 875–876]. Thus a parabolic Lipschitz graph need not itself be parabolic rectifiable.

The deformation maps are the C^1 chronomorphisms of Section 7.1. For open sets $U, U' \subset \mathbb{P}^n$, these are Euclidean C^1 diffeomorphisms $\Psi : U \to U'$ of the form

$$\Psi(x, t) = (f(x, t), h(t)), \qquad f \in C^1(U; \mathbb{R}^n),$$

where $h : \pi_t(U) \to \pi_t(U')$ is a C^1 diffeomorphism between the temporal projections. The inverse has temporal component h^{-1}. Thus the spatial deformation may depend on time, while the change of time is the same at all points of each time slice.

Section 7.5 gives a measure-theoretic reason for this choice. For a general Euclidean C^1 diffeomorphism $\Psi(x, t) = (f(x, t), h(x, t))$ between open subsets of $\mathbb{P}^n$, Proposition 7.5.1 proves that preservation of $\mathscr{H}^q_p$-finiteness on every compact set is equivalent to $D_x h = 0$. The same equivalence holds with preservation of $\mathscr{H}^q_p$-nullity in place of finiteness, and it suffices in either case to test compact countably $\mathbb{P}$-rectifiable sets. Such a map is locally bi-Lipschitz for the parabolic metric. If the nonempty domain is connected and each nonempty time slice is connected, Corollary 7.5.1 identifies these maps globally as chronomorphisms. On such domains, chronomorphisms therefore form the largest class of Euclidean C^1 diffeomorphisms that preserves finiteness of parabolic Hausdorff measure on all compact countably $\mathbb{P}$-rectifiable sets. Our motivation for this characterization comes from Golo's work [17, arXiv version, Proposition 5.1]: Golo proves that a C^2 diffeomorphism of the first Heisenberg group preserving finiteness of intrinsic perimeter for every measurable set in every open set must be contact.

Spatial chronomorphisms generated by $(X, 0)$, with $X \in C^1_c(\Omega; \mathbb{R}^n)$, give deformations supported in a compact subset of Ω. Differentiating their localized parabolic mass change

yields the spatial first chronovariation

$$\delta^p_{\mathrm{chr}} W(X,0) = \int_{\Omega\times\mathbb{P}G} \mathrm{tr}(P_L D_x X(a))\, dW(a, L\times\mathbb{R}_t),$$

as proved in Proposition 8.4.1. Here P_L is the spatial orthogonal projection onto L. Local boundedness is the order-zero estimate (8.4.5); equivalently, this functional is represented by a vector-valued Radon measure (Proposition 8.4.2). A nonzero temporal component depending only on time cannot be compactly supported in an open spacetime domain, as explained in Section 8.5. We therefore formulate the temporal hypothesis directly through local increments of the weight.

The half-time condition expresses first-order control at the parabolic scale. For $s \in \mathbb{R}$, the time translation $\tau_s : \mathbb{P}^n \to \mathbb{P}^n$, $\tau_s(x,t) = (x, t+s)$, displaces each point by $|s|^{1/2}$ in the metric d_p. Accordingly, (3.0.2) normalizes the absolute increment $\left|\int(\phi\circ\tau_s - \phi)\, d\mu\right|$ by $|s|^{1/2}\mu(B_p(a,r))$. The supremum is taken over $\phi \in \mathcal{A}_{x,r}(a)$, the spatially controlled test class in (3.0.1), and $0 < |s| \le c_0 r^2$. The condition requires this supremum to tend uniformly to zero as $r \downarrow 0$ through radii satisfying

$$B_p(a,5r) \Subset \Omega, \qquad \mu(B_p(a,5r)) \le C\mu(B_p(a,r)),$$

for each fixed $1 \le C < \infty$, with the parameters and support conditions specified in Section 3. The normalization uses the first power of the parabolic displacement. The required vanishing supplies the limiting temporal equation in Lemma 9.2.2 and the time-translation invariance in Lemma 9.2.3. Proposition 8.6.2 also gives a sufficient condition: locally finite temporal variation, in the sense of Definition 8.6.1, implies $\mathrm{HT}(\mu, a)$ for μ-almost every $a \in \Omega$.

A quantitative theory of parabolic geometry arose from the study of the heat equation on time-varying domains. Hofmann, Lewis and Nyström introduced parabolic uniform rectifiability through a Carleson condition measuring approximation by vertical planes [20, Section 1, pp. 355–358]. The regular parabolic Lipschitz graphs used in this theory have defining functions whose half-order time derivative belongs to parabolic BMO [5, arXiv version, Definition 2.11]. For parabolic Ahlfors–David regular sets, Bortz, Hoffman, Hofmann, Luna-Garcia and Nyström characterize parabolic uniform rectifiability by corona decompositions with respect to these graphs [5, arXiv version, Theorem 1.2]. Mattila's Example 8.2 gives a parabolic q-rectifiable set of finite positive $\mathscr{H}^q_p$-measure whose intersection with every regular parabolic Lipschitz graph has zero $\mathscr{H}^q_p$-measure [26, p. 876]. Thus the qualitative rectifiability used here does not require a covering by regular graphs.

For $n \ge 2$ and domains above entire parabolic Lipschitz graphs, Bortz, Hofmann, Martell and Nyström establish the necessity of this fractional-time regularity for L^p solvability of the Dirichlet problem for the heat equation for some finite p [6, Theorem 1.2 and the discussion following it, p. 168]. Their condition concerns the graph function; the condition $\mathrm{HT}(\mu, a)$ concerns localized temporal increments of the weight μ.

After beginning this project from the Mattila–Allard motivation, we learned of Liu and Workman's work on Brakke flows. Their construction of a canonical spacetime-Grassmann measure [23, Theorem A] and their proof of vertical parabolic rectifiability of the support of its weight [24, Theorem A] provide a further, independent motivation. They direct attention to the geometry of the spacetime traces of geometric evolutions and to the regularity questions one can ask of those traces. The theory developed here concerns abstract parabolic varifolds with vertical planes of spatial codimension one. Its hypotheses are imposed on the spacetime measure and its plane distribution, and its main conclusion is parabolic rectifiability.

The proof begins by deriving geometric information from the two variation hypotheses. Theorem 1 establishes absolute continuity of μ_+ with respect to $\mathscr{H}^q_p$, positive finite two-sided density bounds, and concentration of the plane distribution at a single plane V_a, at μ_+-almost every point a. The projected equations and compactness argument in Lemmas 10.1

and 10.3 adapt the method of De Philippis, De Rosa and Ghiraldin [10, arXiv version, proof of Lemma 3.2], with HT providing the temporal control. Their tangent-measure approach to anisotropic rectifiability avoids reliance on the isotropic monotonicity formula [10, journal version, Section 1, pp. 1125–1126]. At μ_+-almost every a, every geometric tangent of the weight is invariant under translations by V_a and is a finite positive sum of restrictions of $\mathscr{H}^q_p$ to affine planes parallel to V_a.

Theorem 2 turns this information into a rectifiable representation. The geometric argument extracts a compact graph piece of positive measure from every Borel remainder of positive μ_+-measure. It uses the graph-extraction theorem of Merlo, Mourgoglou and Puliatti [27, Proposition 4.12], in the specialized form and with the qualifications given in Section 12.4. This is the parabolic version of Preiss's quantitative projection lemma [29, Lemma 5.2]; see also [27, pp. 29–30]. On each extracted graph, the tangent structure reduces to one plane, and [27, Theorem 2.34] together with Mattila's tangent-measure criterion [26, Theorem 4.12] gives the required rectifiability; see Proposition 12.2.2. A countable covering and identification of the tangent plane then give the representation of W_+. Finally, Mattila's density theorem [26, Theorem 4.9] supplies the finite-piece density used in Theorem 2(2). Theorem 2(3) proves convergence of the mass and varifold rescalings through all radii tending to zero. The density limit and the unique flat limits are thus conclusions of the variational hypotheses.

2. Project evolution and AI Disclosure.

Written by the first author.

This section is entirely human written and edited—all typos and errors, etc. are mine. Aside from this section, this manuscript is entirely written using gpt-6-astra in codex, with quite a bit of guidance and steering. The mathematical content is in part AI-free and in part significantly AI-assisted. The project's evolution, from purely human to AI-assisted, is detailed below. In this story, the way AI was used is recorded as explicitly as we found reasonable. It is hoped other will find this story interesting or useful (or entertaining).

To begin, this project has roots in me initiating a varifold theory in the Heisenberg group to study the Pansu conjecture. (While writing this statement, gpt-6-astra revealed to me that the Pansu conjecture for the first Heisenberg group had already been solved using AI.) The foundations of the Heisenberg group varifold theory were developed by myself over the course of a year and half, starting in the Fall of 2024. Admittedly, I did try using ChatGPT during that time, but I strongly disagreed with its responses and ignored them. The main hurdle in developing this theory appeared when the I attempted to establish an Allard-type rectifiability theorem for Heisenberg group varifolds. After many failed attempts, I eventually became aware of the anisotropic varifold theory of De Philippis-De Rosa-Ghiraldin (DDG), where they circumvented similar issues.

Totally unrelated, the second author and I began work on a geometric measure theory problem set in the Heisenberg group. Eventually, the second author shared with me parabolic geometry. The similarities between parabolic geometry and the Heisenberg group geometry were of course clear; however, parabolic geometry is nicer in many respects. As such, I proposed we investigate a varifold theory for parabolic space. Around this time, the disproof of the Jacobian conjecture was announced. Out of curiosity (absolute concern) and desire to finish the project, I proposed we let ChatGPT-5.6-Sol try to complete the parabolic varifold theory. The first test was if Sol could produce a result of this form: If a parabolic varifold has "controlled" "variation", then it must be a parabolic rectifiable parabolic varifold. The main task was identifying suitable parabolic analogues of "controlled" and "variation." It is important to the story to note that Euclidean rectifiability implies parabolic rectifiability.

This led to approximately two weeks of Pro Effort prompting ChatGPT-5.6-Sol, approximately once an hour for around 12 hours a day. The process was as follows. ChatGPT-5.6-Sol was informed of the goal (a parabolic varifold theory with an Allard-type rectifiability theorem), informed about my work on the incomplete Heisenberg group varifold theory, and informed about my suspicion that DDG's anisotropic varifold theory (and hence Preiss' work) would be of use. It was also asked to generate a rigorous and robust workflow to follow. Part of this workflow consisted of Sol keeping various ledgers containing, for example, all prompts used (amassing several hundreds of thousands of lines of prompt), derived results accepted as cannon, result dependencies and notation and constants. At a high level, the process was basically: 1. Sol writes a prompt, 2. Sol runs prompt, 3. Sol performs hostile audit on output from 2., 4. Sol makes repairs on output and returns to 3. until no more repairs are necessary.

During these two weeks, we naively expected that a locally bounded first chronovariation might be enough to conclude the Allard-type rectifiability result. Sol proved by counterexample that this is insufficient due to a lack of temporal control. As such, Sol was asked to seek a natural temporal control. Eventually it effectively suggested what one might consider a "full-time derivative" control on the varifold. In hindsight, it should have been obvious that this is too strong of an assumption since time derivatives are of order 2 in parabolic geometry. But alas, Sol proposed a complete proof of an Allard-type rectifiability theorem under this full-time derivative assumption and we were temporarily satisfied.

So, we set out to carefully verify Sol's proposed proof and we convinced ourselves, near the beginning of August 2026, that Sol was indeed correct. During the manuscript preparation,

I met with the second author virtually while I was visiting Oahu, Hawaii. During our meeting, we realized that Sol's proposed full-time derivative control was probably strong enough to conclude not only parabolic rectifiability, but Euclidean rectifiability. Recall: Euclidean rectifiability implies parabolic rectifiability. I remember this moment distinctly: not only did we not actually achieve the desired theorem after all, but Hurricane Lala would also soon cause my hotel to lose its power. Thus, with my laptop powered by a solar battery, I brainstormed with Sol in finding a suitable fix, namely, a "half-time derivative" control condition on the varifold that would imply parabolic rectifiability, but not necessarily Euclidean rectifiability. Sol eventually proposed suitable half-time derivative conditions which achieved this.

Around this same time, I became familiar with agentic systems such as Danus [22]. For sake of having an always-alive terminal process to run codex processes, I repurposed a 2011 HP Windows machine into a Gentoo server hosting a codex sandbox to run Danus in. Under the guidance of ChatGPT, I fed the parabolic varifold results to Danus and had it finalize the half-time derivative control condition. The current manuscript reflects these finalized findings. Danus also serialized the results into a directed acyclic fact graph and a relatively decently written manuscript.

Around this time, OpenAI released gpt-6-astra and also a resolution of the Navier-Stokes problem. In response to complaints about how AI writes mathematics, we decided to take care to make an effort to produce a manuscript which is at least readable and reasonably cites the literature. As such, we took the outputs of Danus, fed it to codex with model gpt-6-astra set to max, and painstakingly guided Astra to write this present manuscript page-by-page. Eventually two subagents were constructed: one would check for places a citation may be needed or useful, and one would remove any AI-isms in the exposition. Several hostile audits were performed by Astra Ultra, Fable 5.1 and Opus 5.5. Regardless, this system is of course imperfect and we will continually update the manuscript, especially with citations and exposition as necessary.

It is now important to emphasize that the project's evolution is incomplete. This is in part the reason for the versioning name of this paper: Parabolic Varifolds v0.1. In a sense, this is a beta release with expected patches.

In closing, I state a roadmap for this project. The purpose is not for claiming result ownership. Instead, I hope it may inspire conversation and collaboration, and possibly prevent others from wasting credits.

v0.2 : exposition improvements and optimization
v0.3 : arbitrary codimension (already generated by Astra)
v0.4 : a parabolic analogue of Allard's compactness theorem (currently in progress)
v0.5 et al : a continuation of analogizing Allard's annals paper.

3. Main Results

Fix $0 < \kappa < 1$ and $c_0 > 0$ with $\kappa^2 + c_0 < 1$. Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, $n \geq 1$, and let $a \in \Omega$. Write B_p and U_p for the closed and open square-sum parabolic balls, respectively. For $r > 0$ with $B_p(a, r) \Subset \Omega$ and $\mu(B_p(a, r)) > 0$, let

$$\mathcal{A}_{x,r}(a) = \left\{ \phi \in C_c^1(U_p(a, \kappa r)) : \|\phi\|_\infty \leq r, \quad \|\nabla_x \phi\|_\infty \leq 1 \right\}. \tag{3.0.1}$$

Each $\phi \in \mathcal{A}_{x,r}(a)$ is extended by zero to $\mathbb{P}^n$. Writing $\tau_s(x, t) = (x, t + s)$, define

$$I_\mu(a, r) = \sup_{\substack{0<|s|\leq c_0 r^2 \\ \phi \in \mathcal{A}_{x,r}(a)}} \frac{\left| \int (\phi \circ \tau_s - \phi)\, d\mu \right|}{|s|^{1/2} \mu(B_p(a, r))}. \tag{3.0.2}$$

We say that the *half-time condition* $\mathrm{HT}(\mu, a)$ holds if, for every fixed $1 \leq C < \infty$, $I_\mu(a, r) \to 0$ uniformly as $r \downarrow 0$ through such radii satisfying

$$B_p(a, 5r) \Subset \Omega, \qquad \mu(B_p(a, 5r)) \leq C \mu(B_p(a, r)). \tag{3.0.3}$$

For the theorems below, let W be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, $n \geq 1$, as defined in Section 6. Put $q = n + 1$ and $\mu = \|W\|$, and set

$$\begin{aligned} E_+ &= \left\{ a \in \Omega : \liminf_{r \downarrow 0} \frac{\mu(B_p(a, r))}{(2r)^q} > 0 \right\}, \\ \mu_+ &= \mu \llcorner E_+, \qquad W_+ = W \llcorner (E_+ \times \mathbb{P}G). \end{aligned} \tag{3.0.4}$$

The set E_+ is Borel by Lemma 5.3.1, and W_+ has weight μ_+ by (6.1.1).

Main Theorem. *Suppose that W has locally bounded spatial first chronovariation and that $\mathrm{HT}(\mu, a)$ holds for μ_+-almost every $a \in \Omega$. Then W_+ is a rectifiable parabolic varifold.*

The proof is divided into two stages. Theorem 1 establishes the density bounds, the Dirac form of the plane distribution, and the translation invariance of tangent measures. Theorem 2 proves the rectifiable representation, identifies the exact densities, and proves convergence of the mass and varifold rescalings as $r \downarrow 0$. The Main Theorem follows from Theorem 2(1) and Definition 6.3.1.

Theorem 1. *Suppose that W has locally bounded spatial first chronovariation and that $\mathrm{HT}(\mu, a)$ holds for μ_+-almost every $a \in \Omega$. Then*

(1) E_+ is Borel and $\mathscr{H}_p^q \llcorner E_+$ is σ-finite.
(2) $\mu_+ \ll \mathscr{H}_p^q$.
(3) $\mu_+(\{a \in \Omega : \Theta_{p*}^q(\mu, a) = \infty\}) = 0$.
(4) For μ_+-almost every $a \in \Omega$,

$$0 < \Theta_{p*}^q(\mu, a) \leq \Theta_p^{q*}(\mu, a) < \infty.$$

(5) There is a Borel field $a \mapsto V_a \in \mathbb{P}G$, defined μ_+-almost everywhere, such that

$$W_+ = \mu_+ \otimes \delta_{V_a}.$$

For μ_+-almost every $a \in \Omega$,
(a) every $\tau \in \mathrm{Tan}_p^q(\mu, a)$ is V_a-translation invariant and has the form

$$\tau = \sum_{j=1}^N c_j \mathscr{H}_p^q \llcorner (V_a + z_j),$$

where $1 \leq N < \infty$, $c_j > 0$ and the $z_j \in V_a^\perp$ are distinct;
(b) every $\Upsilon \in \mathrm{Tan}_p^q(W, a)$ has the form $\Upsilon = \tau \otimes \delta_{V_a}$ for some $\tau \in \mathrm{Tan}_p^q(\mu, a)$.

Theorem 2. *Under the hypotheses of Theorem 1, there are a Borel countably $\mathbb{P}$-rectifiable set $K \subset E_+$ and a Borel function $\theta : K \to (0,\infty)$ such that*

(1) $\mu_+ = \theta \mathscr{H}^q_p \llcorner K$ and

$$W_+ = (\theta \mathscr{H}^q_p \llcorner K) \otimes \delta_{\mathrm{recTan}(K,a)};$$

(2) for μ_+-almost every $a \in K$, $V_a = \mathrm{recTan}(K,a)$ and

$$\Theta^q_p(\mu, a) = \Theta^q_p(\mu_+, a) = v(q)\theta(a);$$

(3) at the same points, as $r \downarrow 0$,

$$\mu_{a,r},\ (\mu_+)_{a,r} \rightharpoonup \theta(a)\mathscr{H}^q_p \llcorner V_a,$$
$$W_{a,r},\ (W_+)_{a,r} \rightharpoonup \theta(a)(\mathscr{H}^q_p \llcorner V_a) \otimes \delta_{V_a}.$$

4. Classical Varifold Theory

4.1. **Notation.** Let X be a locally compact metric space and let $U \subset X$ be open. For a finite-dimensional real Hilbert space H, we write

$$C_c(U;H) = \{f \in C(U;H) : \operatorname{supp} f \text{ is compact in } U\}.$$

On $\mathbb{R}^{n\times n}$, $n \geq 1$, we use the Frobenius inner product and norm

$$A : B = \sum_{j,k=1}^{n} A_{jk}B_{jk}, \qquad |A| = (A:A)^{1/2} \qquad (A, B \in \mathbb{R}^{n\times n}).$$

For $n \geq 1$, $A \in \mathbb{R}^{n\times n}$ and an $(n-1)$-dimensional linear subspace $L \subset \mathbb{R}^n$, define the *Euclidean Jacobian* of $A|_L$ by

$$J_{n-1}(A|_L) = \sqrt{\det\big[(Ae_i)\cdot(Ae_j)\big]_{i,j=1}^{n-1}}, \tag{4.1.1}$$

where $e_1, \ldots, e_{n-1}$ is an orthonormal basis of L. The value does not depend on the choice of orthonormal basis. We use the empty-determinant convention $J_0(A|_{\{0\}}) = 1$. This is the Euclidean Jacobian convention of [33, Chapter 2, (3.3) and (4.12)]. We write A^{T} for the transpose of A and I_n for the $n \times n$ identity matrix.

4.2. **Radon Measures.** For every integer $d \geq 1$, let $\mathcal{L}^d$ denote d-dimensional Lebesgue measure on $\mathbb{R}^d$.

For every integer $d \geq 1$, put $\omega_d = \mathcal{L}^d(\{x \in \mathbb{R}^d : |x| \leq 1\})$, and set $\omega_0 = 1$. For integers $1 \leq m \leq d$ and $A \subset \mathbb{R}^d$, the Euclidean m-dimensional Hausdorff measure is

$$\mathscr{H}^m(A) = \liminf_{\delta\downarrow 0} \left\{ \sum_{j=1}^{\infty} \omega_m \left(\frac{\operatorname{diam} E_j}{2}\right)^m : A \subset \bigcup_{j=1}^{\infty} E_j, \quad \operatorname{diam} E_j < \delta \right\},$$

where $E_j \subset \mathbb{R}^d$ and diam is Euclidean diameter, with $\operatorname{diam} \varnothing = 0$. This is the normalization of [33, Chapter 1, (2.1)–(2.2) and Theorem 2.9]; under an isometric identification of an m-plane with $\mathbb{R}^m$, the restriction of $\mathscr{H}^m$ to that plane is $\mathcal{L}^m$.

Definition 4.2.1 (Translation invariance)**.** Let μ be a positive Radon measure on $\mathbb{R}^d$, and let $V \subset \mathbb{R}^d$ be a linear subspace. We say that μ is *invariant under translations by V* if

$$\mu(A + v) = \mu(A) \qquad \text{for every Borel } A \subset \mathbb{R}^d \text{ and every } v \in V. \tag{4.2.1}$$

Equivalently, $(x \mapsto x + v)_{\#}\mu = \mu$ for every $v \in V$. When $V = \mathbb{R}^d$, we call μ *translation invariant*.

For positive Radon measures λ_i and λ on U, we write $\lambda_i \rightharpoonup \lambda$ in U if

$$\lim_{i\to\infty} \int_U f \, d\lambda_i = \int_U f \, d\lambda \qquad \text{for every } f \in C_c(U;\mathbb{R}). \tag{4.2.2}$$

This is the convergence used in [33, Chapter 1, Theorem 4.16]. Suppose instead that each λ_i is a positive Radon measure on an open set $U_i \subset X$, and that every compact set $K \subset U$ is contained in U_i for all sufficiently large i. We write $\lambda_i \rightharpoonup \lambda$ in U if

$$\lim_{i\to\infty} \int_{U_i\cap U} f \, d\lambda_i = \int_U f \, d\lambda \qquad \big(f \in C_c(U;\mathbb{R})\big). \tag{4.2.3}$$

For each f, the integrals on the left are taken only for indices i large enough that $\operatorname{supp} f \subset U_i$.

Let η be an H-valued Radon measure on U, where H is a finite-dimensional real Hilbert space. Its *total variation* is the positive Radon measure $|\eta|$ characterized, for every open $O \subset U$, by

$$|\eta|(O) = \sup\left\{ \int_O \langle f, d\eta\rangle_H : f \in C_c(O;H),\ \sup_{z\in O} |f(z)|_H \le 1 \right\}. \tag{4.2.4}$$

See [33, Chapter 1, Theorem 4.14 and its proof]. For matrix-valued measures we use the Frobenius inner product.

For $d \ge 1$, a compactly supported finite H-valued Radon measure η on $\mathbb{R}^d$, and $k \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, write

$$(\eta * k)(z) = \int_{\mathbb{R}^d} k(z-w)\, d\eta(w) \qquad \text{for } \mathcal{L}^d\text{-almost every } z \in \mathbb{R}^d. \tag{4.2.5}$$

For a compactly supported $f \in L^1(\mathbb{R}^d;H)$, put $f * k = (f\mathcal{L}^d) * k$.

For an open set $\Omega \subset \mathbb{R}^n_x \times \mathbb{R}_t$, $n \ge 1$, and an $\mathbb{R}^{n\times n}$-valued Radon measure M on Ω, define its *spatial distributional divergence* by

$$\langle \operatorname{div}_x M, \varphi\rangle = -\int_\Omega D_x\varphi : dM, \qquad \varphi \in C_c^\infty(\Omega;\mathbb{R}^n), \tag{4.2.6}$$

where $(D_x\varphi)_{ij} = \partial_{x_j}\varphi_i$ for $1 \le i,j \le n$.

Definition 4.2.2 (Lower Euclidean density)**.** Let $1 \le m \le d$, let $U \subset \mathbb{R}^d$ be open, and let λ be a positive Radon measure on U. For $x \in \mathbb{R}^d$ and $r > 0$, put $B_E(x,r) = \{y \in \mathbb{R}^d : |y-x| < r\}$. The *lower Euclidean m-density* of λ at $x \in U$ is

$$\Theta^m_{E*}(\lambda, x) = \liminf_{r\downarrow 0} \frac{\lambda(B_E(x,r)\cap U)}{\omega_m r^m} \in [0,\infty]. \tag{4.2.7}$$

This is the normalization in [10, arXiv version, Section 2.1]. For each $r > 0$, the function $x \mapsto \lambda(B_E(x,r)\cap U)$ is Borel by [14, Theorem 2.36]. Continuity from below in r allows the lower limit in (4.2.7) to be taken over positive rational radii, so the density is Borel in x.

4.3. **Varifolds.** For the classical background in Sections 4.3–4.5, see Allard [1], Brakke [7, Sections 2.5–2.9] and Tonegawa [38, Sections 1.2–1.5].

Fix integers $1 \le m \le d$ and an open set $U \subset \mathbb{R}^d$. Let $G(d,m)$ be the Grassmannian of m-dimensional linear subspaces of $\mathbb{R}^d$. For $S \in G(d,m)$, write P_S for the orthogonal projection onto S. We equip $G(d,m)$ with the metric $(S,T) \mapsto |P_S - P_T|$, using the Frobenius norm. The space $G(d,m)$ is compact; see [33, Chapter 8, Section 1, pp. 205–206].

Definition 4.3.1 (Varifold)**.** An *m-varifold* on U is a positive Radon measure V on $U \times G(d,m)$. Its *weight* is the Radon measure

$$\|V\| = \pi_\# V, \qquad \pi(x,S) = x, \qquad \|V\|(A) = V(A \times G(d,m)) \tag{4.3.1}$$

for every Borel set $A \subset U$. The weight is Radon because π is continuous and proper.

Definition 4.3.2 (Countable rectifiability and approximate tangents)**.** A Borel set $M \subset U$ is *countably m-rectifiable* if there are m-dimensional embedded C^1 submanifolds $M_j \subset \mathbb{R}^d$, $j \geq 1$, such that

$$\mathscr{H}^m\left(M \setminus \bigcup_{j=1}^{\infty} M_j\right) = 0. \tag{4.3.2}$$

This agrees with the definition by countably many Lipschitz images in [33, Chapter 3, (1.1) and Lemma 1.2].

Let $\theta : M \to (0,\infty)$ be Borel and suppose that $\mu = \theta\mathscr{H}^m \llcorner M$ is Radon on U. An *approximate tangent plane with respect to* θ at $x \in M$ is a plane $T_xM \in G(d,m)$ such that

$$\lim_{r\downarrow 0} r^{-m} \int_M \varphi\left(\frac{y-x}{r}\right)\theta(y)\,d\mathscr{H}^m(y) = \theta(x)\int_{T_xM}\varphi\,d\mathscr{H}^m \tag{4.3.3}$$

for every $\varphi \in C_c(\mathbb{R}^d;\mathbb{R})$. For countably m-rectifiable M, this plane exists and is unique at $\mathscr{H}^m$-almost every $x \in M$ by [33, Chapter 3, Definition 1.7 and Theorem 1.9]. The field $x \mapsto T_xM$ has a Borel representative: on a countable C^1 covering as in (4.3.2), it agrees almost everywhere with $x \mapsto T_xM_j$ on $M \cap M_j$; see [33, Chapter 3, Remark 1.8(1) and Section 2]. We use such a representative and assign an arbitrary fixed plane on a Borel exceptional null set.

Definition 4.3.3 (Rectifiable varifold)**.** An m-varifold V on U is *rectifiable* if there are a Borel countably m-rectifiable set $M \subset U$ and a Borel function $\theta : M \to (0,\infty)$, with $\theta\mathscr{H}^m \llcorner M$ Radon on U, such that

$$\int_{U\times G(d,m)} F(x,S)\,dV(x,S) = \int_M F(x,T_xM)\theta(x)\,d\mathscr{H}^m(x) \tag{4.3.4}$$

for every nonnegative Borel function $F : U \times G(d,m) \to [0,\infty]$. We write $V = \mathbf{v}(M,\theta)$; then $\|V\| = \theta\mathscr{H}^m \llcorner M$. Changes to the tangent field on an $\mathscr{H}^m$-null set do not change V, and $M = \varnothing$ gives the zero varifold. This is the rectifiable-varifold representation in [33, Chapter 8, Section 1, p. 206].

4.4. **Image Varifolds.** Let V be an m-varifold on $U \subset \mathbb{R}^d$, let $U' \subset \mathbb{R}^d$ be open, and let $f \in C^1(U;U')$. For $x \in U$ and $S \in G(d,m)$, define the *tangential Jacobian*

$$J_Sf(x) = \sqrt{\det\big[(Df(x)e_i)\cdot(Df(x)e_j)\big]_{i,j=1}^{m}}, \tag{4.4.1}$$

where $e_1,\dots,e_m$ is an orthonormal basis of S. The value is independent of that basis. Put

$$\mathcal{G}_f^+ = \{(x,S) \in U\times G(d,m) : J_Sf(x) > 0\}.$$

For $(x,S) \in \mathcal{G}_f^+$, the image $Df(x)[S]$ belongs to $G(d,m)$.

Definition 4.4.1 (Image varifold)**.** Suppose that f is proper on $\operatorname{supp}\|V\|$: for every compact $K \subset U'$, the set $f^{-1}(K)\cap\operatorname{supp}\|V\|$ is compact in U. The *image varifold* $f_\#V$ is the positive Radon measure on $U'\times G(d,m)$ defined by

$$\int_{U'\times G(d,m)} F\,d(f_\#V) = \int_{\mathcal{G}_f^+} F\big(f(x),Df(x)[S]\big)J_Sf(x)\,dV(x,S) \tag{4.4.2}$$

for every $F \in C_c(U'\times G(d,m);\mathbb{R})$. Thus pairs with $J_Sf(x) = 0$ contribute no mass, and no m-plane is assigned to their lower-dimensional images. See [33, Chapter 8, (2.1)].

To check local finiteness in (4.4.2), let $K \subset U'$ be the projection of $\operatorname{supp} F$. The integrand vanishes V-almost everywhere outside $(f^{-1}(K)\cap\operatorname{supp}\|V\|)\times G(d,m)$. This set is compact, and $J_Sf(x)$ is bounded there. The right-hand side therefore defines a positive linear functional on $C_c(U'\times G(d,m);\mathbb{R})$, which gives a Radon measure by [14, Theorem 7.2].

4.5. **The First Variation.** Let V be an m-varifold on an open set $U \subset \mathbb{R}^d$. For $X \in C^1_c(U;\mathbb{R}^d)$ and $S \in G(d,m)$, define

$$\operatorname{div}_S X(x) = \operatorname{tr}\big(P_S DX(x)\big) = \sum_{i=1}^{m} e_i \cdot DX(x) e_i, \tag{4.5.1}$$

where $e_1,\dots,e_m$ is an orthonormal basis of S and $(DX)_{ij} = \partial_{x_j} X_i$.

Definition 4.5.1 (First variation)**.** The *first variation* of V is the real linear functional on $C^1_c(U;\mathbb{R}^d)$ given by

$$\delta V(X) = \int_{U\times G(d,m)} \operatorname{div}_S X(x)\, dV(x,S). \tag{4.5.2}$$

See [33, Chapter 8, (2.3)–(2.4)].

For sufficiently small $|s|$, the maps $f_s(x) = x + sX(x)$ are C^1 diffeomorphisms of U onto itself, equal to the identity outside $\operatorname{supp} X$. The classical first-variation formula [33, Chapter 8, (2.2)–(2.4)] reads

$$\delta V(X) = \frac{d}{ds}\bigg|_{s=0} \int_{U\times G(d,m)} \big(J_S f_s(x) - 1\big)\, dV(x,S). \tag{4.5.3}$$

The integral is finite because its integrand vanishes outside $\operatorname{supp} X \times G(d,m)$. Indeed, differentiating (4.4.1) at $s=0$ gives $\partial_s|_{s=0} J_S f_s(x) = \operatorname{div}_S X(x)$; the difference quotients are uniformly bounded on that compact set, so dominated convergence [14, Theorem 2.24] gives (4.5.3).

Definition 4.5.2 (Locally bounded first variation)**.** We say that V has *locally bounded first variation* in U if, for every open $O \Subset U$, there is $0 \le C_O < \infty$ such that

$$|\delta V(X)| \le C_O \,\|X\|_\infty \qquad \big(X \in C^1_c(U;\mathbb{R}^d),\ \operatorname{supp} X \subset O\big). \tag{4.5.4}$$

By [33, Chapter 8, (2.5)–(2.6)], condition (4.5.4) is equivalent to the existence of a unique $\mathbb{R}^d$-valued Radon measure on U, also denoted by δV, such that

$$\delta V(X) = \int_U X \cdot d(\delta V) \qquad \big(X \in C^1_c(U;\mathbb{R}^d)\big). \tag{4.5.5}$$

Its total variation is denoted by $\|\delta V\|$ and satisfies

$$\|\delta V\|(O) = \sup\left\{ |\delta V(X)| : \begin{array}{l} X \in C^1_c(U;\mathbb{R}^d), \quad \operatorname{supp} X \subset O, \\ \|X\|_\infty \le 1 \end{array} \right\} \tag{4.5.6}$$

for every open $O \Subset U$.

4.6. **Allard's Rectifiability Theorem.** We recall the positive-lower-density form of Allard's rectifiability theorem [1], as stated in [10, arXiv version, Section 1]. It is also the area-integrand case of [10, arXiv version, Theorem 1.2(i)]. For the formulation with positive density almost everywhere, see [33, Chapter 8, Theorem 5.5].

Theorem 4.6.1 (Allard)**.** *Let $1 \le m < d$, let $U \subset \mathbb{R}^d$ be open, and let V be an m-varifold on U with locally bounded first variation. Then the m-varifold*

$$V \llcorner \big(\{x \in U : \Theta^m_{E*}(\|V\|,x) > 0\} \times G(d,m)\big) \tag{4.6.1}$$

is rectifiable in the sense of Definition 4.3.3. In particular, if $\Theta^m_{E*}(\|V\|,x) > 0$ for $\|V\|$-almost every $x \in U$, then V is rectifiable.

4.7. **The Anisotropic Rectifiability Theorem.** Fix integers $1 \le m < d$ and an open set $U \subset \mathbb{R}^d$. We use the smooth structure on $G(d,m)$ obtained by identifying S with its orthogonal projection P_S; see [10, arXiv version, Appendix A].

Definition 4.7.1 (Anisotropic energy and first variation)**.** Let $F \in C^1(U \times G(d,m);(0,\infty))$. For an m-varifold V on U and an open set $O \Subset U$, define

$$\mathcal{F}_F(V;O) = \int_{O\times G(d,m)} F(x,S)\,dV(x,S). \tag{4.7.1}$$

For $X \in C^1_c(U;\mathbb{R}^d)$, set $f_s(x) = x + sX(x)$ for sufficiently small $|s|$. The *anisotropic first variation* of V is

$$\delta_F V(X) = \frac{d}{ds}\Big|_{s=0} \int_{U\times G(d,m)} \big[F\big(f_s(x), Df_s(x)[S]\big) J_S f_s(x) - F(x,S)\big]\,dV(x,S). \tag{4.7.2}$$

The integrand vanishes outside $\operatorname{supp} X \times G(d,m)$, so this definition requires only local finiteness of V.

For $(x,S) \in U \times G(d,m)$, define $B_F(x,S) \in \mathbb{R}^{d\times d}$ by the identities

$$B_F(x,S) : A = F(x,S)\operatorname{tr}(P_S A) + \frac{d}{ds}\Big|_{s=0} F\big(x,(I_d + sA)[S]\big) \qquad (A \in \mathbb{R}^{d\times d}). \tag{4.7.3}$$

Here $(I_d + sA)[S] \in G(d,m)$ for sufficiently small $|s|$. The right-hand side is linear in A, and hence determines a unique matrix, by [10, arXiv version, Lemma A.1 and (A.5)]. The first-variation formula [10, arXiv version, Lemma A.2] is

$$\delta_F V(X) = \int_{U\times G(d,m)} \big[\nabla_x F(x,S)\cdot X(x) + B_F(x,S) : DX(x)\big]\,dV(x,S). \tag{4.7.4}$$

We say that V has *locally bounded anisotropic first variation* with respect to F if, for every open $O \Subset U$, there is $0 \le C_O < \infty$ such that

$$|\delta_F V(X)| \le C_O \,\|X\|_\infty \qquad \big(X \in C^1_c(U;\mathbb{R}^d),\ \operatorname{supp} X \subset O\big). \tag{4.7.5}$$

Equivalently, there is a unique $\mathbb{R}^d$-valued Radon measure on U, also denoted by $\delta_F V$, such that

$$\delta_F V(X) = \int_U X \cdot d(\delta_F V) \qquad \big(X \in C^1_c(U;\mathbb{R}^d)\big). \tag{4.7.6}$$

See [10, arXiv version, Section 2.2, p. 8].

Definition 4.7.2 (Atomic condition)**.** Let $F \in C^1(U\times G(d,m);(0,\infty))$, and let $\mathcal{P}(G(d,m))$ denote the Borel probability measures on $G(d,m)$. For $x \in U$ and $\nu \in \mathcal{P}(G(d,m))$, set

$$A_F(x,\nu) = \int_{G(d,m)} B_F(x,S)\,d\nu(S). \tag{4.7.7}$$

The integrand F satisfies the *atomic condition at* x if, for every $\nu \in \mathcal{P}(G(d,m))$,

(i) $\dim\ker A_F(x,\nu) \le d-m$;
(ii) if $\dim\ker A_F(x,\nu) = d-m$, then $\nu = \delta_{S_0}$ for some $S_0 \in G(d,m)$.

This is [10, arXiv version, Definition 1.1].

Theorem 4.7.1 (De Philippis–De Rosa–Ghiraldin)**.** *Let $F \in C^1(U\times G(d,m);(0,\infty))$ satisfy the atomic condition at every $x \in U$. If V is an m-varifold on U with locally bounded anisotropic first variation with respect to F, then*

$$V \llcorner \big(\{x \in U : \Theta^m_{E*}(\|V\|,x) > 0\} \times G(d,m)\big)$$

is a rectifiable m-varifold on U. If F is autonomous, meaning $F(x,S) = F(S)$, the atomic condition is also necessary for this conclusion to hold for every m-varifold with locally bounded anisotropic first variation with respect to F.

This is [10, arXiv version, Theorem 1.2]. For $F \equiv 1$, equation (4.7.3) gives $B_F(x, S) = P_S$ and

$$v \cdot A_1(x, \nu)v = \int_{G(d,m)} |P_S v|^2 \, d\nu(S) \qquad (v \in \mathbb{R}^d). \tag{4.7.8}$$

Let $K = \ker A_1(x, \nu)$. Applying (4.7.8) to a basis of K shows that $K \subset S^\perp$ for ν-almost every S. Thus $\dim K \leq d - m$, and equality forces $S = K^\perp$ for ν-almost every S. This proves the atomic condition for $F \equiv 1$ and recovers Theorem 4.6.1; see also [10, arXiv version, paragraph following Theorem 1.2].

For $m \geq 2$, De Rosa and Kolasiński relate the atomic condition to geometric ellipticity: a C^1 integrand satisfying the atomic condition at a point satisfies strict Almgren ellipticity there [12, arXiv version, Theorem A]. Uniform Almgren ellipticity does not follow in general [12, arXiv version, Remark 9.26].

Theorems 4.6.1 and 4.7.1 impose variation bounds for all $X \in C^1_c(U; \mathbb{R}^d)$, and their conclusions use Euclidean density and rectifiability. For a parabolic varifold W, the bound (8.4.5) controls only the spatial derivatives in (8.4.3). The half-time condition (3.0.2)–(3.0.3) is a condition on temporal translations of the weight μ; the atomic condition instead constrains the integrand F. Lemmas 9.2.1 and 9.2.2 derive the separate spatial and temporal identities for the selected parabolic tangent measures. The conclusions of Theorems 1–2 concern the positive parabolic density set in (3.0.4) and parabolic rectifiability as defined in Definition 6.3.1.

5. Parabolic Geometry

5.1. **Parabolic Space and the Square-Sum Metric.** Let $n \geq 1$ and put $\mathbb{P}^n = \mathbb{R}^n_x \times \mathbb{R}_t$.

For $z = (x, t)$ and $w = (y, s)$ in $\mathbb{P}^n$, define the *parabolic norm* and the *square-sum parabolic metric* by

$$\|z\|_p = (|x|^2 + |t|)^{1/2}, \qquad d_p(z, w) = \|z - w\|_p = (|x - y|^2 + |t - s|)^{1/2}. \tag{5.1.1}$$

For $a = (a_x, a_t) \in \mathbb{P}^n$ and $r > 0$, the closed and open parabolic balls are

$$\begin{aligned} B_p(a, r) &= \{(x, t) \in \mathbb{P}^n : |x - a_x|^2 + |t - a_t| \leq r^2\}, \\ U_p(a, r) &= \{(x, t) \in \mathbb{P}^n : |x - a_x|^2 + |t - a_t| < r^2\}. \end{aligned} \tag{5.1.2}$$

These conventions follow [26, Sections 1–2].

The *temporal projection* is the map

$$\pi_t : \mathbb{P}^n \to \mathbb{R}, \qquad \pi_t(x, t) = t. \tag{5.1.3}$$

For every set $A \subset \mathbb{P}^n$, its *temporal projection* is the image set

$$\pi_t(A) = \{s \in \mathbb{R} : \text{there exists } x \in \mathbb{R}^n \text{ with } (x, s) \in A\}.$$

If $U \subset \mathbb{P}^n$ is open, then $\pi_t(U)$ is open in $\mathbb{R}$. Indeed, for each $t \in \pi_t(U)$ choose $x \in \mathbb{R}^n$ with $(x, t) \in U$. Openness of U gives $\varepsilon > 0$ such that $(x, s) \in U$ whenever $|s - t| < \varepsilon$. Hence $(t - \varepsilon, t + \varepsilon) \subset \pi_t(U)$.

5.2. **Parabolic Hausdorff Measure.** For $E \subset \mathbb{P}^n$, write

$$\operatorname{diam}_p E = \sup\{d_p(z, w) : z, w \in E\}, \qquad \operatorname{diam}_p \varnothing = 0.$$

For $s > 0$ and $A \subset \mathbb{P}^n$, define

$$\mathscr{H}^s_p(A) = \liminf_{\delta \downarrow 0} \left\{ \sum_{j=1}^{\infty} (\operatorname{diam}_p E_j)^s : A \subset \bigcup_{j=1}^{\infty} E_j, \quad \operatorname{diam}_p E_j < \delta \right\}. \tag{5.2.1}$$

The covering sets E_j are arbitrary subsets of $\mathbb{P}^n$. We use the normalization of [26, Section 2, p. 857].

Lemma 5.2.1 (5r covering lemma [25, Theorem 2.1])**.** *Let $\mathcal{B}$ be a family of closed parabolic balls in $\mathbb{P}^n$ with positive, uniformly bounded radii. There is an at most countable pairwise disjoint subfamily $\{B_p(a_i, r_i) : i \in I\} \subset \mathcal{B}$ such that*

$$\bigcup_{B\in\mathcal{B}} B \subset \bigcup_{i\in I} B_p(a_i, 5r_i).$$

Proof. We use the dyadic selection argument in [33, Chapter 1, Lemma 3.4]. Choose $R > 0$ such that $r < R$ for every $B_p(a, r) \in \mathcal{B}$. For each integer $\ell \geq 0$, put

$$\mathcal{B}_\ell = \left\{B_p(a, r) \in \mathcal{B} : 2^{-\ell-1}R \leq r < 2^{-\ell}R\right\}.$$

Inductively choose a subfamily $\mathcal{G}_\ell \subset \mathcal{B}_\ell$ that is maximal with respect to inclusion among pairwise disjoint subfamilies whose members are disjoint from every ball in $\bigcup_{0\leq h<\ell} \mathcal{G}_h$. Then $\mathcal{G} = \bigcup_{\ell=0}^{\infty} \mathcal{G}_\ell$ is pairwise disjoint. Its members have nonempty interiors, so separability of $\mathbb{P}^n$ implies that $\mathcal{G}$ is at most countable.

Let $B_p(a, r) \in \mathcal{B}_\ell \setminus \mathcal{G}$. By construction and maximality, there is a ball $B_p(b, s) \in \mathcal{G}_h$, with $0 \leq h \leq \ell$, that intersects $B_p(a, r)$. The bounds defining the radius classes give $r < 2s$, and intersection gives $d_p(a, b) \leq r + s$. Thus, for every $z \in B_p(a, r)$,

$$d_p(z, b) \leq d_p(z, a) + d_p(a, b) \leq 2r + s < 5s,$$

and therefore $B_p(a, r) \subset B_p(b, 5s)$. Every ball in $\mathcal{G}$ is contained in its own fivefold enlargement. Indexing $\mathcal{G}$ by an at most countable set I proves the assertion. □

5.3. **Densities.** Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and let $q > 0$. For $a \in \Omega$, define the *lower and upper q-densities* of μ at a by

$$\begin{aligned} \Theta^q_{p*}(\mu, a) &= \liminf_{r\downarrow 0} \frac{\mu(B_p(a, r))}{r^q}, \\ \Theta^{q*}_p(\mu, a) &= \limsup_{r\downarrow 0} \frac{\mu(B_p(a, r))}{r^q}. \end{aligned} \tag{5.3.1}$$

The limits use radii with $B_p(a, r) \Subset \Omega$ and take values in $[0, \infty]$. When the two values agree, their common value is denoted by $\Theta^q_p(\mu, a)$.

Lemma 5.3.1. *Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and let $q > 0$. The set*

$$E_+ = \left\{a \in \Omega : \liminf_{r\downarrow 0} \frac{\mu(B_p(a, r))}{(2r)^q} > 0\right\}$$

is Borel, and $\mathscr{H}^q_p \llcorner E_+$ is σ-finite. Moreover, the set

$$E_\infty = \{a \in \Omega : \Theta^{q*}_p(\mu, a) = \infty\} \tag{5.3.2}$$

is Borel and satisfies $\mathscr{H}^q_p(E_\infty) = 0$.

Proof. For each $r > 0$, the function

$$f_r(a) = \mu(B_p(a, r) \cap \Omega) = \int_\Omega \mathbf{1}_{\{d_p(a,z)\leq r\}}\, d\mu(z)$$

is Borel in the variable a, since the integrand is jointly Borel in (a, z); see [14, Proposition 1.5 and Theorem 2.36]. For each $a \in \Omega$ and sufficiently small $r > 0$, continuity from above gives $f_s(a) \to f_r(a)$ as rational radii $s \downarrow r$. Thus

$$2^{-q}\Theta^q_{p*}(\mu, a) = \lim_{\ell\to\infty} \inf_{\substack{r\in\mathbb{Q}\\ 0<r<1/\ell}} \frac{f_r(a)}{(2r)^q},$$

$$2^{-q}\Theta^{q*}_p(\mu, a) = \lim_{\ell\to\infty} \sup_{\substack{r\in\mathbb{Q}\\ 0<r<1/\ell}} \frac{f_r(a)}{(2r)^q}.$$

Both density functions are Borel in the variable a, and hence E_+ and E_∞ are Borel.

Choose compact sets K_j exhausting Ω, and compact sets $K_j' \subset \Omega$ such that $K_j \subset \operatorname{int} K_j'$, where the interior is taken in $\mathbb{P}^n$. Such compact neighborhoods exist because Ω is open and the parabolic metric is proper. Since K_j is compact and contained in the open set $\operatorname{int} K_j'$, its distance from the complement of that open set is positive. We may therefore choose $\rho_j > 0$ with

$$5\rho_j < \operatorname{dist}_p(K_j, \mathbb{P}^n \setminus \operatorname{int} K_j').$$

Then $B_p(a, 5\rho_j) \subset \operatorname{int} K_j' \subset K_j'$ for every $a \in K_j$. Put $r_{jk} = \min\{\rho_j, 1/k\}$ and, for positive integers j, m, k, set

$$E_{jmk} = \left\{a \in K_j : \mu(B_p(a,r)) \geq m^{-1}(2r)^q \text{ for every } 0 < r < r_{jk}\right\}.$$

For a fixed $a \in K_j$, the lower-growth inequality $\mu(B_p(a,r)) \geq m^{-1}(2r)^q$ defining E_{jmk} need only be checked at rational radii $r \in (0, r_{jk})$. Indeed, for any real $0 < r < r_{jk}$, choose rational radii $s_\ell \in (r, r_{jk})$ decreasing to r. If the lower-growth inequality holds at each s_ℓ, continuity from above gives

$$\mu(B_p(a,r)) = \lim_{\ell\to\infty} \mu(B_p(a,s_\ell)) \geq \lim_{\ell\to\infty} m^{-1}(2s_\ell)^q = m^{-1}(2r)^q.$$

Consequently,

$$E_{jmk} = \bigcap_{\substack{s\in\mathbb{Q}\\ 0<s<r_{jk}}} \left\{a \in K_j : f_s(a) \geq m^{-1}(2s)^q\right\}.$$

Each set in the intersection is Borel, since K_j is compact and f_s is Borel in the variable a. Thus E_{jmk} is Borel. A positive lower density gives a bound $\mu(B_p(a,r)) \geq m^{-1}(2r)^q$ for some positive integer m at all sufficiently small radii. Therefore,

$$E_+ = \bigcup_{j,m,k=1}^{\infty} E_{jmk}.$$

For the Hausdorff measure estimate, we use the covering argument for metric density comparison; see [33, Chapter 1, Theorem 3.3]. Fix j, m, k and $\delta > 0$, and set $R = \min\{r_{jk}, \delta/10\}$. Apply Lemma 5.2.1 to the family

$$\mathcal{B} = \{B_p(a,r) : a \in E_{jmk},\ 0 < r < R\}.$$

It gives a countable pairwise disjoint subfamily $B_i = B_p(a_i, r_i)$ such that

$$E_{jmk} \subset \bigcup_i B_p(a_i, 5r_i).$$

The enlarged balls have parabolic diameter $10r_i < \delta$, and their total Hausdorff covering cost satisfies

$$\sum_i (10r_i)^q = 5^q \sum_i (2r_i)^q \leq 5^q m \sum_i \mu(B_i) \leq 5^q m\, \mu(K_j').$$

The last inequality follows from disjointness and $B_i \subset K_j'$. Letting $\delta \downarrow 0$ in the definition of Hausdorff measure gives

$$\mathscr{H}_p^q(E_{jmk}) \leq 5^q m\, \mu(K_j') < \infty.$$

The countable decomposition above proves the asserted σ-finiteness.

For every j, the inclusion $K_j \subset \operatorname{int} K_j'$ gives

$$\Theta_p^{q*}(\mu \llcorner K_j', a) = \infty \qquad (a \in E_\infty \cap K_j).$$

Apply the metric density comparison [33, Chapter 1, Theorem 3.3, pp. 14–15] with $A_1 = E_\infty \cap K_j$ and $A_2 = K_j'$. In the Hausdorff normalization (5.2.1), it gives

$$M \mathscr{H}_p^q(E_\infty \cap K_j) \leq \mu(K_j') < \infty \qquad \text{for every } M > 0.$$

Letting $M \to \infty$ gives $\mathscr{H}_p^q(E_\infty \cap K_j) = 0$ for every j. Since the sets K_j exhaust Ω, $\mathscr{H}_p^q(E_\infty) = 0$. □

Lemma 5.3.2 (Selection of scales)**.** *Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, let $q > 0$, and suppose that $a \in \Omega$ satisfies*

$$\liminf_{r\downarrow 0} \frac{\mu(B_p(a,r))}{(2r)^q} > 0. \tag{5.3.3}$$

The lower limit may be infinite. Let $R_i > 0$ decrease to zero with $B_p(a,R_i) \Subset \Omega$, and let $\varepsilon_i > 0$ decrease to zero. After omitting finitely many indices, there are radii $0 < s_i \le R_i/5$ with $0 < \mu(B_p(a,s_i)) < \infty$ such that

$$\frac{\mu(B_p(a,5s_i))}{\mu(B_p(a,s_i))} \le 5^q(1+\varepsilon_i) \tag{5.3.4}$$

and

$$\frac{\mu(B_p(a,ts_i))}{\mu(B_p(a,s_i))} \ge \frac{t^q}{5^q(1+\varepsilon_i)} \qquad \text{for every } 0 < t \le 1. \tag{5.3.5}$$

Moreover, $s_i \to 0$, and the sequence (s_i) has a strictly decreasing subsequence.

Proof. For $0 < r \le R_1$, put

$$Q(r) = \frac{\mu(B_p(a,r))}{(2r)^q}.$$

By the lower-density hypothesis (5.3.3), there exist $c > 0$ and $0 < r_0 \le R_1$ such that $Q(r) \ge c$ for every $0 < r \le r_0$. After omitting finitely many indices, we may assume that $R_i \le r_0$. Set

$$m_i = \inf_{0<r\le R_i} Q(r).$$

The ball $B_p(a,R_i)$ is compactly contained in Ω, so its μ-measure is finite. Hence

$$0 < c \le m_i \le Q(R_i) < \infty.$$

Since $m_i > 0$ and $\varepsilon_i > 0$, the definition of the infimum provides $\rho_i \in (0, R_i]$ such that

$$Q(\rho_i) < (1+\varepsilon_i)m_i.$$

Set $s_i = \rho_i/5$. Then $0 < s_i \le R_i/5$ and $Q(s_i) \ge m_i > 0$, so $\mu(B_p(a,s_i)) > 0$. Since $B_p(a,s_i) \subset B_p(a,R_i)$, this measure is also finite. Moreover,

$$\frac{\mu(B_p(a,5s_i))}{\mu(B_p(a,s_i))} = 5^q \frac{Q(\rho_i)}{Q(s_i)} \le 5^q(1+\varepsilon_i),$$

which proves (5.3.4).

Fix $0 < t \le 1$. Since $ts_i \le R_i$, we have $Q(ts_i) \ge m_i$. The inclusion $B_p(a,s_i) \subset B_p(a,\rho_i)$ therefore gives

$$\begin{aligned}\frac{\mu(B_p(a,ts_i))}{\mu(B_p(a,s_i))} &\ge \frac{\mu(B_p(a,ts_i))}{\mu(B_p(a,\rho_i))} \\ &= \left(\frac{t}{5}\right)^q \frac{Q(ts_i)}{Q(\rho_i)} \ge \frac{t^q}{5^q(1+\varepsilon_i)}.\end{aligned}$$

This proves (5.3.5) for every $0 < t \le 1$.

It remains to prove that $s_i \to 0$ and that (s_i) has a strictly decreasing subsequence. Since $0 < s_i \le R_i/5$ and $R_i \to 0$, we have $s_i \to 0$. For every index i, positivity of s_i and convergence to zero give an index $k > i$ with $s_k < s_i$. We can therefore choose $i_1 < i_2 < \cdots$ such that $s_{i_{j+1}} < s_{i_j}$ for every $j \ge 1$. The estimates (5.3.4) and (5.3.5) hold along this subsequence with $i = i_j$. □

Remark (Doubling scales). Sequences of doubling cubes shrinking to almost every point, and differentiation along such cubes, are discussed by Tolsa [37, arXiv version, Remark 2.3 and the following paragraph, p. 8] for Radon measures on Euclidean space. Hytönen [21, Lemma 3.3 and Corollary 3.6] gives the corresponding results for balls in geometrically doubling metric spaces, for Borel measures finite on bounded sets. In Lemma 5.3.2, the

approximate-minimum argument gives the fivefold bound (5.3.4) and the inner lower bound (5.3.5) at the same selected radii. The latter holds for every $0 < t \leq 1$ at each selected radius, under the positive lower-density hypothesis (5.3.3), which allows infinite lower density. This inner bound supplies the nonvanishing step in Lemma 9.4.1.

5.4. **Homogeneous Structure.** The parabolic dilations on $\mathbb{P}^n$ are

$$\delta_r(x,t) = (rx, r^2 t), \qquad r > 0.$$

They satisfy $d_p(\delta_r z, \delta_r w) = r\, d_p(z,w)$; see [26, Section 2]. For $a \in \mathbb{P}^n$ and $r > 0$, define

$$T_{a,r}(z) = \delta_{1/r}(z - a). \tag{5.4.1}$$

Thus $T_{a,r}$ sends $B_p(a,r)$ onto $B_p(0,1)$ and $U_p(a,r)$ onto $U_p(0,1)$.

For $a \in \mathbb{P}^n$ and $r > 0$, Fubini's theorem [14, Theorem 2.36] and (5.1.2) give

$$\mathcal{L}^{n+1}(B_p(a,r)) = 2\omega_n \int_0^{r^2} (r^2 - u)^{n/2}\, du = \frac{4\omega_n}{n+2} r^{n+2}. \tag{5.4.2}$$

5.5. **Grassmannians, Vertical Planes and Their Projections.** Put $q = n + 1$, and denote by $G(n, n-1)$ the Grassmannian of $(n-1)$-dimensional linear subspaces of $\mathbb{R}^n$. The vertical planes considered here form the family

$$\mathbb{P}G = \{L \times \mathbb{R}_t : L \in G(n, n-1)\}.$$

For $L \in G(n, n-1)$, let P_L be the Euclidean orthogonal projection onto L. If $V = L \times \mathbb{R}_t$, the projection onto V is $P_V(x,t) = (P_L x, t)$. We equip $\mathbb{P}G$ with the metric

$$d_{\mathbb{P}G}(L \times \mathbb{R}_t, M \times \mathbb{R}_t) = \|P_L - P_M\|_{\mathrm{op}},$$

where $\|\cdot\|_{\mathrm{op}}$ is the Euclidean operator norm. With this metric, $\mathbb{P}G$ is compact; these are the vertical-plane conventions of [26, Section 2]. For $n = 1$, the space $\mathbb{P}G$ consists only of the time axis $\{0\} \times \mathbb{R}_t$.

Definition 5.5.1 (Cylinders over vertical planes)**.** For $W \in \mathbb{P}G$, $z \in \mathbb{P}^n$ and $r > 0$, define the cylinder

$$C_W(z,r) = P_W^{-1}\big(W \cap B_p(P_W z, r)\big) = \{y \in \mathbb{P}^n : d_p(P_W y, P_W z) \leq r\}. \tag{5.5.1}$$

For a Borel probability measure ν on $\mathbb{P}G$, define its *averaged spatial projection* by

$$A(\nu) = \int_{\mathbb{P}G} P_L\, d\nu(L \times \mathbb{R}_t) \in \mathbb{R}^{n \times n}. \tag{5.5.2}$$

5.6. **Parabolic Flat Measures.** Let $V = L \times \mathbb{R}_t \in \mathbb{P}G$, and choose an orthonormal basis $e_1, \ldots, e_{n-1}$ of L. Define Euclidean measure on V by

$$F_V : \mathbb{R}^{n-1} \times \mathbb{R} \longrightarrow V, \qquad F_V(y,t) = \left(\sum_{\alpha=1}^{n-1} y_\alpha e_\alpha, t\right), \tag{5.6.1}$$
$$\lambda_V = (F_V)_{\#}\mathcal{L}^n.$$

The measure λ_V does not depend on the choice of orthonormal basis. When $n = 1$, the spatial sum is empty and λ_V is Lebesgue measure on the time axis.

Lemma 5.6.1 (Measure on a vertical plane)**.** *Put $q = n + 1$ and*

$$b_v(q) = \frac{4\omega_{n-1}}{q}. \tag{5.6.2}$$

There is a finite positive constant $v(q)$, independent of $V \in \mathbb{P}G$, such that

$$\mathscr{H}_p^q \llcorner V = \frac{v(q)}{b_v(q)} \lambda_V \qquad \text{as Radon measures on } V. \tag{5.6.3}$$

For every $a \in V$ and $r > 0$,

$$\lambda_V(V \cap B_p(a,r)) = b_v(q) r^q, \qquad \mathscr{H}_p^q(V \cap B_p(a,r)) = v(q) r^q. \tag{5.6.4}$$

In particular, $v(q) = \mathscr{H}_p^q(V \cap B_p(0,1))$.

Proof. Apply [26, Lemma 2.3, pp. 858–859] with $m = q = n+1$ and the Hausdorff normalization (5.2.1). In the coordinates of (5.6.1), Fubini's theorem [14, Theorem 2.36] gives

$$\lambda_V(V \cap B_p(0,1)) = 2\omega_{n-1} \int_0^1 (1-u)^{(n-1)/2}\, du = \frac{4\omega_{n-1}}{q} = b_v(q). \tag{5.6.5}$$

□

5.7. **Parabolic Rectifiability.**

Definition 5.7.1 (Vertical parabolic Lipschitz graphs)**.** Let $n \geq 1$, put $q = n+1$, and let $V = L \times \mathbb{R}_t \in \mathbb{P}G$. Write $V^\perp = L^\perp \times \{0\}$ for its Euclidean orthogonal complement. For $A \subset V$ and $0 < \Lambda < \infty$, a *vertical parabolic (q,Λ)-Lipschitz graph over V* is a set

$$\Gamma_f = \{v + f(v) : v \in A\}, \tag{5.7.1}$$

where $f : A \to V^\perp$ satisfies

$$d_p(f(v), f(w)) \leq \Lambda\, d_p(v,w) \qquad (v, w \in A). \tag{5.7.2}$$

We call Γ_f a *vertical parabolic Lipschitz graph* if this holds for some $0 < \Lambda < \infty$. This is the vertical case of [26, Definition 3.1, p. 859] with $m = q$.

Definition 5.7.2 (Countable parabolic rectifiability)**.** A set $E \subset \mathbb{P}^n$ is *countably $\mathbb{P}$-rectifiable* if, for every $0 < \Lambda < \infty$, there are vertical parabolic (q,Λ)-Lipschitz graphs $\Gamma_i(\Lambda)$, $i \geq 1$, such that

$$\mathscr{H}_p^q\left(E \setminus \bigcup_{i=1}^{\infty} \Gamma_i(\Lambda)\right) = 0. \tag{5.7.3}$$

The graph family may depend on Λ. This is vertical LG q-rectifiability in [26, Definitions 3.6 and 4.10, pp. 860 and 866].

Lemma 5.7.1 (Finite rectifiable graph pieces)**.** *Let $K \subset \mathbb{P}^n$, $n \geq 1$, be Borel and countably $\mathbb{P}$-rectifiable, and put $q = n+1$. There are pairwise disjoint Borel sets $K_j \subset K$, $j \geq 1$, such that*

$$\mathscr{H}_p^q\left(K \setminus \bigcup_{j \geq 1} K_j\right) = 0, \qquad \mathscr{H}_p^q(K_j) < \infty \quad (j \geq 1). \tag{5.7.4}$$

Each K_j is countably $\mathbb{P}$-rectifiable and is contained in a compact subset of a vertical parabolic Lipschitz graph defined on its entire base plane. In particular, $\mathscr{H}_p^q \llcorner K$ is σ-finite.

Proof. If $\mathscr{H}_p^q(K) = 0$, take $K_j = \varnothing$ for every j. Otherwise choose a countable cover of K, modulo a $\mathscr{H}_p^q$-null set, by vertical parabolic $(q,1)$-Lipschitz graphs $\Gamma_i = \{v + f_i(v) : v \in A_i\}$ over planes $V_i \in \mathbb{P}G$. Discard empty graphs. Since $V_i^\perp$ is one-dimensional, choose a unit vector e_i spanning $V_i^\perp$ and write $f_i(v) = u_i(v) e_i$ for $v \in A_i$, where $u_i : A_i \to \mathbb{R}$ is 1-Lipschitz for d_p. As in [26, paragraph following Definition 3.7, p. 861], extend the scalar function to the whole base by

$$\overline{u}_i(v) = \inf_{w \in A_i} \big(u_i(w) + d_p(v,w)\big) \qquad (v \in V_i). \tag{5.7.5}$$

The function $\overline{u}_i : V_i \to \mathbb{R}$ is 1-Lipschitz and agrees with u_i on A_i. Define $F_i : V_i \to \mathbb{P}^n$ and $\overline{\Gamma}_i$ by

$$F_i(v) = v + \overline{u}_i(v) e_i \quad (v \in V_i), \qquad \overline{\Gamma}_i = F_i(V_i).$$

Orthogonality of V_i and e_i in the spatial variables gives

$$d_p(F_i(v), F_i(w))^2 = d_p(v,w)^2 + |\overline{u}_i(v) - \overline{u}_i(w)|^2 \leq 2d_p(v,w)^2.$$

Thus F_i is $\sqrt{2}$-Lipschitz, and its inverse on $\overline{\Gamma}_i$ is the continuous projection P_{V_i}. If $F_i(v_k) \to z$, then $v_k \to P_{V_i} z$ and continuity of F_i gives $z = F_i(P_{V_i} z)$. Hence $\overline{\Gamma}_i$ is closed. For every integer $m \geq 1$, the set $\overline{\Gamma}_i \cap B_p(0,m)$ is compact and is contained in $F_i(V_i \cap B_p(0,m))$. Since F_i increases diameters by at most $\sqrt{2}$, the definition (5.2.1) and (5.6.4) give

$$\mathscr{H}^q_p(\overline{\Gamma}_i \cap B_p(0,m)) \leq 2^{q/2} v(q) m^q < \infty. \tag{5.7.6}$$

Enumerate the Borel sets $K \cap \overline{\Gamma}_i \cap B_p(0,m)$, $i, m \geq 1$, as $(D_j)_{j \geq 1}$, and put

$$K_j = D_j \setminus \bigcup_{\ell < j} D_\ell \qquad (j \geq 1).$$

These sets are pairwise disjoint and satisfy (5.7.4) by (5.7.6) and the original graph cover. For every $\Lambda > 0$, each graph cover of K in (5.7.3) also covers K_j modulo a null set. Hence each K_j is countably $\mathbb{P}$-rectifiable. □

6. Parabolic Varifolds

Let $\Omega \subset \mathbb{P}^n$ be open. A *parabolic varifold* on Ω is a positive Radon measure W on $\Omega \times \mathbb{P}G$. Its *weight* is the measure

$$\mu = \|W\| = \pi_{\#} W, \qquad \pi(a, P) = a.$$

The projection π is proper because $\mathbb{P}G$ is compact, so μ is Radon.

6.1. **Restriction and Disintegration.** For a Borel set $A \subset \Omega$, the *restriction of W to A* is the measure $W \llcorner (A \times \mathbb{P}G)$ on $\Omega \times \mathbb{P}G$. It is Radon by [14, Theorem 7.8], since its mass on every compact set $K \subset \Omega \times \mathbb{P}G$ is at most $W(K)$. Its weight is

$$\|W \llcorner (A \times \mathbb{P}G)\| = \mu \llcorner A. \tag{6.1.1}$$

Both W and μ are σ-finite, since their underlying spaces are σ-compact.

By the disintegration theorem [9, Theorems 1 and 2(iii)], there is a family of Borel probability measures ν_a on $\mathbb{P}G$, unique for μ-almost every $a \in \Omega$, such that $a \mapsto \nu_a(A)$ is μ-measurable for every Borel set $A \subset \mathbb{P}G$ and

$$\int_{\Omega \times \mathbb{P}G} F(a,P)\, dW(a,P) = \int_\Omega \left(\int_{\mathbb{P}G} F(a,P)\, d\nu_a(P) \right) d\mu(a) \tag{6.1.2}$$

for every nonnegative Borel function F on $\Omega \times \mathbb{P}G$. We write $W = \mu \otimes \nu_a$ for the disintegration (6.1.2).

6.2. **Graph Measure.** Let η be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and let $V : \Omega \to \mathbb{P}G$ be Borel. Write $V_a = V(a)$ for every $a \in \Omega$. Define the *graph map* and its *graph measure* by

$$\begin{aligned} &\mathrm{Gr}(V) : \Omega \longrightarrow \Omega \times \mathbb{P}G, \quad \mathrm{Gr}(V)(a) = (a, V_a), \\ &\eta \otimes \delta_{V_a} = (\mathrm{Gr}(V))_{\#} \eta. \end{aligned} \tag{6.2.1}$$

Thus, for every nonnegative Borel function F on $\Omega \times \mathbb{P}G$,

$$\int_{\Omega \times \mathbb{P}G} F\, d(\eta \otimes \delta_{V_a}) = \int_\Omega F(a, V_a)\, d\eta(a). \tag{6.2.2}$$

For every compact $K \subset \Omega \times \mathbb{P}G$,

$$(\eta \otimes \delta_{V_a})(K) \leq \eta(\pi(K)) < \infty.$$

Consequently, [14, Theorem 7.8] shows that the graph measure is Radon. It is a parabolic varifold with weight η.

6.3. **Rectifiable Parabolic Varifolds.**

Definition 6.3.1 (Rectifiable parabolic varifold)**.** Let $\Omega \subset \mathbb{P}^n$ be open, with $n \geq 1$ and $q = n + 1$. A parabolic varifold W on Ω is *rectifiable* if there are a Borel countably $\mathbb{P}$-rectifiable set $K \subset \Omega$ in the sense of Definition 5.7.2 and a Borel function $\theta : K \to (0, \infty)$ such that $\eta = \theta \mathscr{H}_p^q \llcorner K$ is a Radon measure on Ω and

$$\int_{\Omega \times \mathbb{P}G} F(a, P)\, dW(a, P) = \int_K F(a, \operatorname{recTan}(K, a))\theta(a)\, d\mathscr{H}_p^q(a) \tag{6.3.1}$$

for every nonnegative Borel function $F : \Omega \times \mathbb{P}G \to [0, \infty]$. Here $\operatorname{recTan}(K, \cdot) : K \to \mathbb{P}G$ is the Borel tangent field of Definition 9.1.1.

To express (6.3.1) using the graph measure (6.2.1), fix $P_0 \in \mathbb{P}G$ and define the Borel map $R : \Omega \to \mathbb{P}G$ by

$$R(a) = \begin{cases} \operatorname{recTan}(K, a), & a \in K, \\ P_0, & a \in \Omega \setminus K. \end{cases}$$

Then (6.2.2) gives

$$W = (\theta \mathscr{H}_p^q \llcorner K) \otimes \delta_{R(a)}, \qquad \|W\| = \theta \mathscr{H}_p^q \llcorner K, \tag{6.3.2}$$

as Radon measures on $\Omega \times \mathbb{P}G$ and Ω, respectively. The choice of P_0 does not affect this measure because $\eta(\Omega \setminus K) = 0$. Proposition 9.1.1 and Definition 9.1.1 show that different choices in the construction of $\operatorname{recTan}(K, \cdot)$ agree $\mathscr{H}_p^q$-almost everywhere on K, hence also η-almost everywhere. They therefore leave (6.3.1) unchanged. The zero varifold is included by taking $K = \varnothing$.

Under the hypotheses of Theorem 1, Theorem 2(1) shows that W_+ is a rectifiable parabolic varifold: its carrier and multiplicity satisfy (6.3.1), and its weight $\mu_+ = \theta \mathscr{H}_p^q \llcorner K$ is Radon by (6.1.1).

7. Chronomorphisms, Chronofields and Their Flows

7.1. **Chronomorphisms.** Let $U, U' \subset \mathbb{P}^n$, $n \geq 1$, be open, and set

$$I = \pi_t(U), \qquad I' = \pi_t(U').$$

These sets are open by Section 5.1. A C^1 *chronomorphism* $\Psi : U \to U'$ is a Euclidean C^1 diffeomorphism of the form

$$\Psi(x, t) = (f(x, t), h(t)), \qquad f \in C^1(U; \mathbb{R}^n), \quad h \in C^1(I; I'), \tag{7.1.1}$$

where $h'(t) \neq 0$ for every $t \in I$, and whose inverse has the form

$$\Psi^{-1}(y, s) = (g(y, s), k(s)), \qquad g \in C^1(U'; \mathbb{R}^n), \quad k \in C^1(I'; I). \tag{7.1.2}$$

The inverse identities give $k(h(t)) = t$ for every $t \in I$ and $h(k(s)) = s$ for every $s \in I'$. Thus $h : I \to I'$ is a C^1 diffeomorphism with inverse k.

7.2. **Chrono-Differential.** For a C^1 chronomorphism $\Psi : U \to U'$ and $a = (x, t) \in U$, the *chrono-differential* is the linear map $D_p\Psi(a) : \mathbb{P}^n \to \mathbb{P}^n$ defined by

$$D_p\Psi(a)(v, s) = (D_x f(a)v, h'(t)s), \qquad (v, s) \in \mathbb{R}^n \times \mathbb{R}. \tag{7.2.1}$$

Here $D_x f(a)$ denotes differentiation in the spatial variables. For $P = L \times \mathbb{R}_t \in \mathbb{P}G$, define the induced plane by

$$\Psi_\natural(a, P) = D_x f(a)L \times \mathbb{R}_t. \tag{7.2.2}$$

Lemma 7.2.1. *Let $\Psi : U \to U'$ be a C^1 chronomorphism and let $a \in U$. Then $D_x f(a)$ and $D_p\Psi(a)$ are invertible, and*

$$D_p\Psi(a)(P) = \Psi_\natural(a, P) \in \mathbb{P}G \qquad \text{for every } P \in \mathbb{P}G.$$

Moreover,

$$\lim_{r\downarrow 0} \delta_{1/r}\big(\Psi(a + \delta_r(v, s)) - \Psi(a)\big) = D_p\Psi(a)(v, s), \tag{7.2.3}$$

uniformly for (v, s) in each compact set $K \subset \mathbb{P}^n$; the rescaled maps are defined on K for all sufficiently small $r > 0$. If $\Phi : U' \to U''$ is another C^1 chronomorphism, where $U'' \subset \mathbb{P}^n$ is open, then $\Phi \circ \Psi$ and Ψ^{-1} are C^1 chronomorphisms and

$$D_p(\Phi \circ \Psi)(a) = D_p\Phi(\Psi(a)) \circ D_p\Psi(a), \tag{7.2.4}$$

$$D_p(\Psi^{-1})(\Psi(a)) = (D_p\Psi(a))^{-1}. \tag{7.2.5}$$

Proof. Write $a = (x, t)$. Since Ψ is a C^1 diffeomorphism, the ordinary chain rule [31, Theorem 9.15] implies that $D\Psi(a)$ is invertible. Its block form

$$D\Psi(a) = \begin{pmatrix} D_x f(a) & \partial_t f(a) \\ 0 & h'(t) \end{pmatrix}$$

and $h'(t) \neq 0$ show that $D_x f(a)$ and $D_p\Psi(a)$ are invertible. For $P = L \times \mathbb{R}_t \in \mathbb{P}G$, the space $D_x f(a)L$ has dimension $n - 1$, and $h'(t)\mathbb{R} = \mathbb{R}$. Hence

$$D_p\Psi(a)(P) = D_x f(a)L \times \mathbb{R}_t = \Psi_\natural(a, P) \in \mathbb{P}G,$$

where membership follows from the product definition of $\mathbb{P}G$ in Section 5.5.

Fix a nonempty compact set $K \subset \mathbb{P}^n$, and put

$$M_K = \max\left\{1, \sup_{(v,s)\in K} (|v| + |s|)\right\}.$$

Choose $\rho > 0$ such that $a + w \in U$ and $t + u \in I$ whenever $w \in \mathbb{R}^{n+1}$, $u \in \mathbb{R}$, and $|w|, |u| < \rho$. For $0 < r < \min\{1, \rho/M_K\}$ and $(v, s) \in K$,

$$|(rv, r^2 s)| \le M_K r < \rho, \qquad |r^2 s| \le M_K r^2 < \rho, \tag{7.2.6}$$

so $a + \delta_r K \subset U$.

Define the Taylor remainders $R_f : \{w \in \mathbb{R}^{n+1} : |w| < \rho\} \to \mathbb{R}^n$ and $R_h : (-\rho, \rho) \to \mathbb{R}$ by

$$R_f(w) = f(a + w) - f(a) - Df(a)w, \qquad R_h(u) = h(t + u) - h(t) - h'(t)u. \tag{7.2.7}$$

First-order Taylor expansion [31, Definition 9.11 and Remarks 9.13(a)] gives

$$R_f(w) = o(|w|) \quad (w \to 0), \qquad R_h(u) = o(|u|) \quad (u \to 0). \tag{7.2.8}$$

Consequently, (7.2.6) and (7.2.8) yield the uniform estimate

$$\begin{aligned} e_r &:= \sup_{(v,s)\in K} \left(\frac{|R_f(rv, r^2 s)|}{r} + \frac{|R_h(r^2 s)|}{r^2} \right) \\ &\le M_K \left(\sup_{0<|w|\le M_K r} \frac{|R_f(w)|}{|w|} + \sup_{0<|u|\le M_K r^2} \frac{|R_h(u)|}{|u|} \right) \longrightarrow 0 \qquad (r \downarrow 0). \end{aligned} \tag{7.2.9}$$

The Taylor formulas (7.2.7), after rescaling, give

$$\begin{aligned} &\delta_{1/r}\big(\Psi(a + \delta_r(v, s)) - \Psi(a)\big) \\ &\quad = \left(D_x f(a)v + r\partial_t f(a)s + \frac{R_f(rv, r^2 s)}{r},\ h'(t)s + \frac{R_h(r^2 s)}{r^2} \right). \end{aligned} \tag{7.2.10}$$

Using (7.2.9) in (7.2.10), the square-sum metric gives

$$\begin{aligned}\sup_{(v,s)\in K} d_p\big(\delta_{1/r}\big(\Psi(a+\delta_r(v,s))-\Psi(a)\big), D_p\Psi(a)(v,s)\big)^2 \\ \le \big(rM_K|\partial_t f(a)|+e_r\big)^2+e_r \longrightarrow 0.\end{aligned} \tag{7.2.11}$$

Estimate (7.2.11) proves (7.2.3).

Write $\Phi(y,u)=(F(y,u),\ell(u))$ for $(y,u)\in U'$ and $\Phi^{-1}(z,\theta)=(G(z,\theta),m(\theta))$ for $(z,\theta)\in U''$. Then

$$\begin{aligned}(\Phi\circ\Psi)(x,t)&=\big(F(f(x,t),h(t)),\ell(h(t))\big),\\ (\Phi\circ\Psi)^{-1}(z,\theta)&=\big(g(G(z,\theta),m(\theta)),k(m(\theta))\big).\end{aligned} \tag{7.2.12}$$

The ordinary chain rule gives

$$\begin{aligned}D_x(F\circ\Psi)(a)&=D_yF(\Psi(a))D_xf(a),\\ (\ell\circ h)'(t)&=\ell'(h(t))h'(t)\neq 0.\end{aligned} \tag{7.2.13}$$

The maps in (7.2.12) are C^1 and have temporal coordinates depending only on time, so (7.2.13) verifies that $\Phi\circ\Psi$ is a chronomorphism. Substituting (7.2.13) into (7.2.1) proves (7.2.4).

Differentiating $h\circ k=\mathrm{id}_{I'}$ gives $k'(u)=1/h'(k(u))\neq 0$ for $u\in I'$, so (7.1.2) shows that Ψ^{-1} is a chronomorphism. Applying (7.2.4) to $\Psi^{-1}\circ\Psi=\mathrm{id}_U$ then yields (7.2.5). □

7.3. **Chrono-Jacobian.** Let $\Psi:U\to U'$ be a C^1 chronomorphism between open subsets of $\mathbb{P}^n$, $n\ge 1$, with $\Psi(x,t)=(f(x,t),h(t))$. Its *chrono-Jacobian* is the function $J_p\Psi:U\times\mathbb{P}G\to[0,\infty)$ defined by

$$J_p\Psi(a,P)=|h'(t)|\,J_{n-1}(D_xf(a)|_L),\qquad a=(x,t)\in U,\quad P=L\times\mathbb{R}_t\in\mathbb{P}G. \tag{7.3.1}$$

Lemma 7.3.1. *Let $\Psi:U\to U'$ be a C^1 chronomorphism. The function $J_p\Psi$ is positive and continuous on $U\times\mathbb{P}G$. For every compact set $K\subset U$ there are constants $0<c_K\le C_K<\infty$ such that*

$$c_K\le J_p\Psi(a,P)\le C_K\qquad \text{for every } (a,P)\in K\times\mathbb{P}G. \tag{7.3.2}$$

If $\Phi:U'\to U''$ is another C^1 chronomorphism, then

$$J_p(\Phi\circ\Psi)(a,P)=J_p\Phi\big(\Psi(a),\Psi_\natural(a,P)\big)\,J_p\Psi(a,P) \tag{7.3.3}$$

for every $(a,P)\in U\times\mathbb{P}G$.

Proof. For $a=(x,t)\in U$ and $P=L\times\mathbb{R}_t\in\mathbb{P}G$, $D_xf(a)$ is invertible by Lemma 7.2.1. Thus its restriction to L is injective and the determinant in (4.1.1) is positive. Together with $h'(t)\neq 0$, this proves $J_p\Psi(a,P)>0$.

For $A\in\mathbb{R}^{n\times n}$ and $L\in G(n,n-1)$, evaluation in an orthonormal basis adapted to $L\oplus L^\perp$ gives

$$J_{n-1}(A|_L)^2=\det\big(P_LA^{\mathsf T}AP_L+I_n-P_L\big). \tag{7.3.4}$$

The right-hand side is continuous in A and P_L. Since $a\mapsto D_xf(a)$ and $t\mapsto h'(t)$ are continuous, (7.3.4) and the projection metric in Section 5.5 give continuity of $J_p\Psi$. Positivity and compactness of $K\times\mathbb{P}G$, with compactness of $\mathbb{P}G$ supplied by Section 5.5, give (7.3.2).

Write $\Phi(y,u)=(F(y,u),\ell(u))$ for $(y,u)\in U'$. Set $A=D_xf(a)$ and $B=D_yF(\Psi(a))$. For $n>1$, apply the Euclidean Jacobian composition formula [33, Chapter 2, (4.11) and its derivation] to $A|_L:L\to AL$ and $B|_{AL}:AL\to\mathbb{R}^n$. The first map is an isomorphism by Lemma 7.2.1, and the formula gives

$$J_{n-1}(BA|_L)=J_{n-1}(B|_{AL})\,J_{n-1}(A|_L). \tag{7.3.5}$$

For $n=1$, (7.3.5) follows from the convention $J_0=1$. Equation (7.2.13) gives $D_x(F\circ\Psi)(a)=BA$ and $|(\ell\circ h)'(t)|=|\ell'(h(t))|\,|h'(t)|$. Since $\Psi_\natural(a,P)=AL\times\mathbb{R}_t$ by (7.2.2), substituting these identities and (7.3.5) into (7.3.1) proves (7.3.3). □

7.4. **Chronofields.** Let $\Omega \subset \mathbb{P}^n$, $n \geq 1$, be open, and put $I = \pi_t(\Omega)$. A C^1 *chronofield* on Ω is a vector field $Y \in C^1(\Omega; \mathbb{R}^{n+1})$ of the form

$$Y(x,t) = (X(x,t), \gamma(t)), \qquad X \in C^1(\Omega; \mathbb{R}^n), \quad \gamma \in C^1(I; \mathbb{R}). \tag{7.4.1}$$

It is called *spatial* if $\gamma = 0$.

For $X \in C^1_c(\Omega; \mathbb{R}^n)$, the *spatial flow* of X is the unique map $\Phi \in C^1(\mathbb{R} \times \Omega; \Omega)$, with $\Phi_s(a) = \Phi(s,a)$, satisfying

$$\partial_s \Phi_s(a) = (X(\Phi_s(a)), 0), \qquad \Phi_0(a) = a \qquad (s \in \mathbb{R},\ a \in \Omega). \tag{7.4.2}$$

It fixes the time coordinate: for each $s \in \mathbb{R}$,

$$\Phi_s(x,t) = (F_s(x,t), t), \qquad F_s \in C^1(\Omega; \mathbb{R}^n). \tag{7.4.3}$$

For every $r, s \in \mathbb{R}$,

$$\Phi_{s+r} = \Phi_s \circ \Phi_r, \qquad \Phi_s^{-1} = \Phi_{-s}. \tag{7.4.4}$$

Each $\Phi_s : \Omega \to \Omega$ is a C^1 chronomorphism in the sense of Section 7.1, and

$$\Phi_s(a) = a \qquad (s \in \mathbb{R},\ a \in \Omega \setminus \operatorname{supp} X). \tag{7.4.5}$$

To obtain this flow, extend $(X, 0)$ by zero to $\mathbb{R}^{n+1}$. The extended field is bounded and C^1, so [36, Theorems 2.17 and 6.1] give a unique global C^1 flow with the composition and inverse identities (7.4.4). Uniqueness fixes every point outside $\operatorname{supp} X$; together with the inverse identity, this shows that each flow map sends Ω onto Ω. The zero temporal component keeps the time coordinate fixed. Restricting the flow to Ω therefore gives (7.4.2)–(7.4.5).

7.5. **Characterization of Chronomorphisms.** Within the class of Euclidean C^1 diffeomorphisms, chronomorphisms form the largest class of deformations that preserve finiteness of $\mathscr{H}^{n+1}_p$ on every compact countably $\mathbb{P}$-rectifiable set, provided the domain is connected and each nonempty time slice is connected; see Corollary 7.5.1. The same characterization holds with preservation of nullity in place of finiteness. Proposition 7.5.1 establishes the corresponding local characterization on arbitrary open sets.

Our motivation for establishing this result comes from Golo's work [17, arXiv version, Proposition 5.1]. Golo proves that a C^2 diffeomorphism of the first Heisenberg group which preserves finiteness of intrinsic perimeter for every measurable set in every open set must be contact. The proof below uses horizontal slices and projections onto vertical planes.

Proposition 7.5.1 (Preservation of parabolic nullity and finiteness)**.** *Let $\Omega, \Omega' \subset \mathbb{P}^n$, $n \geq 1$, be open, put $q = n+1$, and let $\Psi : \Omega \to \Omega'$ be a Euclidean C^1 diffeomorphism, written as $\Psi(x,t) = (f(x,t), h(x,t))$. The following conditions are equivalent:*

(i) For every compact $E \subset \Omega$,

$$\mathscr{H}^q_p(E) < \infty \quad \Longrightarrow \quad \mathscr{H}^q_p(\Psi(E)) < \infty. \tag{7.5.1}$$

(ii) For every compact $E \subset \Omega$,

$$\mathscr{H}^q_p(E) = 0 \quad \Longrightarrow \quad \mathscr{H}^q_p(\Psi(E)) = 0. \tag{7.5.2}$$

(iii) The temporal component satisfies

$$D_x h(x,t) = 0 \qquad ((x,t) \in \Omega). \tag{7.5.3}$$

Under these conditions, Ψ is locally bi-Lipschitz for d_p. The equivalence remains valid if the compact sets in (i) *and* (ii) *are required to be countably $\mathbb{P}$-rectifiable in the sense of Definition 5.7.2.*

Proof. Assume (7.5.3). Fix $a_0 = (x_0, t_0) \in \Omega$ and choose a cylinder $Q = B_E(x_0, \rho) \times I_0 \Subset \Omega$, where I_0 is an open interval containing t_0 of length at most 1. For each $t \in I_0$, the function $x \mapsto h(x,t)$ has zero derivative on the connected ball $B_E(x_0, \rho)$. Thus $h(x,t) = \widehat{h}(t)$ on Q, where $\widehat{h}(t) = h(x_0, t)$ belongs to $C^1(I_0; \mathbb{R})$. The derivative of Ψ on Q has the form

$$D\Psi(x,t) = \begin{pmatrix} D_x f(x,t) & \partial_t f(x,t) \\ 0 & \widehat{h}'(t) \end{pmatrix}. \tag{7.5.4}$$

Since Ψ and its inverse are C^1, the chain rule [31, Theorem 9.15] implies that this matrix is invertible. In particular, $\widehat{h}'(t) \neq 0$ on I_0, and $\widehat{h}$ is strictly monotone there.

Choose $M \geq 1$ bounding $\|D_x f\|_{\mathrm{op}}$, $|\partial_t f|$ and $|\widehat{h}'|$ on Q. For $z = (x,t), w = (y,s) \in Q$, convexity of Q gives

$$|f(z) - f(w)| \leq M(|x-y| + |t-s|) \leq 2M d_p(z,w), \qquad |\widehat{h}(t) - \widehat{h}(s)| \leq M|t-s|.$$

Here $|t-s| \leq 1$, so $|t-s| \leq |t-s|^{1/2}$. Consequently,

$$d_p(\Psi(z), \Psi(w)) \leq (4M^2 + M)^{1/2} d_p(z,w) \qquad (z, w \in Q). \tag{7.5.5}$$

On $\Psi(Q)$, the temporal component of Ψ^{-1} is $\widehat{h}^{-1}$. Choose a small Euclidean ball $U' \Subset \Psi(Q)$ about $\Psi(a_0)$. The inverse is C^1 and its temporal component depends only on time, so the calculation in (7.5.5), after shrinking U' so that its time differences have magnitude at most 1, applies to Ψ^{-1} on U'. Together with (7.5.5) on $\Psi^{-1}(U') \subset Q$, this proves local bi-Lipschitz continuity.

For a compact $E \subset \Omega$, choose a finite open cover $U_1, \ldots, U_N$ of E on which Ψ has parabolic Lipschitz constants $L_1, \ldots, L_N$. Applying (5.2.1) to the images of diameter covers gives

$$\mathscr{H}^q_p(\Psi(E)) \leq \sum_{j=1}^N L_j^q \mathscr{H}^q_p(E \cap U_j) \leq \left(\sum_{j=1}^N L_j^q \right) \mathscr{H}^q_p(E). \tag{7.5.6}$$

This proves both (7.5.1) and (7.5.2).

Conversely, suppose that $D_x h(a_0) \neq 0$ at some $a_0 = (x_0, t_0) \in \Omega$. The map $D_x\Psi(a_0)$ has rank n, and its image

$$T_0 = D_x\Psi(a_0)[\mathbb{R}^n]$$

is not $\mathbb{R}^n \times \{0\}$. Hence some spatial coordinate vector $(e_i, 0)$, $1 \leq i \leq n$, is not in T_0. Let

$$V = \{(y,s) \in \mathbb{P}^n : y_i = 0\} \in \mathbb{P}G, \qquad P(y,s) = (y - y_i e_i, s).$$

Thus P is the vertical projection of Section 5.5. For $z = (x,t)$ and $w = (y,s)$, (5.1.1) gives

$$d_p(Pz, Pw)^2 = d_p(z,w)^2 - |x_i - y_i|^2 \leq d_p(z,w)^2. \tag{7.5.7}$$

Since $\ker P = \operatorname{span}\{(e_i, 0)\}$ intersects T_0 only at zero, the map

$$A_0 = D_x(P \circ \Psi)(a_0) : \mathbb{R}^n \longrightarrow V$$

is an isomorphism. Set $\sigma = \left\|A_0^{-1}\right\|_{\mathrm{op}}^{-1} > 0$. Continuity allows us to choose $\delta, \tau, \eta > 0$ such that

$$C = \overline{B_E(x_0, \delta)} \times [t_0 - \tau, t_0 + \tau] \Subset \Omega$$

and, throughout C,

$$|\det D_x(P \circ \Psi)(x,t)| \geq \eta, \qquad \|D_x(P \circ \Psi)(x,t) - A_0\|_{\mathrm{op}} \leq \frac{\sigma}{2}. \tag{7.5.8}$$

The determinant is computed in orthonormal coordinates on V, as in (5.6.1).

For $|t - t_0| \leq \tau$, define

$$g_t : \overline{B_E(x_0, \delta)} \longrightarrow V, \qquad g_t(x) = P\Psi(x,t).$$

For $u, v \in \overline{B_E(x_0, \delta)}$, integration along their segment and (7.5.8) give

$$g_t(u) - g_t(v) - A_0(u - v) = \int_0^1 \big[D_x g_t(v + \theta(u - v)) - A_0\big](u - v)\, d\theta,$$

$$|g_t(u) - g_t(v)| \geq \sigma|u - v| - \frac{\sigma}{2}|u - v| = \frac{\sigma}{2}|u - v|.$$

Thus g_t is injective. Its derivative is bounded on the spatial ball, so it is also Lipschitz. The Euclidean area formula [33, Chapter 2, (3.2)–(3.3), p. 53] and (7.5.8) yield

$$\lambda_V\big(g_t(\overline{B_E(x_0, \delta)})\big) = \int_{\overline{B_E(x_0,\delta)}} |\det D_x g_t(x)|\, dx \geq \omega_n \eta \delta^n. \tag{7.5.9}$$

Put

$$S_t = \overline{B_E(x_0, \delta)} \times \{t\}, \qquad m_0 = \frac{v(q)}{b_v(q)} \omega_n \eta \delta^n > 0.$$

Lemma 5.6.1, (7.5.7) and (7.5.9) imply

$$\mathscr{H}_p^q(\Psi(S_t)) \geq \mathscr{H}_p^q(P\Psi(S_t)) = \frac{v(q)}{b_v(q)} \lambda_V(g_t(\overline{B_E(x_0, \delta)})) \geq m_0 \qquad (|t - t_0| \leq \tau). \tag{7.5.10}$$

Each S_t is compact. On a horizontal slice the metric d_p is Euclidean, and $q = n + 1 > n$, so

$$\mathscr{H}_p^q(S_t) = 0 \qquad (|t - t_0| \leq \tau). \tag{7.5.11}$$

Equations (7.5.10) and (7.5.11) contradict (7.5.2). To contradict (7.5.1), set

$$t_j = t_0 + \frac{\tau}{j} \quad (j \geq 1), \qquad E = S_{t_0} \cup \bigcup_{j=1}^{\infty} S_{t_j}.$$

The set E is the product of the closed spatial ball with the compact set $\{t_0\} \cup \{t_j : j \geq 1\}$, so $E \Subset \Omega$ is compact. Equation (7.5.11) gives $\mathscr{H}_p^q(E) = 0$. The sets $\Psi(S_{t_j})$ are pairwise disjoint compact sets, since Ψ is continuous and injective. By countable additivity and (7.5.10),

$$\mathscr{H}_p^q(\Psi(E)) \geq \sum_{j=1}^{\infty} \mathscr{H}_p^q(\Psi(S_{t_j})) \geq \sum_{j=1}^{\infty} m_0 = \infty.$$

This proves the necessity of (7.5.3) for each measure-preservation condition.

Finally, every $\mathscr{H}_p^q$-null set is countably $\mathbb{P}$-rectifiable by Definition 5.7.2. Thus S_t and E are admissible tests even when (i) and (ii) are restricted to compact countably $\mathbb{P}$-rectifiable sets. The sufficiency argument already applies to every compact set. □

Corollary 7.5.1 (Characterization of chronomorphisms)**.** *Let $\Omega \subset \mathbb{P}^n$ be nonempty, open and connected, and suppose that each nonempty set*

$$\{x \in \mathbb{R}^n : (x, t) \in \Omega\}, \qquad t \in \mathbb{R},$$

is connected. Let $\Omega' \subset \mathbb{P}^n$ be open and let $\Psi : \Omega \to \Omega'$ be a Euclidean C^1 diffeomorphism. Then Ψ is a C^1 chronomorphism in the sense of Section 7.1 if and only if it satisfies the equivalent conditions of Proposition 7.5.1. In particular, this characterization applies when $\Omega = U \times I$ with $U \subset \mathbb{R}^n$ nonempty, open and connected and I an open interval.

Proof. A chronomorphism satisfies (7.5.3), so Proposition 7.5.1 proves one implication. Conversely, assume the equivalent conditions of that proposition, and write $\Psi(x, t) = (f(x, t), h(x, t))$. For each $t \in I = \pi_t(\Omega)$, the function $x \mapsto h(x, t)$ has zero derivative on the connected open slice at t, and hence is constant there. Denote this value by $\widehat{h}(t)$. Given $t_0 \in I$, choose x_0 with $(x_0, t_0) \in \Omega$. For some interval J about t_0, openness gives $(x_0, t) \in \Omega$ for every $t \in J$, and $\widehat{h}(t) = h(x_0, t)$ on J. Thus $\widehat{h} \in C^1(I; \mathbb{R})$.

The set I is open by Section 5.1 and connected as the continuous image of Ω; hence it is an interval. Invertibility of $D\Psi$ in (7.5.4) gives $\widehat{h}'(t) \neq 0$ for every $t \in I$. Its continuous derivative has one sign on I, so $\widehat{h}$ is strictly monotone. Since Ψ maps Ω onto Ω', the image of $\widehat{h}$ is $I' = \pi_t(\Omega')$. Its local inverses, as in the proof of Proposition 7.5.1, show that the inverse $k : I' \to I$ belongs to $C^1(I'; I)$. The temporal component of $\Psi^{-1}(y, s)$ is therefore $k(s)$. Since Ψ^{-1} is C^1, both (7.1.1) and (7.1.2) hold, as required. □

8. The First Chronovariation

8.1. **Image Parabolic Varifolds.** Let $U, U' \subset \mathbb{P}^n$, $n \geq 1$, be open, let $\Psi : U \to U'$ be a C^1 chronomorphism with $\Psi(x,t) = (f(x,t), h(t))$, and let W be a parabolic varifold on U. The *image parabolic varifold* $\Psi^p_\# W$ is the positive Radon measure on $U' \times \mathbb{P}G$ defined by

$$\int_{U' \times \mathbb{P}G} F \, d(\Psi^p_\# W) = \int_{U \times \mathbb{P}G} F\big(\Psi(a), \Psi_\natural(a,P)\big) J_p\Psi(a,P) \, dW(a,P) \tag{8.1.1}$$

for every $F \in C_c(U' \times \mathbb{P}G; \mathbb{R})$. Compare the Euclidean image-varifold construction in [33, Chapter 8, (2.1)].

Since $a \mapsto D_x f(a)$ is continuous and $D_x f(a)$ is invertible by Lemma 7.2.1, the induced-plane map in (7.2.2) is continuous. For $F \in C_c(U' \times \mathbb{P}G; \mathbb{R})$, let $C \subset U'$ be the projection of $\operatorname{supp} F$ onto U', and set $K = \Psi^{-1}(C)$. Continuity of Ψ^{-1} and Section 5.5 give compactness of $K \times \mathbb{P}G$. By Lemma 7.3.1, the integrand on the right of (8.1.1) is continuous and supported in $K \times \mathbb{P}G$. The integral is therefore finite, linear in F, and nonnegative when $F \geq 0$. The Riesz representation theorem [14, Theorem 7.2] gives the unique positive Radon measure $\Psi^p_\# W$ on the locally compact Hausdorff space $U' \times \mathbb{P}G$.

Let $U'' \subset \mathbb{P}^n$ be open and let $\Phi : U' \to U''$ be a C^1 chronomorphism. As Radon measures on $U'' \times \mathbb{P}G$,

$$\Phi^p_\#(\Psi^p_\# W) = (\Phi \circ \Psi)^p_\# W. \tag{8.1.2}$$

Indeed, (7.2.2) and (7.2.4) give

$$(\Phi \circ \Psi)_\natural(a,P) = \Phi_\natural\big(\Psi(a), \Psi_\natural(a,P)\big) \qquad ((a,P) \in U \times \mathbb{P}G). \tag{8.1.3}$$

For $F \in C_c(U'' \times \mathbb{P}G; \mathbb{R})$, applying (8.1.1) twice and using (8.1.3) and (7.3.3) yields

$$\begin{aligned} &\int_{U'' \times \mathbb{P}G} F \, d\big(\Phi^p_\#(\Psi^p_\# W)\big) \\ &\quad = \int_{U \times \mathbb{P}G} F\big((\Phi \circ \Psi)(a), (\Phi \circ \Psi)_\natural(a,P)\big) J_p(\Phi \circ \Psi)(a,P) \, dW(a,P) \\ &\quad = \int_{U'' \times \mathbb{P}G} F \, d\big((\Phi \circ \Psi)^p_\# W\big). \end{aligned}$$

This proves (8.1.2).

8.2. **The Infinitesimal Chrono-Jacobian.**

Lemma 8.2.1. *Let $\Omega \subset \mathbb{P}^n$, $n \geq 1$, be open, let $X \in C^1_c(\Omega; \mathbb{R}^n)$, and let Φ_s be its spatial flow from Section 7.4. For every $a \in \Omega$ and $P = L \times \mathbb{R}_t \in \mathbb{P}G$,*

$$\left.\frac{d}{ds}\right|_{s=0} J_p\Phi_s(a,P) = \operatorname{tr}\big(P_L D_x X(a)\big). \tag{8.2.1}$$

Writing $K = \operatorname{supp} X$, there are $s_0 > 0$ and $0 \leq C_X < \infty$ such that

$$\left|\frac{J_p\Phi_s(a,P) - 1}{s}\right| \leq C_X \mathbf{1}_K(a) \qquad (0 < |s| \leq s_0, \ (a,P) \in \Omega \times \mathbb{P}G). \tag{8.2.2}$$

Proof. We adapt the first-order Jacobian calculation in [33, Chapter 2, derivation of (5.3), pp. 62–63]. Write $\Phi_s(x,t) = (F_s(x,t),t)$ as in (7.4.3), and set $A_s(a) = D_xF_s(a)$. Differentiating the integral form of (7.4.2) in x gives

$$A_s(a) = I_n + \int_0^s D_xX(\Phi_\sigma(a))A_\sigma(a)\,d\sigma. \tag{8.2.3}$$

Consequently $\partial_s A_s(a) = D_xX(\Phi_s(a))A_s(a)$ is continuous in (s,a), with $A_0(a) = I_n$ and $\partial_s A_s(a)|_{s=0} = D_xX(a)$.

Since the time coordinate is fixed, (7.3.1) and (7.3.4) give

$$J_p\Phi_s(a,P)^2 = \det\big(P_L A_s(a)^{\mathsf{T}} A_s(a) P_L + I_n - P_L\big).$$

Differentiating at $s=0$ yields

$$\begin{aligned}\left.\frac{d}{ds}\right|_{s=0} J_p\Phi_s(a,P) &= \tfrac{1}{2}\operatorname{tr}\big(P_L(D_xX(a)^{\mathsf{T}} + D_xX(a))P_L\big)\\ &= \operatorname{tr}\big(P_L D_xX(a)\big),\end{aligned}$$

which proves (8.2.1).

Equations (8.2.3) and (7.3.4), together with positivity from Lemma 7.3.1, show that $\partial_s J_p\Phi_s(a,P)$ is continuous in (s,a,P). Fix $s_0 > 0$. Compactness of $K \times \mathbb{P}G$ (Section 5.5) gives a finite constant $C_X \geq 0$ bounding $|\partial_s J_p\Phi_s|$ on $[-s_0,s_0]\times K\times \mathbb{P}G$. By (7.4.5), $J_p\Phi_s = 1$ outside $K \times \mathbb{P}G$. The mean value theorem therefore gives (8.2.2). □

8.3. **The Spatial Moment Matrix.** Let W be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, $n \geq 1$, with weight μ. Its *spatial moment measure* is the $\mathbb{R}^{n\times n}$-valued Radon measure M_W on Ω defined by

$$M_W(E) = \int_{E\times\mathbb{P}G} P_L\,dW(z, L\times\mathbb{R}_t) \tag{8.3.1}$$

for every relatively compact Borel set $E \subset \Omega$, with the integral taken entrywise. Compare the averaging of projection matrices for the area integrand in [10, arXiv version, Definition 1.1 and the discussion following Theorem 1.2].

Lemma 8.3.1. *The measure M_W is well defined, and $M_W(E)$ is symmetric and positive semidefinite for every relatively compact Borel set $E \subset \Omega$. As Radon measures on Ω,*

$$\operatorname{tr} M_W = (n-1)\mu, \qquad |M_W| \leq C(n)\mu, \tag{8.3.2}$$

where $C(n) < \infty$ depends only on n.

Proof. For $1 \leq j,k \leq n$, define positive Borel measures on Ω by

$$\lambda^{\pm}_{jk}(E) = \int_{E\times\mathbb{P}G} \max\{\pm(P_L)_{jk}, 0\}\,dW(z, L\times\mathbb{R}_t) \qquad (E \subset \Omega \text{ Borel}).$$

The entries $(P_L)_{jk}$ are continuous in $L \in G(n,n-1)$ and have absolute value at most one. Hence $\lambda^{\pm}_{jk}(K) \leq \mu(K) < \infty$ for every compact $K \subset \Omega$. These measures are Radon by [14, Theorem 7.8], since Ω is locally compact and second countable. On relatively compact Borel sets, their difference defines $(M_W)_{jk}$.

The symmetry of P_L gives symmetry of $M_W(E)$. For every $e \in \mathbb{R}^n$ and every relatively compact Borel set $E \subset \Omega$,

$$e \cdot M_W(E)e = \int_{E\times\mathbb{P}G} |P_L e|^2\,dW(z, L\times\mathbb{R}_t) \geq 0.$$

Since $\operatorname{tr} P_L = n-1$, taking traces in the defining integral gives $\operatorname{tr} M_W(E) = (n-1)\mu(E)$. This proves the trace identity on Ω by exhaustion.

Choose $C(n) \geq 1$ such that $|P_L| \leq C(n)$ for every $L \in G(n, n-1)$; such a constant exists by the compactness in Section 5.5. For every open $O \subset \Omega$ and $\Phi \in C_c(O; \mathbb{R}^{n\times n})$ with $\sup_{z\in O} |\Phi(z)| \leq 1$,

$$\left|\int_O \Phi : dM_W\right| = \left|\int_{O\times\mathbb{P}G} \Phi(z) : P_L \, dW(z, L \times \mathbb{R}_t)\right| \leq C(n)\mu(O).$$

Taking the supremum in (4.2.4) gives $|M_W|(O) \leq C(n)\mu(O)$. Outer regularity extends this inequality to every Borel subset of Ω. □

8.4. **The First Spatial Chronovariation Formula.** Let W be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, $n \geq 1$, with weight μ. For $X \in C^1_c(\Omega; \mathbb{R}^n)$, let Φ_s be the spatial flow in Section 7.4. For s sufficiently close to zero, define the *localized mass change* by

$$M^p_{W,X}(s) = \int_{\Omega\times\mathbb{P}G} \big(J_p\Phi_s(a, P) - 1\big)\, dW(a, P). \tag{8.4.1}$$

The *spatial first chronovariation* of W in the direction $(X, 0)$ is

$$\delta^p_{\mathrm{chr}} W(X, 0) = \left.\frac{d}{ds}\right|_{s=0} M^p_{W,X}(s), \tag{8.4.2}$$

whenever this derivative exists.

Proposition 8.4.1. *The integral in* (8.4.1) *is finite for all sufficiently small s, and its derivative at zero exists. For every $X \in C^1_c(\Omega; \mathbb{R}^n)$,*

$$\begin{aligned}\delta^p_{\mathrm{chr}} W(X, 0) &= \int_{\Omega\times\mathbb{P}G} \mathrm{tr}\big(P_L D_x X(a)\big)\, dW(a, L \times \mathbb{R}_t)\\ &= \int_\Omega D_x X : dM_W.\end{aligned} \tag{8.4.3}$$

In particular, $X \mapsto \delta^p_{\mathrm{chr}} W(X, 0)$ is real linear. For every open $U \Subset \Omega$ containing $\operatorname{supp} X$*, the integral in* (8.4.1) *can be restricted to $U \times \mathbb{P}G$. If $\mu(\Omega) < \infty$, then*

$$M^p_{W,X}(s) = \|(\Phi_s)^p_\# W\|(\Omega) - \mu(\Omega) \tag{8.4.4}$$

for all sufficiently small s.

Proof. We adapt the Euclidean first-variation argument in [33, Chapter 8, (2.2)–(2.4)]. Put $K = \operatorname{supp} X$. For $0 < |s| \leq s_0$, (8.2.2) gives

$$\left|\frac{J_p\Phi_s(a, P) - 1}{s}\right| \leq C_X \mathbf{1}_{K\times\mathbb{P}G}(a, P) \qquad \big((a, P) \in \Omega \times \mathbb{P}G\big).$$

Since $W(K \times \mathbb{P}G) = \mu(K) < \infty$, the integral in (8.4.1) is finite for small s. The dominated convergence theorem [14, Theorem 2.24] and (8.2.1) give the first equality in (8.4.3). Since P_L is symmetric, $\mathrm{tr}(P_L D_x X) = D_x X : P_L$; the definition (8.3.1) gives the second equality and linearity. The same support bound proves the assertion about U.

If $\mu(\Omega) < \infty$, apply (8.1.1) with $F_j(a, P) = \chi_j(a)$, where $\chi_j \in C_c(\Omega; \mathbb{R})$ and $0 \leq \chi_j \uparrow 1$. Monotone convergence gives

$$\|(\Phi_s)^p_\# W\|(\Omega) = \int_{\Omega\times\mathbb{P}G} J_p\Phi_s \, dW = \mu(\Omega) + M^p_{W,X}(s) < \infty.$$

This proves (8.4.4). □

For $n \geq 2$, the spatial tangential-divergence integrand in (8.4.3) also appears in the spacetime first variation of Buet, Leonardi, Masnou and Sagueni [8, Definition 5.1, pp. 39–40]. Taking their plane dimension to be $n-1$, their spatial planes correspond to our vertical planes $L \times \mathbb{R}_t$; this comparison uses vector fields supported in the interior of their time interval.

We say that W has *locally bounded spatial first chronovariation* if, for every open $U \Subset \Omega$, there is $0 \le C_U < \infty$ such that

$$|\delta^p_{\mathrm{chr}} W(X,0)| \le C_U \, \|X\|_\infty \qquad \big(X \in C^1_c(\Omega;\mathbb{R}^n),\ \operatorname{supp} X \subset U\big). \tag{8.4.5}$$

Proposition 8.4.2. *The varifold W has locally bounded spatial first chronovariation if and only if there is a unique $\mathbb{R}^n$-valued Radon measure m_{chr} on Ω such that*

$$\int_\Omega D_x X : dM_W = \int_\Omega X \cdot dm_{\mathrm{chr}} \qquad \big(X \in C^1_c(\Omega;\mathbb{R}^n)\big). \tag{8.4.6}$$

In that case,

$$-\operatorname{div}_x M_W = m_{\mathrm{chr}} \qquad \text{as distributions on } \Omega. \tag{8.4.7}$$

For an open set $U \subset \Omega$, the parabolic varifold $W_U = W \llcorner (U \times \mathbb{P}G)$ on U also has locally bounded spatial first chronovariation, and its representing measure $m_{\mathrm{chr},U}$ satisfies

$$M_{W_U} = M_W \llcorner U, \qquad m_{\mathrm{chr},U} = m_{\mathrm{chr}} \llcorner U \qquad \text{as Radon measures on } U. \tag{8.4.8}$$

Proof. We follow the Euclidean representation argument in [33, Chapter 8, (2.5)–(2.6)]. Suppose (8.4.5) holds. For $Y \in C_c(\Omega;\mathbb{R}^n)$, choose an open $U \Subset \Omega$ containing $\operatorname{supp} Y$ and a sequence $X_j \in C^\infty_c(\Omega;\mathbb{R}^n)$ converging uniformly to Y, with all supports in one compact subset of U; see [14, Proposition 8.17 and Lemma 8.18]. The bound (8.4.5) makes

$$\Lambda(Y) = \lim_{j\to\infty} \delta^p_{\mathrm{chr}} W(X_j, 0)$$

well defined and independent of the approximating sequence and of U. It extends $\delta^p_{\mathrm{chr}} W(\,\cdot\,,0)$ to a real linear functional on $C_c(\Omega;\mathbb{R}^n)$, with

$$|\Lambda(Y)| \le C_U \, \|Y\|_\infty \qquad \big(Y \in C_c(\Omega;\mathbb{R}^n),\ \operatorname{supp} Y \subset U\big).$$

The Riesz representation theorem [33, Chapter 1, Theorem 4.14] gives an $\mathbb{R}^n$-valued Radon measure m_{chr} on Ω satisfying

$$\Lambda(Y) = \int_\Omega Y \cdot dm_{\mathrm{chr}} \qquad \big(Y \in C_c(\Omega;\mathbb{R}^n)\big).$$

Together with (8.4.3), this proves (8.4.6). The same uniform approximation shows that (8.4.6) determines m_{chr} uniquely.

Conversely, if (8.4.6) holds, then, for $U \Subset \Omega$ open and $X \in C^1_c(\Omega;\mathbb{R}^n)$ with $\operatorname{supp} X \subset U$, total variation gives

$$|\delta^p_{\mathrm{chr}} W(X,0)| \le \|X\|_\infty \, |m_{\mathrm{chr}}|(\overline{U}).$$

Since $|m_{\mathrm{chr}}|(\overline{U}) < \infty$, this is (8.4.5). The sign in (8.4.7) follows from (4.2.6).

For an open $U \subset \Omega$, (8.3.1) gives $M_{W_U} = M_W \llcorner U$. Applying (8.4.6) to the zero extension of any $X \in C^1_c(U;\mathbb{R}^n)$ gives the representation on U with $m_{\mathrm{chr}} \llcorner U$. The equivalence and uniqueness just proved give local boundedness for W_U and (8.4.8). □

8.5. **The Full First Chronovariation.** Let W be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, with weight $\mu = \|W\|$. If $\Xi(x,t) = (X(x,t), \gamma(t))$ is a chronofield as in (7.4.1) and $\gamma \not\equiv 0$ on $\pi_t(\Omega)$, then Ξ cannot be compactly supported in Ω. Indeed, if $\gamma(t_0) \neq 0$, then Ξ is nonzero on the entire slice $S = \Omega \cap (\mathbb{R}^n_x \times \{t_0\})$. This slice is closed in Ω and is not compact, since its spatial projection is a nonempty open subset of $\mathbb{R}^n$. Thus $S \subset \operatorname{supp} \Xi$ rules out compact support in Ω.

For $\gamma \in C^1_c(\pi_t(\Omega);\mathbb{R})$, the temporal term $\int_\Omega \gamma'(t)\, d\mu(x,t)$ in (8.5.2) below also integrates over entire spatial slices. Compact support of γ in time and local finiteness of μ therefore do not by themselves ensure that this integral is absolutely convergent. To ensure absolute convergence, we impose the additional condition

$$\mu\big(\Omega \cap (\mathbb{R}^n_x \times I)\big) < \infty \qquad \text{for every compact interval } I \subset \mathbb{R}. \tag{8.5.1}$$

Definition 8.5.1 (Full first chronovariation)**.** For $X \in C^1_c(\Omega;\mathbb{R}^n)$ and $\gamma \in C^1_c(\pi_t(\Omega);\mathbb{R})$, define

$$\delta^p_{\mathrm{chr}} W(X,\gamma) = \int_{\Omega\times\mathbb{P}G} \big[\operatorname{tr}(P_L D_x X(x,t)) + \gamma'(t)\big]\, dW\big((x,t), L\times\mathbb{R}_t\big). \tag{8.5.2}$$

The integral in (8.5.2) is absolutely convergent. Indeed, let $K = \operatorname{supp} X$ and choose a compact interval $I \subset \mathbb{R}$ containing $\operatorname{supp}\gamma'$. Then

$$\int_{\Omega\times\mathbb{P}G} |\operatorname{tr}(P_L D_x X)|\, dW \le n\, \|D_x X\|_\infty\, \mu(K) < \infty,$$

$$\int_{\Omega\times\mathbb{P}G} |\gamma'(t)|\, dW \le \|\gamma'\|_\infty\, \mu\big(\Omega \cap (\mathbb{R}^n_x \times I)\big) < \infty.$$

The first bound uses the Radon property, and the second uses (8.5.1). Thus (8.5.2) defines a real linear functional. By (8.4.3), it extends the spatial first chronovariation through the identity

$$\delta^p_{\mathrm{chr}} W(X,\gamma) = \delta^p_{\mathrm{chr}} W(X,0) + \int_\Omega \gamma'(t)\, d\mu(x,t). \tag{8.5.3}$$

8.6. **Temporal Variation.** The temporal term in (8.5.3) tests μ against functions of time alone. To formulate a local condition, we allow tests depending on both space and time and control their spatial derivatives. We shall prove that finiteness of this temporal variation implies the half-time condition defined in (3.0.1)–(3.0.3).

Definition 8.6.1 (Temporal variation)**.** Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$. For an open set $U \subset \Omega$ and a parameter $\rho > 0$, define

$$\mathcal{T}_{\mu;U,\rho} = \sup\left\{ \left| -\int_U \partial_t\phi\, d\mu \right| : \begin{array}{l} \phi \in C^1_c(U;\mathbb{R}), \\ \max\{\rho\,\|\nabla_x\phi\|_\infty, \|\phi\|_\infty\} \le 1 \end{array} \right\}. \tag{8.6.1}$$

We say that μ has *locally finite temporal variation* if, for every open $U \Subset \Omega$, there is a $\rho > 0$ such that $\mathcal{T}_{\mu;U,\rho} < \infty$.

Proposition 8.6.1. *For μ, U and ρ as in Definition 8.6.1, the condition $\mathcal{T}_{\mu;U,\rho} < \infty$ is equivalent to the existence of a finite $\mathbb{R}^n$-valued Radon measure J and a finite signed Radon measure σ on U satisfying*

$$-\int_U \partial_t\phi\, d\mu - \int_U \nabla_x\phi \cdot dJ = \int_U \phi\, d\sigma \qquad \big(\phi \in C^1_c(U;\mathbb{R})\big). \tag{8.6.2}$$

In the finite case,

$$\mathcal{T}_{\mu;U,\rho} = \min\big\{\rho^{-1}|J|(U) + |\sigma|(U) : (J,\sigma) \text{ satisfies (8.6.2)}\big\}, \tag{8.6.3}$$

where the minimum is over finite measures of the indicated types. For every $\rho, \rho' > 0$,

$$\min\{1, \rho/\rho'\}\mathcal{T}_{\mu;U,\rho} \le \mathcal{T}_{\mu;U,\rho'} \le \max\{1, \rho/\rho'\}\mathcal{T}_{\mu;U,\rho}. \tag{8.6.4}$$

In particular, finiteness for one positive parameter is equivalent to finiteness for every positive parameter.

Proof. We use the Hahn–Banach theorem [14, Theorem 5.6] and the Riesz representation theorem [33, Chapter 1, Theorem 4.14]. For a related Hahn–Banach argument giving an attained minimum, see Federer's flat-norm representation [13, 4.1.12, pp. 367–368]; the test norm here controls spatial derivatives only. Give the real vector space $\mathcal{E} = C_c(U;\mathbb{R}^n) \times C_c(U;\mathbb{R})$ the norm

$$\|(p,h)\|_\rho = \max\{\rho\,\|p\|_\infty, \|h\|_\infty\}. \tag{8.6.5}$$

Define the spatial first-jet map

$$\mathcal{J} : C^1_c(U;\mathbb{R}) \longrightarrow \mathcal{E}, \qquad \mathcal{J}\phi = (\nabla_x\phi, \phi).$$

Its second component makes $\mathcal{J}$ injective, so

$$\ell(\mathcal{J}\phi) = -\int_U \partial_t\phi \, d\mu$$

defines a real linear functional on the subspace $\mathcal{J}(C_c^1(U;\mathbb{R}))$. If $\mathcal{T}_{\mu;U,\rho} < \infty$, then (8.6.1) gives $\|\ell\| = \mathcal{T}_{\mu;U,\rho}$. Hahn–Banach extends ℓ to a functional Λ on $\mathcal{E}$ with the same norm. Applying Riesz representation to $p \mapsto \Lambda(p,0)$ and $h \mapsto \Lambda(0,h)$ gives Radon measures J and σ on U such that

$$\Lambda(p,h) = \int_U p \cdot dJ + \int_U h \, d\sigma \qquad \big((p,h) \in \mathcal{E}\big). \tag{8.6.6}$$

The bound on Λ and (4.2.4) give $|J|(U) \le \rho\,\|\Lambda\|$ and $|\sigma|(U) \le \|\Lambda\|$, so both measures are finite. Moreover, the unit ball in (8.6.5) allows p and h to vary independently under the bounds $\|p\|_\infty \le \rho^{-1}$ and $\|h\|_\infty \le 1$. Taking the two suprema in (4.2.4) therefore gives

$$\rho^{-1}|J|(U) + |\sigma|(U) = \|\Lambda\| = \mathcal{T}_{\mu;U,\rho}.$$

Evaluating (8.6.6) at $(p,h) = \mathcal{J}\phi$ proves (8.6.2).

Conversely, every finite pair satisfying (8.6.2) obeys

$$\left|-\int_U \partial_t\phi \, d\mu\right| \le |J|(U)\,\|\nabla_x\phi\|_\infty + |\sigma|(U)\,\|\phi\|_\infty \qquad \big(\phi \in C_c^1(U;\mathbb{R})\big).$$

Taking the supremum over the tests in (8.6.1) gives $\mathcal{T}_{\mu;U,\rho} \le \rho^{-1}|J|(U) + |\sigma|(U)$. Together with the constructed pair, this proves the equivalence and the attained minimum (8.6.3).

Finally, (8.6.5) gives

$$\|(p,h)\|_{\rho'} \le \max\{1, \rho'/\rho\}\,\|(p,h)\|_\rho\,, \qquad \|(p,h)\|_\rho \le \max\{1, \rho/\rho'\}\,\|(p,h)\|_{\rho'}\,.$$

Rescaling the admissible tests in (8.6.1) proves (8.6.4), including when a supremum is infinite. □

Proposition 8.6.1 is a standard application of the Hahn–Banach and Riesz representation theorems. Its identity (8.6.2) is the weak form of the measure-valued continuity equation $\partial_t\mu + \operatorname{div}_x J = \sigma$ on U. The study of such equations is closely connected with the decomposition of acyclic normal one-dimensional currents into curves, developed by Smirnov [34, Theorem C] and extended to metric spaces by Paolini and Stepanov [28, arXiv version, Theorem 5.1]. Stepanov and Trevisan [35, arXiv version, Theorem 3.4] establish a superposition principle for narrowly continuous families of finite measures solving the homogeneous continuity equation, under their integrability and test-algebra hypotheses. These representations express measures as superpositions of curves; they do not by themselves assert concentration on a countably rectifiable set of a prescribed dimension.

Geometric restrictions imposed by linear PDEs are studied by De Philippis and Rindler [11, arXiv version, Theorem 1.1 and Remark 1.3], including equations with a measure-valued right-hand side. Building on that work, Arroyo-Rabasa, De Philippis, Hirsch and Rindler [4, arXiv version, Theorem 1.5] prove Euclidean rectifiability on the part of a PDE-constrained measure with positive upper density in the prescribed dimension, under their wave-cone hypothesis on the operator. In the present argument, (8.6.2) supplies the half-time condition through Proposition 8.6.2 below.

Proposition 8.6.2 (Temporal variation implies HT)**.** *Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and let $U \subset \Omega$ be open. Suppose that a finite $\mathbb{R}^n$-valued Radon measure J and a finite signed Radon measure σ on U satisfy* (8.6.2)*. For $a \in U$ and $r > 0$ with $B_p(a,r) \Subset U$ and $\mu(B_p(a,r)) > 0$,*

$$I_\mu(a,r) \le \sqrt{c_0}\left[r\frac{|J|(B_p(a,r))}{\mu(B_p(a,r))} + r^2\frac{|\sigma|(B_p(a,r))}{\mu(B_p(a,r))}\right]. \tag{8.6.7}$$

Consequently, $\mathrm{HT}(\mu, a)$ *holds for* μ*-almost every* $a \in U$*. If* μ *has locally finite temporal variation on* Ω*, then* $\mathrm{HT}(\mu, a)$ *holds for* μ*-almost every* $a \in \Omega$*.*

Proof. Fix a and r as in the statement, and let $\phi \in \mathcal{A}_{x,r}(a)$, with the test class defined in (3.0.1). Extend ϕ by zero to $\mathbb{P}^n$. For $|u| \le c_0 r^2$, every point $z = (x, t)$ in the support of $\phi \circ \tau_u$ satisfies

$$|x - a_x|^2 + |t - a_t| \le |x - a_x|^2 + |t + u - a_t| + |u| < (\kappa^2 + c_0) r^2 < r^2.$$

The translated supports thus lie in a common compact subset of $U_p(a, r)$. The same holds for their first derivatives. Define

$$g(u) = \int_U \phi \circ \tau_u \, d\mu \qquad (|u| \le c_0 r^2).$$

The difference quotients are bounded by $\|\partial_t \phi\|_\infty$ on that compact set. Dominated convergence [14, Theorem 2.24] permits differentiation under the integral and shows that g is continuously differentiable. Applying (8.6.2) to $\phi \circ \tau_u \in C^1_c(U; \mathbb{R})$ yields

$$g'(u) = -\int_U \nabla_x(\phi \circ \tau_u) \cdot dJ - \int_U \phi \circ \tau_u \, d\sigma.$$

The bounds in (3.0.1) and integration from 0 to s give

$$|g(s) - g(0)| \le |s| \big[|J|(B_p(a, r)) + r|\sigma|(B_p(a, r)) \big] \qquad (|s| \le c_0 r^2). \tag{8.6.8}$$

For $s \neq 0$, divide (8.6.8) by $|s|^{1/2} \mu(B_p(a, r))$ and use $|s|^{1/2} \le \sqrt{c_0}\, r$. All tests are supported in U, so their integrals against $\mu \llcorner U$ and μ agree. Taking the suprema in (3.0.2) proves (8.6.7).

Apply Lemma 11.3 on U, with reference measure $\mu \llcorner U$, to $|J|$ and $|\sigma|$. Let $N_U \subset U$ be the common μ-null set outside which (11.0.9) holds for both measures and every $1 \le C < \infty$. Fix $a \in U \setminus N_U$ and such a C. For sufficiently small r, the ball $B_p(a, 5r)$ is compactly contained in U. Hence the radii satisfying (3.0.3) in Ω also belong to $\mathcal{R}_C(a)$ of (11.0.1) for $\mu \llcorner U$ on U. Equation (11.0.9) provides finite constants $M_{a,C}$ and $r_{a,C} > 0$ such that both ratios on the right of (8.6.7) are at most $M_{a,C}$ for all these radii with $0 < r < r_{a,C}$. It follows that

$$I_\mu(a, r) \le \sqrt{c_0}\, M_{a,C}(r + r^2) \longrightarrow 0$$

uniformly through the radii required by (3.0.3). This proves $\mathrm{HT}(\mu, a)$ on $U \setminus N_U$.

Now suppose that μ has locally finite temporal variation on Ω. Choose a countable cover $(U_k)_{k \ge 1}$ of Ω by open sets $U_k \Subset \Omega$. For each k, Definition 8.6.1 and Proposition 8.6.1 supply one pair (J_k, σ_k) of finite measures satisfying (8.6.2) on U_k. Apply the preceding argument to this pair, and let $N_k \subset U_k$ be its exceptional μ-null set. The union $N = \bigcup_{k \ge 1} N_k$ is μ-null, and every $a \in \Omega \setminus N$ belongs to some $U_k \setminus N_k$. Thus $\mathrm{HT}(\mu, a)$ holds at every such point. □

9. Parabolic Tangents

Let W be a parabolic varifold on Ω with weight μ. For $a \in \Omega$ and $r > 0$, define

$$\widehat{T}_{a,r}(z, P) = (T_{a,r}(z), P), \qquad (z, P) \in \Omega \times \mathbb{P}G.$$

For $a \in \Omega$, $r > 0$ and $A_0 > 0$, put $\Omega' = T_{a,r}(\Omega)$ and define the rescaled measure

$$W' = A_0 (\widehat{T}_{a,r})_\# W \qquad \text{on } \Omega' \times \mathbb{P}G. \tag{9.0.1}$$

Proposition 9.1. *The measure* W' *is a parabolic varifold on* Ω'*, and*

$$\|W'\| = A_0 (T_{a,r})_\# \mu, \qquad M_{W'} = A_0 (T_{a,r})_\# M_W \tag{9.0.2}$$

as Radon measures on Ω'*. If* W *has locally bounded spatial first chronovariation, with representing measure* m_{chr}*, then* W' *has the same property and its representing measure is*

$$m'_{\mathrm{chr}} = A_0 r (T_{a,r})_\# m_{\mathrm{chr}} \qquad \text{on } \Omega'. \tag{9.0.3}$$

Proof. The map $\widehat{T}_{a,r} : \Omega \times \mathbb{P}G \to \Omega' \times \mathbb{P}G$ is a homeomorphism, so W' is a positive Radon measure. For every relatively compact Borel set $E \subset \Omega'$, the weight definition in Section 6 and (8.3.1) give

$$\|W'\|(E) = A_0 W\big(T_{a,r}^{-1}(E) \times \mathbb{P}G\big) = A_0 \mu\big(T_{a,r}^{-1}(E)\big),$$
$$M_{W'}(E) = A_0 \int_{T_{a,r}^{-1}(E)\times \mathbb{P}G} P_L \, dW(z, L \times \mathbb{R}_t) = A_0 M_W\big(T_{a,r}^{-1}(E)\big).$$

This proves (9.0.2).

For the variation formula, we adapt the Euclidean calculation in [10, arXiv version, proof of Lemma 2.4, p. 8]. Assume that W has locally bounded spatial first chronovariation. For $X' \in C_c^1(\Omega'; \mathbb{R}^n)$, define

$$X(z) = rX'(T_{a,r}(z)), \qquad z \in \Omega.$$

Then $X \in C_c^1(\Omega; \mathbb{R}^n)$, and (5.4.1) gives

$$D_x X(z) = (D_x X')(T_{a,r}(z)), \qquad z \in \Omega.$$

Using (9.0.2) and (8.4.6), we obtain

$$\begin{aligned}
\int_{\Omega'} D_x X' : dM_{W'} &= A_0 \int_{\Omega} (D_x X')(T_{a,r}(z)) : dM_W(z) \\
&= A_0 \int_{\Omega} D_x X : dM_W \\
&= A_0 \int_{\Omega} X \cdot dm_{\mathrm{chr}} \\
&= A_0 r \int_{\Omega'} X' \cdot d\big((T_{a,r})_{\#} m_{\mathrm{chr}}\big).
\end{aligned}$$

The measure $A_0 r (T_{a,r})_{\#} m_{\mathrm{chr}}$ is Radon on Ω', since $T_{a,r} : \Omega \to \Omega'$ is a homeomorphism. Proposition 8.4.2 now gives local boundedness for W' and (9.0.3). □

Given a sequence $s_i \downarrow 0$ with $B_p(a, s_i) \Subset \Omega$ and $\mu(B_p(a, s_i)) > 0$, define the *mass-normalized rescalings*

$$\mu_i = \frac{(T_{a,s_i})_{\#}\mu}{\mu(B_p(a, s_i))}, \qquad W_i = \frac{(\widehat{T}_{a,s_i})_{\#} W}{\mu(B_p(a, s_i))}. \tag{9.0.4}$$

The measures μ_i and W_i are defined on $T_{a,s_i}(\Omega)$ and $T_{a,s_i}(\Omega) \times \mathbb{P}G$, respectively.

Let μ be a positive Radon measure on Ω, and let $a \in \Omega$. A nonzero positive Radon measure τ on $U_p(0,1)$ is a *mass-normalized limit of μ at a* if there is a sequence $s_i \downarrow 0$ such that $B_p(a, s_i) \Subset \Omega$, $0 < \mu(B_p(a, s_i)) < \infty$ for every i, and

$$\frac{(T_{a,s_i})_{\#}\mu}{\mu(B_p(a, s_i))} \rightharpoonup \tau \qquad \text{in } U_p(0,1). \tag{9.0.5}$$

Convergence is understood in the sense of (4.2.2).

For $a \in \Omega$ and $r > 0$, define the *geometric rescalings*

$$\mu_{a,r} = r^{-q} (T_{a,r})_{\#}\mu, \qquad W_{a,r} = r^{-q} (\widehat{T}_{a,r})_{\#} W, \tag{9.0.6}$$

where $q = n + 1$. The measures $\mu_{a,r}$ and $W_{a,r}$ are defined on $T_{a,r}(\Omega)$ and $T_{a,r}(\Omega) \times \mathbb{P}G$, respectively.

Let η be a positive Radon measure on Ω, and let $a \in \Omega$. A nonzero positive Radon measure τ on $\mathbb{P}^n$ is a *geometric q-tangent of η at a* if there is a sequence $r_i > 0$ decreasing to zero such that

$$r_i^{-q} (T_{a,r_i})_{\#}\eta \rightharpoonup \tau \qquad \text{in } \mathbb{P}^n. \tag{9.0.7}$$

We denote the class of these measures by $\mathrm{Tan}_p^q(\eta, a)$.

Definition 9.1 (Parabolic tangent measures)**.** Let η be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and let $a \in \Omega$. A nonzero positive Radon measure σ on $\mathbb{P}^n$ is a *parabolic tangent measure of η at a* if there are sequences $r_i \downarrow 0$ and $b_i > 0$ such that

$$b_i (T_{a,r_i})_{\#}\eta \rightharpoonup \sigma \qquad \text{in } \mathbb{P}^n. \tag{9.0.8}$$

We denote this class by $\mathrm{Tan}_p(\eta, a)$. The measures on the left are defined on $T_{a,r_i}(\Omega)$, and convergence has the meaning of (4.2.3). We use the parabolic dilations and normalization convention of [26, Definition 4.4, p. 864].

For a parabolic varifold W on Ω, a nonzero positive Radon measure Υ on $\mathbb{P}^n \times \mathbb{P}G$ is a *geometric q-tangent of W at a* if there is a sequence $r_i > 0$ decreasing to zero such that

$$r_i^{-q} (\widehat{T}_{a,r_i})_{\#} W \rightharpoonup \Upsilon \qquad \text{in } \mathbb{P}^n \times \mathbb{P}G. \tag{9.0.9}$$

We denote this class by $\mathrm{Tan}_p^q(W, a)$.

Since $a \in \Omega$, every compact set $K \subset \mathbb{P}^n$ is contained in $T_{a,r_i}(\Omega)$ for all sufficiently large i. The limits in (9.0.7) and (9.0.9) are understood in the sense of (4.2.3).

Suppose that W has locally bounded spatial first chronovariation, with representing measure m_{chr}. By Proposition 9.1, the mass-normalized rescalings W_i in (9.0.4) have weight μ_i and locally bounded spatial first chronovariation. Writing $m_{\mathrm{chr},i}$ for their representing measures, we have

$$\begin{aligned} M_{W_i} &= \frac{(T_{a,s_i})_{\#} M_W}{\mu(B_p(a, s_i))}, \\ m_{\mathrm{chr},i} &= \frac{s_i (T_{a,s_i})_{\#} m_{\mathrm{chr}}}{\mu(B_p(a, s_i))} \end{aligned} \tag{9.0.10}$$

as Radon measures on $T_{a,s_i}(\Omega)$ for every i.

For $a \in \Omega$ and $r > 0$, the geometric rescaling $W_{a,r}$ in (9.0.6) has weight $\mu_{a,r}$ and locally bounded spatial first chronovariation. Its representing measure $m_{\mathrm{chr},a,r}$ satisfies

$$M_{W_{a,r}} = r^{-q} (T_{a,r})_{\#} M_W, \qquad m_{\mathrm{chr},a,r} = r^{1-q} (T_{a,r})_{\#} m_{\mathrm{chr}} \tag{9.0.11}$$

as Radon measures on $T_{a,r}(\Omega)$, where $q = n + 1$.

Lemma 9.1 (Recentering geometric tangents)**.** *Let ϕ be a finite positive Radon measure on $\mathbb{P}^n$, $n \geq 1$, and put $q = n + 1$. Suppose that*

$$0 < \Theta^q_{p*}(\phi, a) \leq \Theta^{q*}_p(\phi, a) < \infty \qquad \textit{for } \phi\textit{-almost every } a \in \mathbb{P}^n. \tag{9.0.12}$$

Then, for ϕ-almost every $a \in \mathbb{P}^n$, every $\tau \in \mathrm{Tan}^q_p(\phi, a)$ satisfies

$$s^{-q} (T_{z,s})_{\#} \tau \in \mathrm{Tan}^q_p(\phi, a) \qquad \big(z \in \operatorname{supp} \tau,\ s > 0\big). \tag{9.0.13}$$

Proof. Apply [27, Proposition 2.13(iv), p. 15] with $h = q$ and the homogeneous metric d_p. The density and tangent definitions in [27, Definitions 2.10 and 2.12, p. 14] use the same r^{-q} normalization as (5.3.1) and (9.0.7). Moreover,

$$(T_{z,s})_{\#} \tau(A) = \tau(z + \delta_s A) \qquad (A \subset \mathbb{P}^n \text{ Borel}),$$

which is the measure rescaling in its Definition 2.12. The measure in (9.0.13) is nonzero, since $\tau \neq 0$ and $T_{z,s}$ is a bijection. □

9.1. **Rectifiable tangent planes.** The following lemma is a consequence of Mattila's results [26, Theorems 4.9 and 4.12, pp. 866–867]. We give an explicit Borel representative of the tangent plane.

Lemma 9.1.1 (Borel tangent planes on a finite graph piece)**.** *Let $V \in \mathbb{P}G$, let $g : V \to V^{\perp}$ be Λ-Lipschitz for d_p, where $0 \leq \Lambda < \infty$, and set $\Gamma = \{v + g(v) : v \in V\}$. Let $E \subset \Gamma$ be*

Borel and countably $\mathbb{P}$-rectifiable with $\mathscr{H}_p^q(E) < \infty$, where $q = n+1$. Put $\nu = \mathscr{H}_p^q \llcorner E$, regarded as a finite Radon measure on $\mathbb{P}^n$. Then

$$0 < \Theta^q_{p*}(\nu, a) \le \Theta^{q*}_p(\nu, a) < \infty \qquad \text{for } \nu\text{-almost every } a \in E. \tag{9.1.1}$$

There is a Borel map $P_E : E \to \mathbb{P}G$ such that, for ν-almost every $a \in E$,

$$\varnothing \ne \operatorname{Tan}^q_p(\nu, a) \subset \operatorname{Tan}_p(\nu, a) = \{c\, \mathscr{H}_p^q \llcorner P_E(a) : c > 0\}. \tag{9.1.2}$$

The plane $P_E(a)$ is uniquely determined at those points.

Proof. By Definition 5.7.2, E is vertically LG q-rectifiable, and $\mathscr{H}_p^q(E) < \infty$. Thus [26, Theorem 4.9, p. 866] gives (9.1.1); its density uses $(2r)^{-q}$, so our raw densities are 2^q times those in that theorem. The vertical implication (2)⇒(4) of [26, Theorem 4.12, p. 867] gives, for ν-almost every $a \in E$, a plane $\widetilde{P}_a \in \mathbb{P}G$ with

$$\operatorname{Tan}_p(\nu, a) = \{c\, \mathscr{H}_p^q \llcorner \widetilde{P}_a : c > 0\}. \tag{9.1.3}$$

The metric, Hausdorff normalization and arbitrary positive tangent normalization agree with those in that theorem. The plane is unique because (5.6.4) identifies it with the support of each measure in (9.1.3).

For $a \in E$ and $r > 0$, define the rescaled measure on $\mathbb{P}^n$ by

$$\nu_{a,r} = r^{-q} (T_{a,r})_{\#} \nu. \tag{9.1.4}$$

Choose a Borel ν-null set $N \subset E$ containing the exceptional points of (9.1.1) and (9.1.3). For $a \in E \setminus N$, (9.1.1) gives $c_-, c_+, r_0 > 0$ with

$$c_- r^q \le \nu(B_p(a, r)) \le c_+ r^q \qquad (0 < r < r_0). \tag{9.1.5}$$

For every sequence $r_i \downarrow 0$, (9.1.5) gives uniform mass bounds on compact sets for $(\nu_{a,r_i})_i$. By [33, Chapter 1, Theorem 4.16, pp. 37–38], a subsequence converges weakly to a Radon measure τ on $\mathbb{P}^n$, and [25, Theorem 1.24(1), p. 19] gives

$$\tau(B_p(0,1)) \ge \limsup_i \nu_{a,r_i}(B_p(0,1)) \ge c_- > 0.$$

Thus every such limit is a nonzero geometric tangent and belongs to the ray (9.1.3). For this compactness argument, compare [25, proof of Theorem 14.3 and Remark 14.4(2), p. 187].

It remains to choose the plane field measurably. If $n = 1$, take the constant field equal to the time axis. Suppose $n \ge 2$ and set $\psi(z) = \max\{0, 1 - d_p(z, 0)\}$ for $z \in \mathbb{P}^n$. For $a \in E$ and $r > 0$, define

$$\begin{aligned} M_r(a) &= \int_{\mathbb{P}^n} \psi(x,t)\, xx^{\mathsf{T}}\, d\nu_{a,r}(x,t), \\ d_r(a) &= \int_{\mathbb{P}^n} \psi(x,t) |x|^2\, d\nu_{a,r}(x,t) = \operatorname{tr} M_r(a), \\ R_r(a) &= \begin{cases} M_r(a)/d_r(a), & d_r(a) > 0, \\ 0, & d_r(a) = 0. \end{cases} \end{aligned} \tag{9.1.6}$$

For fixed r, the integrands, written using (9.1.4), are bounded Borel functions of (a, z) integrated against the fixed finite measure ν. Hence $R_r : E \to \mathbb{R}^{n\times n}$ is Borel by [14, Theorem 2.36, pp. 66–67].

For $a \in E \setminus N$, write $\widetilde{P}_a = L_a \times \mathbb{R}_t$. Spatial rotations preserving L_a and (5.6.3) give

$$\begin{aligned} \int_{\widetilde{P}_a} \psi(x,t) xx^{\mathsf{T}}\, d\mathscr{H}_p^q(x,t) &= \beta_a P_{L_a}, \\ \beta_a &= \frac{1}{n-1} \int_{\widetilde{P}_a} \psi(x,t) |x|^2\, d\mathscr{H}_p^q(x,t) > 0. \end{aligned} \tag{9.1.7}$$

The moment integrands belong to $C_c(\mathbb{P}^n;\mathbb{R})$ entrywise. Along any subsequence with $\nu_{a,r_i} \rightharpoonup c\,\mathscr{H}^q_p \llcorner \widetilde{P}_a$, equations (9.1.6) and (9.1.7) imply $(n-1)R_{r_i}(a) \to P_{L_a}$. Every sequence of radii tending to zero has such a subsequence, so

$$(n-1)R_r(a) \longrightarrow P_{L_a} \qquad (r \downarrow 0,\ a \in E \setminus N). \tag{9.1.8}$$

Let $\mathcal{Q} = \{P_L : L \in G(n,n-1)\}$ and define $J : \mathbb{P}G \to \mathcal{Q}$ by $J(L \times \mathbb{R}_t) = P_L$. By Section 5.5, $\mathcal{Q}$ is compact and J^{-1} is continuous. The set

$$D = \{a \in E : ((n-1)R_{1/k}(a))_{k\ge 1} \text{ converges to a member of } \mathcal{Q}\}$$

is Borel by the Cauchy criterion and closedness of $\mathcal{Q}$. Fix $P_0 \in \mathbb{P}G$ and define

$$P_E(a) = \begin{cases} J^{-1}\Big(\lim\limits_{k\to\infty}(n-1)R_{1/k}(a)\Big), & a \in D, \\ P_0, & a \in E \setminus D. \end{cases} \tag{9.1.9}$$

This map is Borel and agrees with $\widetilde{P}_a$ on $E \setminus N$ by (9.1.8), proving (9.1.2). □

Proposition 9.1.1 (Compatibility of rectifiable tangent planes)**.** *Let $E, F \subset \mathbb{P}^n$ satisfy the hypotheses of Lemma 9.1.1, possibly on different full vertical Lipschitz graphs. Let $P_E : E \to \mathbb{P}G$ and $P_F : F \to \mathbb{P}G$ be the Borel fields in* (9.1.2)*. Then*

$$P_E(a) = P_F(a) \qquad \text{for } \mathscr{H}^q_p\text{-almost every } a \in E \cap F. \tag{9.1.10}$$

Proof. Put $B = E \cap F$, $\nu_E = \mathscr{H}^q_p \llcorner E$ and $\nu_F = \mathscr{H}^q_p \llcorner F$. Both measures are finite and satisfy (9.1.1). Apply [27, Proposition 2.13(ii)–(iii) and (2.7), pp. 14–15] to each measure with the integrable Borel function $\mathbf{1}_B$. The geometric tangent normalization is the same r^{-q} as in (9.0.7). Since $\nu_E \llcorner B = \nu_F \llcorner B = \mathscr{H}^q_p \llcorner B$, we obtain

$$\operatorname{Tan}^q_p(\nu_E, a) = \operatorname{Tan}^q_p(\mathscr{H}^q_p \llcorner B, a) = \operatorname{Tan}^q_p(\nu_F, a) \qquad \text{for } \mathscr{H}^q_p\text{-almost every } a \in B. \tag{9.1.11}$$

Choose a Borel $\mathscr{H}^q_p$-null set $N \subset B$ containing the exceptional points in B of (9.1.11) and of (9.1.2) for both measures. Fix $a \in B \setminus N$. By (9.1.2), choose a nonzero $\tau \in \operatorname{Tan}^q_p(\nu_E, a)$. The same equation and (9.1.11) imply

$$\tau = c\,\mathscr{H}^q_p \llcorner P_E(a) = c'\,\mathscr{H}^q_p \llcorner P_F(a) \qquad \text{for some } c, c' > 0.$$

The support of each flat measure is its plane by (5.6.4). Taking supports proves (9.1.10). □

Definition 9.1.1 (Rectifiable tangent plane)**.** Let $\Omega \subset \mathbb{P}^n$ be open, and let $K \subset \Omega$ be Borel and countably $\mathbb{P}$-rectifiable, with $q = n+1$. Choose the pairwise disjoint Borel pieces K_j in Lemma 5.7.1 and the Borel maps $P_j : K_j \to \mathbb{P}G$ in Lemma 9.1.1. Choose a Borel $\mathscr{H}^q_p$-null set $N \subset K$ containing $K \setminus \bigcup_j K_j$ and the exceptional sets of (9.1.2) on all the pieces. Fix $P_0 \in \mathbb{P}G$. The *rectifiable tangent plane field* $\operatorname{recTan}(K,\cdot) : K \to \mathbb{P}G$ is a Borel representative defined by

$$\operatorname{recTan}(K,a) = \begin{cases} P_j(a), & a \in K_j \setminus N, \\ P_0, & a \in N. \end{cases} \tag{9.1.12}$$

The map is Borel because its domains are pairwise disjoint Borel sets and each P_j is Borel. Proposition 9.1.1, applied to the countably many intersections of pieces from two decompositions, shows that the field is independent of all these choices $\mathscr{H}^q_p$-almost everywhere on K. Proposition 9.1.1, applied to pieces of $K \cap U$ and $K \cap U'$, also gives

$$\operatorname{recTan}(K \cap U, a) = \operatorname{recTan}(K \cap U', a) \quad \text{for } \mathscr{H}^q_p\text{-almost every } a \in K \cap U \cap U', \tag{9.1.13}$$

whenever $U, U' \subset \Omega$ are open.

9.2. **Tangent Translation Invariance.**

Lemma 9.2.1 (Limiting spatial equation)**.** *Let $W = \mu \otimes \nu_z$ be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, $n \geq 1$, with locally bounded spatial first chronovariation and representing measure m_{chr}. Put $q = n+1$ and $U = U_p(0,1)$. For μ-almost every $a \in \Omega$ satisfying*

$$\liminf_{r \downarrow 0} \frac{\mu(B_p(a,r))}{(2r)^q} > 0, \tag{9.2.1}$$

the following holds. Let $R_i \downarrow 0$ and $\varepsilon_i \downarrow 0$ be arbitrary sequences of positive numbers with $B_p(a,R_i) \Subset \Omega$. Choose radii s_i by applying Lemma 5.3.2 at a to these sequences. After omitting finitely many indices and passing to a subsequence, relabel R_i, ε_i, s_i together so that s_i decreases strictly to zero. Let μ_i, W_i be the rescalings in (9.0.4)*, and let $m_{\mathrm{chr},i}$ be the representing measures of W_i in* (9.0.10)*. Then*

$$|m_{\mathrm{chr},i}|(K) \longrightarrow 0 \qquad \text{for every compact } K \subset U. \tag{9.2.2}$$

If τ is a positive Radon measure on U such that

$$\mu_i \rightharpoonup \tau \quad \text{in } U, \qquad W_i \rightharpoonup \tau \otimes \nu_a \quad \text{in } U \times \mathbb{P}G, \tag{9.2.3}$$

then, for every $X \in C^1_c(U;\mathbb{R}^n)$,

$$\delta^p_{\mathrm{chr}}(\tau \otimes \nu_a)(X,0) = \int_U D_x X : dM_{\tau \otimes \nu_a} = 0. \tag{9.2.4}$$

Proof. We adapt the Euclidean argument in [10, arXiv version, proof of Lemma 2.4, p. 8]. Apply Lemma 11.3 with $\lambda = |m_{\mathrm{chr}}|$. Write $|m_{\mathrm{chr}}| = g\mu + \lambda_s$, where $g \geq 0$ and $\lambda_s \perp \mu$, and let N be the exceptional μ-null set in that lemma. Fix $a \in \Omega \setminus N$ satisfying (9.2.1), and choose R_i, ε_i, s_i as in the statement. Lemma 5.3.2 gives $B_p(a,5s_i) \Subset \Omega$ and $0 < \mu(B_p(a,s_i)) < \infty$. Since $\varepsilon_i \leq 1$ for all sufficiently large i, (5.3.4) gives $s_i \in \mathcal{R}_{2\cdot 5^q}(a)$ for those indices (see (11.0.1)). Equation (11.0.9) therefore yields

$$\frac{|m_{\mathrm{chr}}|(B_p(a,s_i))}{\mu(B_p(a,s_i))} \longrightarrow g(a) < \infty. \tag{9.2.5}$$

For every compact $K \subset U$, the rescaling map satisfies $T^{-1}_{a,s_i}(K) \subset B_p(a,s_i)$. The total variation in (4.2.4) and the scaling formula (9.0.10) give

$$\begin{aligned} |m_{\mathrm{chr},i}|(K) &\leq \frac{s_i}{\mu(B_p(a,s_i))} |m_{\mathrm{chr}}|\big(T^{-1}_{a,s_i}(K)\big) \\ &\leq s_i \frac{|m_{\mathrm{chr}}|(B_p(a,s_i))}{\mu(B_p(a,s_i))} \longrightarrow 0, \end{aligned}$$

where the limit follows from (9.2.5) and $s_i \to 0$. This proves (9.2.2).

Now suppose (9.2.3) holds. Fix $X \in C^1_c(U;\mathbb{R}^n)$, set $K = \operatorname{supp} X$, and extend X by zero to $\mathbb{P}^n$. Since $U \subset T_{a,s_i}(\Omega)$, the first-variation formula (8.4.3) and its representation (8.4.6) give

$$\begin{aligned} \left| \int_{U \times \mathbb{P}G} \operatorname{tr}(P_L D_x X(z))\, dW_i(z, L \times \mathbb{R}_t) \right| &= \left| \int_U X \cdot dm_{\mathrm{chr},i} \right| \\ &\leq \|X\|_\infty\, |m_{\mathrm{chr},i}|(K) \longrightarrow 0 \end{aligned}$$

by (9.2.2). The map $(z, L \times \mathbb{R}_t) \mapsto \operatorname{tr}(P_L D_x X(z))$ belongs to $C_c(U \times \mathbb{P}G;\mathbb{R})$: it is continuous and supported in $K \times \mathbb{P}G$, which is compact by Section 5.5. Passing to the limit in (9.2.3) gives

$$\int_{U \times \mathbb{P}G} \operatorname{tr}(P_L D_x X(z))\, d(\tau \otimes \nu_a)(z, L \times \mathbb{R}_t) = 0.$$

Equation (8.4.3), applied to $\tau \otimes \nu_a$ on U, proves (9.2.4). □

Lemma 9.2.2 (Limiting temporal equation)**.** *Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and put $q = n+1$ and $U = U_p(0,1)$. Suppose that $a \in \Omega$ satisfies* (5.3.3) *and* $\mathrm{HT}(\mu, a)$*. Let $R_i \downarrow 0$ and $\varepsilon_i \downarrow 0$ be positive sequences with $B_p(a, R_i) \Subset \Omega$, and choose s_i by Lemma 5.3.2. After omitting finitely many indices and passing to a subsequence, relabel R_i, ε_i, s_i together so that s_i decreases strictly to zero. Let μ_i be the mass-normalized rescalings in* (9.0.4)*. Suppose that τ is a positive Radon measure on U and*

$$\mu_i \rightharpoonup \tau \qquad \text{in } U. \tag{9.2.6}$$

Then

$$\int_{U_p(0,\kappa)} \partial_t \phi \, d\tau = 0 \qquad \big(\phi \in C^1_c(U_p(0,\kappa); \mathbb{R})\big). \tag{9.2.7}$$

Proof. Since $\varepsilon_i \le 1$ for all sufficiently large i, (5.3.4) makes the selected radii satisfy (3.0.3) with $C = 2 \cdot 5^q$. The half-time condition therefore gives

$$I_\mu(a, s_i) \longrightarrow 0. \tag{9.2.8}$$

Fix $\phi \in C^1_c(U_p(0,\kappa); \mathbb{R})$ and extend it by zero to $\mathbb{P}^n$. Put

$$C_\phi = \max\{1, \|\phi\|_\infty, \|\nabla_x \phi\|_\infty\}, \qquad \Phi_i(z) = \frac{s_i}{C_\phi} \phi(T_{a,s_i}(z)).$$

Then $\Phi_i \in \mathcal{A}_{x,s_i}(a)$ by (3.0.1). For $|h| \le c_0$, all the functions $\phi \circ \tau_h$ are supported in the compact set

$$K = \{\tau_{-h} z : z \in \operatorname{supp} \phi, \ |h| \le c_0\} \subset U.$$

Indeed, if $z \in \operatorname{supp} \phi$ and $|h| \le c_0$, then

$$\|\tau_{-h} z\|_p^2 \le \|z\|_p^2 + |h| \le \kappa^2 + c_0 < 1.$$

The maps satisfy

$$T_{a,s_i} \circ \tau_{s_i^2 h} = \tau_h \circ T_{a,s_i}.$$

Thus, for $0 < |h| \le c_0$, the rescaling formula (9.0.4) and (3.0.2) give

$$\begin{aligned} \left| \int_U (\phi \circ \tau_h - \phi) \, d\mu_i \right| &= \frac{C_\phi}{s_i \mu(B_p(a, s_i))} \left| \int_\Omega (\Phi_i \circ \tau_{s_i^2 h} - \Phi_i) \, d\mu \right| \\ &\le C_\phi I_\mu(a, s_i) |h|^{1/2} \longrightarrow 0. \end{aligned} \tag{9.2.9}$$

The limit follows from (9.2.8). The function $\phi \circ \tau_h - \phi$ belongs to $C_c(U; \mathbb{R})$. Passing to the limit in (9.2.6) yields

$$\int_U (\phi \circ \tau_h - \phi) \, d\tau = 0 \qquad (0 < |h| \le c_0). \tag{9.2.10}$$

Finally,

$$\frac{\phi(x, t+h) - \phi(x,t)}{h} = \int_0^1 \partial_t \phi(x, t + \theta h) \, d\theta \longrightarrow \partial_t \phi(x,t)$$

uniformly on K as $h \to 0$, by continuity of $\partial_t \phi$. All these functions vanish outside K, and $\tau(K) < \infty$. Dividing (9.2.10) by h and passing to the limit proves (9.2.7). □

Lemma 9.2.3 (Translation invariance of geometric tangents)**.** *Let $W = \mu \otimes \nu_z$ be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, $n \ge 1$, with locally bounded spatial first chronovariation, and put $q = n+1$. For μ-almost every $a \in \Omega$ satisfying*

$$0 < \Theta^q_{p*}(\mu, a) \le \Theta^{q*}_p(\mu, a) < \infty, \qquad \mathrm{HT}(\mu, a), \qquad \nu_a = \delta_V, \tag{9.2.11}$$

where $V = L \times \mathbb{R}_t \in \mathbb{P}G$, every $\tau \in \operatorname{Tan}^q_p(\mu, a)$ is invariant under translations by V:

$$(z \mapsto z + v)_\# \tau = \tau \qquad (v \in V) \tag{9.2.12}$$

as Radon measures on $\mathbb{P}^n$.

Proof. We adapt the spatial argument in [10, arXiv version, proof of Lemma 2.4, p. 8], as in Lemma 9.2.1. Let m_{chr} represent the spatial first chronovariation of W, and write $|m_{\mathrm{chr}}| = g\mu + \lambda_s$, where $g \geq 0$ and $\lambda_s \perp \mu$. Choose a Borel μ-null set $N \subset \Omega$ containing the exceptional sets for Lemma 9.3.2 and for Lemma 11.3 applied to $\lambda = |m_{\mathrm{chr}}|$. Fix $a \in \Omega \setminus N$ satisfying (9.2.11). Choose $0 < c_- \leq c_+ < \infty$ and $r_0 > 0$ with $B_p(a, r_0) \Subset \Omega$ so that (9.3.9) holds. For $0 < s < r_0/5$,

$$\frac{\mu(B_p(a,5s))}{\mu(B_p(a,s))} \leq 5^q \frac{c_+}{c_-} =: C,$$

so $s \in \mathcal{R}_C(a)$ by (11.0.1). Lemma 11.3 and the half-time condition give

$$\frac{|m_{\mathrm{chr}}|(B_p(a,s))}{\mu(B_p(a,s))} \longrightarrow g(a) < \infty, \qquad I_\mu(a,s) \longrightarrow 0 \qquad (s \downarrow 0). \tag{9.2.13}$$

Fix $\tau \in \mathrm{Tan}^q_p(\mu, a)$ and choose $r_i > 0$ decreasing to zero with $\mu_{a,r_i} \rightharpoonup \tau$ in $\mathbb{P}^n$. By (9.3.13) in the proof of Lemma 9.3.2, the same sequence satisfies

$$W_{a,r_i} \rightharpoonup \tau \otimes \delta_V \qquad \text{in } \mathbb{P}^n \times \mathbb{P}G. \tag{9.2.14}$$

Put $\Omega_i = T_{a,r_i}(\Omega)$. By (9.0.11), the representing measure of W_{a,r_i} on Ω_i is

$$m_i = r_i^{1-q} (T_{a,r_i})_{\#} m_{\mathrm{chr}}.$$

For every fixed $R > 0$ and all sufficiently large i, (4.2.4) and (9.3.9) give

$$\begin{aligned} |m_i|(B_p(0,R)) &\leq r_i^{1-q} |m_{\mathrm{chr}}|(B_p(a, Rr_i)) \\ &\leq c_+ R^q r_i \frac{|m_{\mathrm{chr}}|(B_p(a,Rr_i))}{\mu(B_p(a,Rr_i))} \longrightarrow 0, \end{aligned} \tag{9.2.15}$$

by (9.2.13). For $X \in C^1_c(\mathbb{P}^n; \mathbb{R}^n)$, the function $(y, L' \times \mathbb{R}_t) \mapsto \mathrm{tr}(P_{L'} D_x X(y))$ belongs to $C_c(\mathbb{P}^n \times \mathbb{P}G; \mathbb{R})$. The representation (8.4.6), the bundle limit (9.2.14) and (9.2.15) therefore yield

$$\int_{\mathbb{P}^n} P_L : D_x X \, d\tau = \lim_{i \to \infty} \int_{\Omega_i} X \cdot dm_i = 0 \qquad \big(X \in C^1_c(\mathbb{P}^n; \mathbb{R}^n)\big). \tag{9.2.16}$$

For $e \in L$ and $\phi \in C^1_c(\mathbb{P}^n; \mathbb{R})$, take $X = e\phi$ in (9.2.16). Since $P_L e = e$,

$$\int_{\mathbb{P}^n} e \cdot \nabla_x \phi \, d\tau = 0.$$

Apply this identity to $\phi_s(x,t) = \phi(x + se, t)$ for $0 \leq s \leq 1$. The functions ϕ_s have their supports in the compact set

$$K = \{(x - se, t) : (x,t) \in \mathrm{supp}\, \phi, \ 0 \leq s \leq 1\}.$$

The fundamental theorem of calculus and Fubini's theorem give

$$\int_{\mathbb{P}^n} \big(\phi(x+e,t) - \phi(x,t)\big) \, d\tau(x,t) = \int_0^1 \int_{\mathbb{P}^n} e \cdot \nabla_x \phi(x + se, t) \, d\tau(x,t) \, ds = 0. \tag{9.2.17}$$

We next prove invariance under every time translation. Fix $\phi \in C^1_c(\mathbb{P}^n; \mathbb{R})$ and $h \in \mathbb{R} \setminus \{0\}$. Choose $R \geq 1$ such that $\mathrm{supp}\, \phi \subset U_p(0, \kappa R)$ and $|h| \leq c_0 R^2$. Define

$$C_\phi = \max\{1, \|\phi\|_\infty, \|\nabla_x \phi\|_\infty\}, \qquad \Phi_i(z) = \frac{r_i}{C_\phi} \phi(T_{a,r_i}(z)) \quad (z \in \mathbb{P}^n).$$

For all sufficiently large i, $\Phi_i \in \mathcal{A}_{x,Rr_i}(a)$: its support is in $U_p(a, \kappa R r_i)$, its supremum norm is at most $r_i \leq R r_i$, and $\|\nabla_x \Phi_i\|_\infty \leq 1$. Moreover, ϕ and $\phi \circ \tau_h$ are supported in $U_p(0,R)$, since

$$\|\tau_{-h} y\|_p^2 \leq \|y\|_p^2 + |h| < (\kappa^2 + c_0) R^2 < R^2 \qquad (y \in \mathrm{supp}\, \phi).$$

Using $T_{a,r_i} \circ \tau_{r_i^2 h} = \tau_h \circ T_{a,r_i}$, the rescaling formula (9.0.6) and the definition (3.0.2) give

$$\begin{aligned} \left| \int_{\Omega_i} (\phi \circ \tau_h - \phi)\, d\mu_{a,r_i} \right| &= \frac{C_\phi}{r_i^{q+1}} \left| \int_\Omega (\Phi_i \circ \tau_{r_i^2 h} - \Phi_i)\, d\mu \right| \\ &\le C_\phi |h|^{1/2} I_\mu(a, Rr_i) \frac{\mu(B_p(a, Rr_i))}{r_i^q} \\ &\le C_\phi c_+ R^q |h|^{1/2} I_\mu(a, Rr_i) \longrightarrow 0. \end{aligned} \tag{9.2.18}$$

The last limit follows from (9.2.13). Passing to $\mu_{a,r_i} \rightharpoonup \tau$ gives

$$\int_{\mathbb{P}^n} \phi(x, t+h)\, d\tau(x,t) = \int_{\mathbb{P}^n} \phi\, d\tau \qquad \big(\phi \in C_c^1(\mathbb{P}^n; \mathbb{R}),\ h \in \mathbb{R}\big). \tag{9.2.19}$$

Combining (9.2.17) and (9.2.19) proves

$$\int_{\mathbb{P}^n} \phi(z+v)\, d\tau(z) = \int_{\mathbb{P}^n} \phi\, d\tau \qquad \big(\phi \in C_c^1(\mathbb{P}^n; \mathbb{R}),\ v \in V\big).$$

Uniform approximation by smooth functions with a common compact support and uniqueness of Radon measures [14, Proposition 8.17, Lemma 8.18 and Theorem 7.2] now give (9.2.12). □

9.3. **Fiber Freezing.**

Lemma 9.3.1 (Fiber freezing)**.** *Let $W = \mu \otimes \nu_z$ be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, and put $q = n+1$ and $U = U_p(0,1)$. For μ-almost every $a \in \Omega$ satisfying*

$$\liminf_{r \downarrow 0} \frac{\mu(B_p(a,r))}{(2r)^q} > 0, \tag{9.3.1}$$

the following holds. Let $R_i \downarrow 0$ and $\varepsilon_i \downarrow 0$ be arbitrary sequences of positive numbers such that $B_p(a, R_i) \Subset \Omega$. Choose radii s_i by applying Lemma 5.3.2 at a to these sequences. After omitting finitely many indices and passing to a subsequence, relabel R_i, ε_i, s_i together so that s_i decreases strictly to zero. Let μ_i and W_i be the mass-normalized rescalings in (9.0.4)*. Suppose that τ and Z are positive Radon measures on U and $U \times \mathbb{P}G$, respectively, and that*

$$\mu_i \rightharpoonup \tau \quad \text{in } U, \qquad W_i \rightharpoonup Z \quad \text{in } U \times \mathbb{P}G. \tag{9.3.2}$$

Then

$$Z = \tau \otimes \nu_a \qquad \text{on } U \times \mathbb{P}G.$$

Proof. Compare the Euclidean fiber-freezing result in [10, arXiv version, Lemma 2.3, p. 6]. We use a countable family in $C(\mathbb{P}G; \mathbb{R})$ and differentiation, as in [33, Chapter 8, proof of Lemma 1.3]; the differentiation along parabolic radii is supplied here by Lemma 11.2. Since $\mathbb{P}G$ is a compact metric space, choose a sequence $(h_k)_{k \ge 1}$ dense in $C(\mathbb{P}G; \mathbb{R})$ for the uniform norm, with $h_1 = 1$. Define

$$g_k(z) = \int_{\mathbb{P}G} h_k(P)\, d\nu_z(P), \qquad z \in \Omega. \tag{9.3.3}$$

The function g_k is μ-measurable in z, and $|g_k(z)| \le \|h_k\|_\infty$ because ν_z is a probability measure. Thus $g_k \in L^1_{\mathrm{loc}}(\Omega, \mu)$. For each k, let N_k be a μ-null set outside which the conclusion of Lemma 11.2 holds for g_k and every $1 \le C < \infty$. Set

$$N = \bigcup_{k=1}^{\infty} N_k.$$

Then $\mu(N) = 0$, and N is independent of the point and of the subsequent choice of radii.

Fix $a \in \Omega \setminus N$ satisfying (9.3.1), and choose R_i, ε_i, s_i as in the statement. Lemma 5.3.2 gives $B_p(a, 5s_i) \Subset \Omega$ and $0 < \mu(B_p(a, s_i)) < \infty$. Choose i_0 so that $\varepsilon_i \le 1$ for $i \ge i_0$.

The estimate (5.3.4) then implies $s_i \in \mathcal{R}_{2\cdot 5^q}(a)$ (see (11.0.1)) for $i \geq i_0$. Since $a \notin N_k$, Lemma 11.2 yields

$$\frac{1}{\mu(B_p(a,s_i))}\int_{B_p(a,s_i)} |g_k(z)-g_k(a)|\,d\mu(z) \longrightarrow 0 \qquad \text{for every } k\geq 1. \tag{9.3.4}$$

Fix $f \in C_c(U;\mathbb{R})$ and $k \geq 1$. We shall prove that

$$\int_{U\times\mathbb{P}G} f(y)h_k(P)\,dW_i(y,P) - \left(\int_U f\,d\mu_i\right)\left(\int_{\mathbb{P}G} h_k\,d\nu_a\right) \longrightarrow 0 \qquad (i\to\infty).$$

Extend f by zero outside U. The disintegration formula (6.1.2), together with (9.0.4) and (9.3.3), gives

$$\begin{aligned}
&\left|\int_{U\times\mathbb{P}G} f(y)h_k(P)\,dW_i(y,P) - g_k(a)\int_U f\,d\mu_i\right| \\
&\quad= \frac{1}{\mu(B_p(a,s_i))}\left|\int_{B_p(a,s_i)} f(T_{a,s_i}(z))(g_k(z)-g_k(a))\,d\mu(z)\right| \qquad (9.3.5)\\
&\quad\leq \frac{\|f\|_\infty}{\mu(B_p(a,s_i))}\int_{B_p(a,s_i)} |g_k(z)-g_k(a)|\,d\mu(z).
\end{aligned}$$

The upper bound in (9.3.5) tends to zero by (9.3.4). Since $\mathbb{P}G$ is compact, the function $(y,P)\mapsto f(y)h_k(P)$ belongs to $C_c(U\times\mathbb{P}G;\mathbb{R})$. Passing to the limits in (9.3.2) therefore gives

$$\int_{U\times\mathbb{P}G} f(y)h_k(P)\,dZ(y,P) = \left(\int_U f\,d\tau\right)\left(\int_{\mathbb{P}G} h_k\,d\nu_a\right). \tag{9.3.6}$$

For $h\in C(\mathbb{P}G;\mathbb{R})$, choose $h_{k_j}\to h$ uniformly. With $K=\operatorname{supp} f$, both $Z(K\times\mathbb{P}G)$ and $\tau(K)$ are finite. Uniform convergence thus permits passage to the limit in (9.3.6) with $k=k_j$, so that the identity also holds with h in place of h_k.

It remains to prove that $Z=\tau\otimes\nu_a$ as Radon measures on $U\times\mathbb{P}G$. Let $F\in C_c(U\times\mathbb{P}G;\mathbb{R})$. Choose an open set $V \Subset U$ such that $\operatorname{supp} F \subset V\times\mathbb{P}G$, and put $K=\overline{V}$. By [14, Proposition 7.21], there are finite sums

$$F_j(y,P)=\sum_{\ell=1}^{m_j} f_{j\ell}(y)h_{j\ell}(P), \qquad f_{j\ell}\in C_c(U;\mathbb{R}),\quad h_{j\ell}\in C(\mathbb{P}G;\mathbb{R}),$$

such that $\operatorname{supp} F_j\subset V\times\mathbb{P}G$ and $\|F_j-F\|_\infty\to 0$. Applying (9.3.6) with $f=f_{j\ell}$ and h_k replaced by $h_{j\ell}$, and summing over $1\leq\ell\leq m_j$, gives $\int F_j\,dZ=\int F_j\,d(\tau\otimes\nu_a)$. Consequently,

$$\left|\int F\,dZ-\int F\,d(\tau\otimes\nu_a)\right| \leq \|F-F_j\|_\infty\big(Z(K\times\mathbb{P}G)+\tau(K)\big)\longrightarrow 0.$$

The measures Z and $\tau\otimes\nu_a$ are Radon and agree on $C_c(U\times\mathbb{P}G;\mathbb{R})$, so they are equal [14, Theorems 7.2 and 7.20(c)]. □

Lemma 9.3.2 (Geometric fiber freezing)**.** *Let $W=\mu\otimes\nu_z$ be a parabolic varifold on an open set $\Omega\subset\mathbb{P}^n$, and put $q=n+1$. For μ-almost every $a\in\Omega$ satisfying*

$$0<\Theta^q_{p*}(\mu,a)\leq\Theta^{q*}_p(\mu,a)<\infty \tag{9.3.7}$$

and $\nu_a=\delta_V$ for some $V\in\mathbb{P}G$, one has

$$\operatorname{Tan}^q_p(W,a)=\{\tau\otimes\delta_V:\tau\in\operatorname{Tan}^q_p(\mu,a)\}. \tag{9.3.8}$$

Proof. We use the differentiation argument in Lemma 9.3.1 with the geometric normalization; compare [10, arXiv version, Lemma 2.3, p. 6]. Let $(h_k)_{k\geq1}$ and g_k be the dense sequence in $C(\mathbb{P}G;\mathbb{R})$ and the functions in (9.3.3), respectively. Choose a Borel μ-null set $N\subset\Omega$ containing the exceptional set constructed in that proof for Lemma 11.2, simultaneously

for every g_k and every $1 \le C < \infty$. Fix $a \in \Omega \setminus N$ satisfying (9.3.7) and $\nu_a = \delta_V$. Then $g_k(a) = h_k(V)$ for every k.

By (9.3.7), choose $0 < c_- \le c_+ < \infty$ and $r_0 > 0$ with $B_p(a, r_0) \Subset \Omega$ such that

$$c_- s^q \le \mu(B_p(a,s)) \le c_+ s^q \qquad (0 < s < r_0). \tag{9.3.9}$$

For $0 < s < r_0/5$,

$$\frac{\mu(B_p(a,5s))}{\mu(B_p(a,s))} \le 5^q \frac{c_+}{c_-} =: C.$$

Thus $s \in \mathcal{R}_C(a)$, as defined in (11.0.1). Lemma 11.2 gives

$$\lim_{s \downarrow 0} \frac{1}{\mu(B_p(a,s))} \int_{B_p(a,s)} |g_k(z) - g_k(a)| \, d\mu(z) = 0 \qquad (k \ge 1). \tag{9.3.10}$$

Let $r_i > 0$ decrease to zero and put $\Omega_i = T_{a,r_i}(\Omega)$. For each fixed $R > 0$, (9.3.9) gives, for all sufficiently large i,

$$W_{a,r_i}(B_p(0,R) \times \mathbb{P}G) = \mu_{a,r_i}(B_p(0,R)) = \frac{\mu(B_p(a,Rr_i))}{r_i^q} \le c_+ R^q. \tag{9.3.11}$$

Fix $f \in C_c(\mathbb{P}^n; \mathbb{R})$ and $k \ge 1$, and choose $R > 0$ with $\operatorname{supp} f \subset B_p(0,R)$. The disintegration formula (6.1.2) and the rescalings (9.0.6) imply

$$\begin{aligned} &\left| \int_{\Omega_i \times \mathbb{P}G} f(y) h_k(P) \, dW_{a,r_i}(y,P) - h_k(V) \int_{\Omega_i} f \, d\mu_{a,r_i} \right| \\ &\quad \le \frac{\|f\|_\infty}{r_i^q} \int_{B_p(a,Rr_i)} |g_k(z) - g_k(a)| \, d\mu(z) \\ &\quad \le c_+ R^q \, \|f\|_\infty \, \frac{1}{\mu(B_p(a,Rr_i))} \int_{B_p(a,Rr_i)} |g_k(z) - g_k(a)| \, d\mu(z) \longrightarrow 0. \end{aligned} \tag{9.3.12}$$

The limit follows from (9.3.10). The uniform density of (h_k) in $C(\mathbb{P}G; \mathbb{R})$, the bound (9.3.11) and product approximation [14, Proposition 7.21] extend (9.3.12) to

$$\int_{\Omega_i \times \mathbb{P}G} F(y,P) \, dW_{a,r_i}(y,P) - \int_{\Omega_i} F(y,V) \, d\mu_{a,r_i}(y) \longrightarrow 0 \qquad \big(F \in C_c(\mathbb{P}^n \times \mathbb{P}G; \mathbb{R})\big). \tag{9.3.13}$$

Here the approximating finite sums of products have their supports in one fixed compact subset of $\mathbb{P}^n \times \mathbb{P}G$, as in the proof of Lemma 9.3.1.

If $\mu_{a,r_i} \rightharpoonup \tau$ in $\mathbb{P}^n$, then (9.3.13) gives $W_{a,r_i} \rightharpoonup \tau \otimes \delta_V$ in $\mathbb{P}^n \times \mathbb{P}G$. This proves the inclusion from right to left in (9.3.8). Conversely, suppose that $W_{a,r_i} \rightharpoonup \Upsilon$ in $\mathbb{P}^n \times \mathbb{P}G$, with $\Upsilon \ne 0$. Since $\mathbb{P}G$ is compact, $\tau = \pi_\# \Upsilon$ is a nonzero positive Radon measure on $\mathbb{P}^n$. For every $f \in C_c(\mathbb{P}^n; \mathbb{R})$,

$$\int_{\Omega_i} f \, d\mu_{a,r_i} = \int_{\Omega_i \times \mathbb{P}G} f(y) \, dW_{a,r_i}(y,P) \longrightarrow \int_{\mathbb{P}^n \times \mathbb{P}G} f(y) \, d\Upsilon(y,P) = \int_{\mathbb{P}^n} f \, d\tau.$$

Thus $\tau \in \operatorname{Tan}^q_p(\mu, a)$, and the already proved implication gives $\Upsilon = \tau \otimes \delta_V$ as Radon measures on $\mathbb{P}^n \times \mathbb{P}G$. □

9.4. **Mass Normalized Tangents.**

Lemma 9.4.1 (Existence of nonzero mass-normalized limits)**.** *Let $W = \mu \otimes \nu_z$ be a parabolic varifold on an open set $\Omega \subset \mathbb{P}^n$, and put $q = n + 1$ and $U = U_p(0,1)$. For μ-almost every $a \in \Omega$ satisfying*

$$\liminf_{r \downarrow 0} \frac{\mu(B_p(a,r))}{(2r)^q} > 0, \tag{9.4.1}$$

the following holds. Let $R_i \downarrow 0$ and $\varepsilon_i \downarrow 0$ be arbitrary sequences of positive numbers with $B_p(a,R_i) \Subset \Omega$. Choose radii s_i by applying Lemma 5.3.2 at a to these sequences. After

omitting finitely many indices and passing to a subsequence, relabel R_i, ε_i, s_i together so that s_i decreases strictly to zero. Let μ_i and W_i be the mass-normalized rescalings in (9.0.4). *There are indices $i_j \uparrow \infty$ and a positive Radon measure τ on U such that*

$$\mu_{i_j} \rightharpoonup \tau \quad \text{in } U, \qquad W_{i_j} \rightharpoonup \tau \otimes \nu_a \quad \text{in } U \times \mathbb{P}G. \tag{9.4.2}$$

Moreover,

$$\tau(B_p(0,t)) \geq 5^{-q} t^q \qquad \text{for every } 0 < t < 1. \tag{9.4.3}$$

In particular, $\tau \neq 0$.

Proof. The compactness and nonvanishing steps parallel the tangent-measure arguments in [25, proof of Theorem 14.3 and Remark 14.4(2)]. Compare also the Euclidean nonvanishing result in [10, arXiv version, Lemma 2.1, pp. 4–5]. Let N be the μ-null set constructed in the proof of Lemma 9.3.1. Fix $a \in \Omega \setminus N$ satisfying (9.4.1), and choose R_i, ε_i, s_i as in the statement. Lemma 5.3.2 gives $B_p(a, 5s_i) \Subset \Omega$ and $0 < \mu(B_p(a, s_i)) < \infty$. Thus $U \subset T_{a,s_i}(\Omega)$ for every i. By (9.0.4) and the identity $\mu = \pi_{\#} W$,

$$W_i(U \times \mathbb{P}G) = \mu_i(U) = \frac{\mu(U_p(a, s_i))}{\mu(B_p(a, s_i))} \leq 1 \qquad \text{for every } i. \tag{9.4.4}$$

We first extract a common weakly convergent subsequence of (μ_i) and (W_i) on U and $U \times \mathbb{P}G$, respectively. Since the parabolic metric is proper, U is locally compact, and

$$U = \bigcup_{m=2}^{\infty} B_p(0, 1 - m^{-1}).$$

Each ball in this union is compact, as is $\mathbb{P}G$. Hence U and $U \times \mathbb{P}G$ are σ-compact locally compact metric spaces. The bound (9.4.4) gives

$$\sup_i \mu_i(K) \leq 1, \qquad \sup_i W_i(H) \leq 1$$

for every compact $K \subset U$ and every compact $H \subset U \times \mathbb{P}G$. The compactness theorem for Radon measures [33, Chapter 1, Theorem 4.16], applied to $(\mu_i \llcorner U)$, gives indices $k_j \uparrow \infty$ and a positive Radon measure τ on U with $\mu_{k_j} \rightharpoonup \tau$ in U. Applying the same theorem to $(W_{k_j} \llcorner (U \times \mathbb{P}G))$ gives a further subsequence (i_j) of (k_j) and a positive Radon measure Z on $U \times \mathbb{P}G$ such that

$$\mu_{i_j} \rightharpoonup \tau \quad \text{in } U, \qquad W_{i_j} \rightharpoonup Z \quad \text{in } U \times \mathbb{P}G.$$

Lemma 9.3.1, applied with the sequences (R_{i_j}), (ε_{i_j}) and (s_{i_j}), gives $Z = \tau \otimes \nu_a$. This proves (9.4.2).

It remains to prove (9.4.3). Fix $0 < t < 1$. By (9.0.4) and (5.3.5),

$$\mu_{i_j}(B_p(0,t)) = \frac{\mu(B_p(a, t s_{i_j}))}{\mu(B_p(a, s_{i_j}))} \geq \frac{t^q}{5^q(1 + \varepsilon_{i_j})}.$$

The ball $B_p(0,t)$ is a compact subset of U. The compact-set Portmanteau inequality [25, Theorem 1.24(1)] therefore gives

$$\begin{aligned} \tau(B_p(0,t)) &\geq \limsup_{j \to \infty} \mu_{i_j}(B_p(0,t)) \\ &\geq \lim_{j \to \infty} \frac{t^q}{5^q(1 + \varepsilon_{i_j})} = 5^{-q} t^q. \end{aligned}$$

Taking $t = 1/2$ yields $\tau(B_p(0, 1/2)) \geq 10^{-q} > 0$, so $\tau \neq 0$. □

Lemma 9.4.2 (Concentration at one plane)**.** *Let $n \geq 1$, put $q = n + 1$ and $U = U_p(0,1)$, and fix $0 < \kappa < 1$. Let ν be a Borel probability measure on $\mathbb{P}G$, and let τ be a positive Radon measure on U satisfying*

$$\tau(B_p(0,r)) \geq 5^{-q} r^q \qquad (0 < r < 1). \tag{9.4.5}$$

Suppose that

$$\begin{aligned} \int_U A(\nu) : D_x X \, d\tau = 0 \quad \big(X \in C^1_c(U;\mathbb{R}^n)\big), \\ \int_{U_p(0,\kappa)} \partial_t \phi \, d\tau = 0 \quad \big(\phi \in C^1_c(U_p(0,\kappa);\mathbb{R})\big). \end{aligned} \tag{9.4.6}$$

Then $\nu = \delta_{L\times\mathbb{R}_t}$ for some $L \in G(n, n-1)$.

Proof. We adapt the argument comparing invariance with a lower mass bound in [10, arXiv version, proof of Lemma 3.1, p. 9]. For $n = 1$, the space $\mathbb{P}G$ is a singleton by Section 5.5, so the conclusion holds. Suppose that $n \geq 2$ and that ν is not a Dirac measure. Put $A = A(\nu)$. By (5.5.2), A is symmetric and

$$e \cdot Ae = \int_{\mathbb{P}G} |P_L e|^2 \, d\nu(L \times \mathbb{R}_t) \qquad (e \in \mathbb{R}^n). \tag{9.4.7}$$

If $e \neq 0$ and $Ae = 0$, then (9.4.7) gives $P_L e = 0$ for ν-almost every $L \times \mathbb{R}_t$. Every such L equals $e^\perp$, since $\dim L = n-1$. This would imply $\nu = \delta_{e^\perp \times \mathbb{R}_t}$. Therefore A is invertible.

For $\psi \in C^1_c(U;\mathbb{R})$ and $j = 1, \ldots, n$, use the field $X = A^{-1} e_j \psi$ in (9.4.6), where e_j is the jth standard basis vector of $\mathbb{R}^n$. It gives

$$0 = \int_U A : D_x(A^{-1} e_j \psi) \, d\tau = \int_U \partial_{x_j} \psi \, d\tau.$$

Together with the temporal identity in (9.4.6), this shows that every distributional coordinate derivative of τ vanishes on the connected open set $U_p(0,\kappa)$. The constancy argument in [33, Chapter 6, proof of Theorem 2.34, pp. 142–143] therefore gives

$$\tau = c\mathcal{L}^{n+1} \qquad \text{as Radon measures on } U_p(0,\kappa)$$

for some finite $c \geq 0$. For $0 < r < \kappa$, (5.4.2) and (9.4.5) now imply

$$5^{-q} r^q \leq \tau(B_p(0,r)) = c\frac{4\omega_n}{n+2} r^{n+2}, \qquad 5^{-q} \leq c\frac{4\omega_n}{n+2} r.$$

Letting $r \downarrow 0$ gives a contradiction. Hence ν is a Dirac measure. □

10. The Projected Blow-up Theorem

Fix $V = L \times \mathbb{R}_t \in \mathbb{P}G$, put $m = n-1$, and use the coordinates $(y,t) \in \mathbb{R}^m \times \mathbb{R}$ from (5.6.1). For a positive Radon measure μ on an open set $U \subset \mathbb{P}^n$ and $\chi \in C^1_c(U;\mathbb{R})$ with $0 \leq \chi \leq 1$, define

$$u = (P_V)_\#(\chi\mu), \qquad u(E) = \int_U \chi(z) \mathbf{1}_E(P_V z) \, d\mu(z) \quad (E \subset V \text{ Borel}). \tag{10.0.1}$$

This is a finite positive Radon measure on V, supported in $P_V(\operatorname{supp}\chi)$.

For a finite signed Radon measure η on V, define

$$\|\eta\|_{\mathcal{N}_{\mathrm{sp}}} = \sup\left\{ \left| \int_V \psi \, d\eta \right| : \psi \in C^1_c(V;\mathbb{R}),\ \|\psi\|_\infty \leq 1,\ \|\nabla_y \psi\|_\infty \leq 1 \right\}. \tag{10.0.2}$$

There is no bound on $\partial_t \psi$. When $n = 1$, the spatial gradient condition is omitted.

Lemma 10.1 (Projected spatial equations)**.** *Let W_i be parabolic varifolds on an open set $U \subset \mathbb{P}^n$, with weights μ_i and locally bounded spatial first chronovariation. Let $m_{\mathrm{chr},i}$ be their representing measures in (8.4.6). Fix $\chi \in C^1_c(U;\mathbb{R})$ with $0 \leq \chi \leq 1$, put $S = \operatorname{supp}\chi$ and $K = P_V(S)$, and set $u_i = (P_V)_\#(\chi\mu_i)$.*

If $n \geq 2$, there are finite $\mathbb{R}^{m\times m}$-valued measures X_i and finite $\mathbb{R}^m$-valued measures h_i, supported in K, such that

$$D_y u_i = \operatorname{div}_y X_i + h_i \qquad \text{in distributions on } V. \tag{10.0.3}$$

They satisfy

$$\begin{aligned} |X_i|(V) &\le n\|\chi\|_\infty \int_{S\times\mathbb{P}G} \|P_{L'} - P_L\|_{\mathrm{op}}\, dW_i(z, L'\times\mathbb{R}_t), \\ |h_i|(V) &\le n\left(\|\chi\|_\infty |m_{\mathrm{chr},i}|(S) + \|\nabla_x\chi\|_\infty \mu_i(S)\right). \end{aligned} \tag{10.0.4}$$

For every $n \ge 1$,

$$u_i(V) \le \|\chi\|_\infty \mu_i(S). \tag{10.0.5}$$

When $n = 1$, there are no spatial equations.

Proof. We adapt the Euclidean projected first-variation calculation in [10, arXiv version, proof of Lemma 3.2, (3.2)–(3.5), pp. 10–11] to the spatial first chronovariation. Assume $n \ge 2$, and write $e_1, \dots, e_m$ for the chosen basis of L. To project measures on the varifold bundle, define

$$\Pi_V : U \times \mathbb{P}G \longrightarrow V, \qquad \Pi_V(z, L'\times\mathbb{R}_t) = P_V z.$$

For $1 \le \alpha, \beta \le m$, put

$$\begin{aligned} (X_i)_{\alpha\beta} &= -(\Pi_V)_\#\big(\chi\,[e_\beta\cdot(P_{L'} - P_L)e_\alpha]\, W_i\big), \\ (h_i)_\alpha &= -(P_V)_\#(\chi\, e_\alpha \cdot m_{\mathrm{chr},i}) \\ &\quad + (\Pi_V)_\#\big((P_{L'}e_\alpha\cdot\nabla_x\chi)\, W_i\big). \end{aligned} \tag{10.0.6}$$

These measures are finite and supported in K.

For $\psi \in C^1_c(V;\mathbb{R})$ and $1 \le \alpha \le m$, apply (8.4.3) and its representation (8.4.6) to the field $Z \in C^1_c(U;\mathbb{R}^n)$ defined by $Z(z) = \chi(z)e_\alpha\psi(P_V z)$. Since $P_L e_\alpha = e_\alpha$, the definitions in (10.0.6) give

$$-\int_V \partial_{y_\alpha}\psi\, du_i = -\sum_{\beta=1}^m \int_V \partial_{y_\beta}\psi\, d(X_i)_{\alpha\beta} + \int_V \psi\, d(h_i)_\alpha. \tag{10.0.7}$$

With the distributional convention (4.2.6) in the y coordinates, this is (10.0.3).

The definitions (10.0.6) and (4.2.4) give

$$\begin{aligned} |X_i| &\le n(\Pi_V)_\#\big(\chi\|P_{L'} - P_L\|_{\mathrm{op}}\, W_i\big), \\ |h_i| &\le n(P_V)_\#\big(\chi|m_{\mathrm{chr},i}|\big) + n(\Pi_V)_\#\big(|\nabla_x\chi|\, W_i\big) \end{aligned} \tag{10.0.8}$$

as inequalities of positive Radon measures on V. Evaluating (10.0.8) on V gives (10.0.4). The mass bound (10.0.5) follows from (10.0.1) for all $n \ge 1$. □

For $h \in \mathbb{R}$, define temporal translation on V by

$$\tau_h^V : V \longrightarrow V, \qquad \tau_h^V(y,t) = (y, t+h).$$

Lemma 10.2 (Time translations of the projected measures)**.** *Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$, and suppose that $a \in \Omega$ satisfies $\mathrm{HT}(\mu, a)$. Let $s_i \downarrow 0$ satisfy (3.0.3) with $r = s_i$ and one constant $1 \le C < \infty$, and suppose $\mu(B_p(a, s_i)) > 0$. Let μ_i be the mass-normalized rescalings in (9.0.4). For $\chi \in C^\infty_c(U_p(0,\kappa);\mathbb{R})$ with $0 \le \chi \le 1$, put $u_i = (P_V)_\#(\chi\mu_i)$ and*

$$C_\chi = 1 + \|\nabla_x\chi\|_\infty + \|\partial_t\chi\|_\infty.$$

Then, for every i and $0 < |h| \le c_0$,

$$\|(\tau_h^V)_\# u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}} \le C_\chi I_\mu(a, s_i)|h|^{1/2} + C_\chi|h|. \tag{10.0.9}$$

In particular,

$$\lim_{\delta\downarrow 0} \limsup_{i\to\infty} \sup_{|h|<\delta} \|(\tau_h^V)_\# u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}} = 0. \tag{10.0.10}$$

Proof. Extend χ by zero to $\mathbb{P}^n$, and take $\psi \in C^1_c(V;\mathbb{R})$ satisfying the bounds in (10.0.2). Put $F(z) = \chi(z)\psi(P_V z)$. Since $P_V \circ \tau_h = \tau_h^V \circ P_V$,

$$\begin{aligned}\int_V \psi \, d\big((\tau_h^V)_\# u_i - u_i\big) &= \int (F \circ \tau_h - F)\, d\mu_i \\ &\quad - \int (\chi \circ \tau_h - \chi)(\psi \circ P_V \circ \tau_h)\, d\mu_i.\end{aligned} \tag{10.0.11}$$

Since $\|F\|_\infty \le 1$ and $\|\nabla_x F\|_\infty \le 1 + \|\nabla_x \chi\|_\infty \le C_\chi$, (3.0.1) gives

$$\Phi_i(z) = \frac{s_i}{C_\chi} F(T_{a,s_i}(z)), \qquad \Phi_i \in \mathcal{A}_{x,s_i}(a).$$

For $0 < |h| \le c_0$, apply (3.0.2) to Φ_i with time shift $s_i^2 h$. Since $T_{a,s_i} \circ \tau_{s_i^2 h} = \tau_h \circ T_{a,s_i}$, (9.0.4) yields

$$\left| \int (F \circ \tau_h - F)\, d\mu_i \right| \le C_\chi I_\mu(a, s_i) |h|^{1/2}. \tag{10.0.12}$$

For $|h| \le c_0$, the function $\chi \circ \tau_h - \chi$ is supported in $U_p(0,1)$, because $\kappa^2 + c_0 < 1$. Since $\mu_i(B_p(0,1)) = 1$,

$$\left| \int (\chi \circ \tau_h - \chi)(\psi \circ P_V \circ \tau_h)\, d\mu_i \right| \le \|\partial_t \chi\|_\infty |h|. \tag{10.0.13}$$

Taking the supremum over ψ in (10.0.11), using (10.0.12) and (10.0.13), proves (10.0.9). Since $\mathrm{HT}(\mu, a)$ gives $I_\mu(a, s_i) \to 0$, (10.0.10) follows from

$$\limsup_{i \to \infty} \sup_{|h| < \delta} \|(\tau_h^V)_\# u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}} \le C_\chi \delta \longrightarrow 0 \qquad (\delta \downarrow 0).$$

□

Lemma 10.3 (Compactness of the projected measures)**.** *Let $K \subset V$ be compact, and let u_i be positive Radon measures supported in K, with $\sup_i u_i(V) < \infty$. Suppose that* (10.0.10) *holds. If $m \ge 1$, suppose also that finite $\mathbb{R}^{m\times m}$-valued measures X_i and finite $\mathbb{R}^m$-valued measures h_i, all supported in K, satisfy* (10.0.3) *and*

$$|X_i|(V) \longrightarrow 0, \qquad \sup_i |h_i|(V) < \infty. \tag{10.0.14}$$

Then there are a subsequence and a nonnegative $\gamma \in L^1_{\mathrm{loc}}(V, \lambda_V)$ such that

$$|u_i - \gamma \lambda_V|(A) \longrightarrow 0 \qquad \text{for every compact } A \subset V. \tag{10.0.15}$$

Proof. We adapt the potential argument in [10, arXiv version, proof of Lemma 3.2, pp. 11–12]. We derive the temporal L^1 translation estimates from (10.0.10). For a related use of spatial compactness and time translations in a weaker norm, see Jacques Simon [32, Section 8, Theorem 5 and Lemma 8, pp. 84–85], who attributes the underlying interpolation argument to J.-L. Lions. Use the coordinates of (5.6.1), so that $\lambda_V = \mathcal{L}^{m+1}$, and write $\|\cdot\|_1$ for the $L^1(V, \lambda_V)$ norm. For a function $f : V \to \mathbb{R}$, set

$$T_h f(y,t) = f(y, t-h), \qquad R_z f(y,t) = f(y - z, t) \quad (h \in \mathbb{R},\ z \in \mathbb{R}^m).$$

The spatial translations R_z are used only when $m \ge 1$.

Choose $\rho \in C^\infty_c(\mathbb{R}^{m+1};\mathbb{R})$ with $\rho \ge 0$, $\operatorname{supp}\rho \subset \{w : |w| < 1\}$ and $\int \rho \, d\mathcal{L}^{m+1} = 1$. For $\varepsilon > 0$, put

$$\rho_\varepsilon(y,t) = \varepsilon^{-(m+1)} \rho(y/\varepsilon, t/\varepsilon).$$

Let $0 < \varepsilon_i < 1$ decrease to zero, with no further restriction, and use (4.2.5) to define

$$v_i = u_i * \rho_{\varepsilon_i}, \qquad Y_i = X_i * \rho_{\varepsilon_i}, \qquad g_i = h_i * \rho_{\varepsilon_i}. \tag{10.0.16}$$

The last two definitions are omitted when $m = 0$. These functions have a common compact support, $v_i \geq 0$ and $\|v_i\|_1 = u_i(V)$. When $m \geq 1$, they satisfy

$$\begin{aligned} \|Y_i\|_1 \leq |X_i|(V) \longrightarrow 0, \qquad \|g_i\|_1 \leq |h_i|(V), \\ D_y v_i = \operatorname{div}_y Y_i + g_i. \end{aligned} \tag{10.0.17}$$

For a test ψ in (10.0.2), the function $\psi * \check{\rho}_{\varepsilon_i}$, where $\check{\rho}(w) = \rho(-w)$, has the same amplitude and spatial gradient bounds. Consequently,

$$\|(T_h v_i - v_i)\lambda_V\|_{\mathcal{N}_{\mathrm{sp}}} \leq \|(\tau_h^V)_{\#} u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}}. \tag{10.0.18}$$

For $m \geq 1$, $k \in L^1_{\mathrm{loc}}(\mathbb{R}^m)$ and $f \in C_c^\infty(V;\mathbb{R})$, define

$$(k *_y f)(y,t) = \int_{\mathbb{R}^m} k(y-z) f(z,t)\, dz.$$

Choose $\vartheta \in C_c^\infty(\mathbb{R}^m;\mathbb{R})$ with $\vartheta \geq 0$, $\operatorname{supp}\vartheta \subset \{|y| < 1\}$ and $\int \vartheta\, dy = 1$. For $0 < \delta < 1$ and compactly supported $f \in L^1(V)$, define

$$\vartheta_\delta(y) = \delta^{-m}\vartheta(y/\delta), \qquad S_\delta f(y,t) = \int_{\mathbb{R}^m} \vartheta_\delta(z) f(y-z,t)\, dz.$$

Set $c_\vartheta = \max\{1, \|\nabla\vartheta\|_{L^1(\mathbb{R}^m)}\}$. By L^1 duality and (10.0.2),

$$\begin{aligned} \|S_\delta f\|_1 \leq c_\vartheta \delta^{-1} \|f\lambda_V\|_{\mathcal{N}_{\mathrm{sp}}}, \\ \|T_h f - f\|_1 \leq 2\|f - S_\delta f\|_1 + c_\vartheta\delta^{-1}\|(T_h f - f)\lambda_V\|_{\mathcal{N}_{\mathrm{sp}}}. \end{aligned} \tag{10.0.19}$$

Suppose first that $m \geq 2$. For $y \neq 0$, let

$$N(y) = \begin{cases} (2\pi)^{-1}\log|y|, & m = 2, \\ -\big(m(m-2)\omega_m\big)^{-1}|y|^{2-m}, & m \geq 3, \end{cases} \qquad \Delta_y N = \delta_0.$$

Applying $N *_y$ to the spatial divergence of (10.0.17) gives

$$\begin{aligned} v_i &= a_i + b_i, \\ a_i &= \sum_{\alpha,\beta=1}^m \partial_{y_\alpha}\partial_{y_\beta}\big(N *_y (Y_i)_{\alpha\beta}\big), \\ b_i &= \sum_{\alpha=1}^m (\partial_{y_\alpha} N) *_y (g_i)_\alpha. \end{aligned} \tag{10.0.20}$$

The derivatives in the definition of a_i are distributional. The operators $f \mapsto \partial_{y_\alpha}\partial_{y_\beta}(N * f)$ on $C_c^\infty(\mathbb{R}^m;\mathbb{R})$ have multipliers $\xi_\alpha\xi_\beta/|\xi|^2$ for $\xi \in \mathbb{R}^m \setminus \{0\}$, smooth away from zero and homogeneous of degree zero. Their weak-$(1,1)$ bounds [18, Theorem 6.2.7, p. 446], applied at each t and integrated in time, give

$$\sup_{s>0} s\lambda_V(\{|a_i| > s\}) \leq C_m\|Y_i\|_1 \longrightarrow 0. \tag{10.0.21}$$

Choose $R > 1$ so that all functions in (10.0.16) are supported in $\{(y,t) : |y| \leq R,\ |t| \leq R\}$. Fix $\theta \in C_c^\infty(\mathbb{R}^m;\mathbb{R})$ with $0 \leq \theta \leq 1$, $\theta = 1$ on $\{|y| \leq R\}$ and $\operatorname{supp}\theta \subset \{|y| < 2R\}$, and put

$$A_i(y,t) = \theta(y)a_i(y,t), \qquad B_i(y,t) = \theta(y)b_i(y,t).$$

Then $v_i = A_i + B_i$, and A_i, B_i have a common compact support. Define

$$(q_i)_\alpha = \sum_{\beta=1}^m (\partial_{y_\beta} N) *_y (Y_i)_{\alpha\beta}, \qquad a_i = \operatorname{div}_y q_i.$$

Integration by parts in (10.0.2) and local integrability of $\nabla_y N$ give

$$\begin{aligned} \|A_i\lambda_V\|_{\mathcal{N}_{\mathrm{sp}}} &\leq (1 + \|\nabla\theta\|_\infty)\|q_i\|_{L^1(\{|y|<2R\}\times\mathbb{R})} \\ &\leq C\|Y_i\|_1 \longrightarrow 0, \end{aligned} \tag{10.0.22}$$

where C depends only on m, R, θ.

With

$$k(y) = \mathbf{1}_{\{|y|<4R\}} \nabla_y N(y) \in L^1(\mathbb{R}^m; \mathbb{R}^m),$$

we have $B_i = \theta \sum_\alpha k_\alpha *_y (g_i)_\alpha$ and

$$\begin{aligned} \|B_i\|_1 &\le \|k\|_{L^1(\mathbb{R}^m)} \|g_i\|_1, \\ \|R_z B_i - B_i\|_1 &\le \|g_i\|_1 \big(\|k(\cdot - z) - k\|_{L^1(\mathbb{R}^m)} + |z| \|\nabla\theta\|_\infty \|k\|_{L^1(\mathbb{R}^m)} \big). \end{aligned} \tag{10.0.23}$$

Translation continuity in L^1 [14, Proposition 8.5] and (10.0.14) imply $\sup_i \|R_z B_i - B_i\|_1 \to 0$ as $z \to 0$. Moreover, (10.0.18) and (10.0.22) give

$$\|(T_h B_i - B_i)\lambda_V\|_{\mathcal{N}_{\mathrm{sp}}} \le \|(\tau_h^V)_\# u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}} + 2C\|Y_i\|_1. \tag{10.0.24}$$

By (10.0.23),

$$\sup_i \|B_i - S_\delta B_i\|_1 \le \sup_i \sup_{|z|<\delta} \|R_z B_i - B_i\|_1 \longrightarrow 0 \qquad (\delta \downarrow 0).$$

Equations (10.0.19), (10.0.24) and (10.0.10) give, for every $0 < \delta < 1$,

$$\lim_{r\downarrow 0} \limsup_{i\to\infty} \sup_{|h|<r} \|T_h B_i - B_i\|_1 \le 2 \sup_i \|B_i - S_\delta B_i\|_1.$$

Letting $\delta \downarrow 0$ yields

$$\lim_{r\downarrow 0} \limsup_{i\to\infty} \sup_{|h|<r} \|T_h B_i - B_i\|_1 = 0. \tag{10.0.25}$$

The uniform L^1 bound, common compact support, and spatial and temporal translation estimates (10.0.23) and (10.0.25) imply that (B_i) is precompact in $L^1(V)$ by the Kolmogorov–Riesz theorem [19, arXiv version, Theorem 5 and Remark 6(II)]. Remark 6(II) permits the $\limsup_i$ in (10.0.25). Equations (10.0.21) and (10.0.22) imply

$$\sup_{s>0} s\lambda_V(\{|A_i| > s\}) \longrightarrow 0, \qquad A_i \longrightarrow 0 \quad \text{in distributions on } V.$$

Thus $v_i = A_i + B_i \ge 0$ satisfies all hypotheses of [10, arXiv version, Lemma 3.3, p. 12], with dimension $m+1$. It follows that (v_i) is precompact in $L^1(V)$, since it has a common compact support.

If $m = 1$, put $w_i = v_i - Y_i$. The functions w_i have common compact support and uniformly bounded L^1 norms. Equation (10.0.17) gives

$$w_i(y,t) = \int_{-\infty}^{y} g_i(z,t)\, dz.$$

Consequently,

$$\begin{aligned} \|R_z w_i - w_i\|_1 &\le |z| \|g_i\|_1, \\ \|(T_h w_i - w_i)\lambda_V\|_{\mathcal{N}_{\mathrm{sp}}} &\le \|(\tau_h^V)_\# u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}} + 2\|Y_i\|_1. \end{aligned} \tag{10.0.26}$$

Equations (10.0.19) and (10.0.26) give L^1 precompactness of (w_i) by [19, arXiv version, Theorem 5 and Remark 6(II)]. Since $\|Y_i\|_1 \to 0$, the same holds for (v_i).

If $m = 0$, (10.0.2) gives $\|\eta\|_{\mathcal{N}_{\mathrm{sp}}} = |\eta|(V)$ for every finite signed Radon measure η on V. Hence (10.0.18) reads

$$\|T_h v_i - v_i\|_1 \le \|(\tau_h^V)_\# u_i - u_i\|_{\mathcal{N}_{\mathrm{sp}}}.$$

The mass bound, common compact support and (10.0.10) give L^1 precompactness of (v_i) by the same theorem.

Return to the measures. For every choice $\varepsilon_i \downarrow 0$, we have proved that $(u_i * \rho_{\varepsilon_i})$ is precompact in $L^1(V)$. By [33, Chapter 1, Theorem 4.16], pass first to $u_i \rightharpoonup u$ on V; the limit is supported in K. Using any one sequence $\varepsilon_i \downarrow 0$, pass further so that $u_i * \rho_{\varepsilon_i} \to \gamma$ in $L^1(V)$. For every $\varphi \in C_c^\infty(V; \mathbb{R})$,

$$\int_V \varphi\gamma\, d\lambda_V = \lim_i \int_V (\varphi * \check{\rho}_{\varepsilon_i})\, du_i = \int_V \varphi\, du.$$

Here $\varphi * \check{\rho}_{\varepsilon_i} \to \varphi$ uniformly by [14, Theorem 8.14(b)], and the masses $u_i(V)$ are bounded. Hence $u = \gamma\lambda_V$ as Radon measures on V, with $\gamma \geq 0$.

For every compactly supported finite signed measure η on V,

$$\lim_{\varepsilon\downarrow 0} \|\eta * \rho_\varepsilon\|_1 = |\eta|(V). \tag{10.0.27}$$

Indeed, $(\eta * \rho_\varepsilon)\lambda_V$ converges to η against every function in $C_c(V;\mathbb{R})$, and (4.2.4) gives

$$\|\eta * \rho_\varepsilon\|_1 \leq |\eta|(V) \leq \liminf_{\varepsilon\downarrow 0} \|\eta * \rho_\varepsilon\|_1.$$

If $|u_i - u|(V)$ did not tend to zero, then for a subsequence and some $b > 0$ it would be at least b. By (10.0.27), choose $\varepsilon_i \downarrow 0$ so that

$$\|(u_i - u) * \rho_{\varepsilon_i}\|_1 \geq b/2.$$

The compactness already proved gives a further subsequence with $u_i * \rho_{\varepsilon_i} \to \gamma$ in $L^1(V)$, since its weak limit is $u = \gamma\lambda_V$. But $\gamma * \rho_{\varepsilon_i} \to \gamma$ in $L^1(V)$ by [14, Theorem 8.14(a)], contradicting this lower bound. This proves (10.0.15). □

Proposition 10.1 (Projected convergence)**.** *Put $q = n + 1$, and let $U \subset \mathbb{P}^n$ be open and contain $B_p(0,1)$. Let W_i be parabolic varifolds on U with weights μ_i and locally bounded spatial first chronovariation, represented by $m_{\mathrm{chr},i}$. Suppose that, on U, the weights μ_i agree with the mass-normalized rescalings* (9.0.4) *of a positive Radon measure μ on an open set $\Omega \subset \mathbb{P}^n$, at a point $a \in \Omega$ satisfying* $\mathrm{HT}(\mu, a)$*. The radii $s_i \downarrow 0$ satisfy $\mu(B_p(a, s_i)) > 0$ and* (3.0.3) *for one fixed $1 \leq C < \infty$, and $U \subset T_{a,s_i}(\Omega)$ for every i. Fix $\chi \in C_c^\infty(U_p(0,\kappa);\mathbb{R})$ with $0 \leq \chi \leq 1$, and put $S = \operatorname{supp}\chi$. Assume*

$$\begin{gathered} \sup_i \mu_i(S) < \infty, \qquad \sup_i |m_{\mathrm{chr},i}|(S) < \infty, \\ \int_{S\times\mathbb{P}G} \|P_{L'} - P_L\|_{\mathrm{op}}\, dW_i(z, L' \times \mathbb{R}_t) \longrightarrow 0. \end{gathered} \tag{10.0.28}$$

Then a subsequence and a nonnegative $\gamma \in L^1_{\mathrm{loc}}(V, \lambda_V)$ satisfy

$$\big|(P_V)_\#(\chi\mu_i) - \gamma\lambda_V\big|(A) \longrightarrow 0 \qquad (A \subset V \text{ compact}). \tag{10.0.29}$$

The limit measure can also be written as

$$\gamma\lambda_V = \frac{b_v(q)}{v(q)}\gamma\, \mathscr{H}^q_p \llcorner V. \tag{10.0.30}$$

Proof. Set $u_i = (P_V)_\#(\chi\mu_i)$ and $K = P_V(S)$. These measures are positive and supported in the compact set K. Lemma 10.1 gives the mass bound (10.0.5) and, when $m \geq 1$, the system (10.0.3). Its bounds (10.0.4), together with (10.0.28), give (10.0.14). Lemma 10.2 supplies (10.0.10) with exactly the norm (10.0.2). Lemma 10.3 therefore proves (10.0.29). Finally, Lemma 5.6.1 gives (10.0.30). □

11. Differentiation Lemmata

Let μ be a positive Radon measure on an open set $\Omega \subset \mathbb{P}^n$. For $a \in \Omega$ and $1 \leq C < \infty$, define

$$\begin{aligned} \mathcal{R}_C(a) = \big\{ r > 0 : B_p(a, 5r) \Subset \Omega, \quad & 0 < \mu(B_p(a,r)), \\ & \mu(B_p(a,5r)) \leq C\mu(B_p(a,r)) \big\}. \end{aligned} \tag{11.0.1}$$

For $a \in \Omega$, $1 \leq C < \infty$, $F : \mathcal{R}_C(a) \to \mathbb{R}$ and $L \in \mathbb{R}$, the notation

$$\lim_{\substack{r\downarrow 0 \\ r\in\mathcal{R}_C(a)}} F(r) = L$$

means that, for every $\varepsilon > 0$, there is a $\rho > 0$ such that

$$|F(r) - L| < \varepsilon \qquad \text{for every } r \in \mathcal{R}_C(a) \cap (0, \rho).$$

This convention does not assert that $\mathcal{R}_C(a)$ contains arbitrarily small radii. If $\mathcal{R}_C(a) \cap (0, \rho_0) = \varnothing$ for some $\rho_0 > 0$, the condition holds for every $L \in \mathbb{R}$.

Lemma 11.1 (Restricted maximal estimate)**.** *Fix $1 \le C < \infty$. Let $U \Subset U' \Subset \Omega$ be open, and choose $r_0 > 0$ with*

$$5r_0 < \operatorname{dist}_p(\overline{U}, \mathbb{P}^n \setminus U').$$

For a finite positive Radon measure σ on U', define

$$\mathcal{M}_{C,U,r_0}\sigma(a) = \sup_{\substack{r \in \mathcal{R}_C(a) \\ 0<r<r_0}} \frac{\sigma(B_p(a,r))}{\mu(B_p(a,r))}, \qquad a \in U,$$

where the supremum of the empty set is taken to be zero. Then, for every $\alpha > 0$,

$$\mu^*\big(\{a \in U : \mathcal{M}_{C,U,r_0}\sigma(a) > \alpha\}\big) \le \frac{C}{\alpha}\sigma(U'),$$

where μ^* denotes the outer measure induced by μ.

Proof. We adapt the maximal-estimate argument in [30, proof of Theorem 5.3] to the balls defining $\mathcal{M}_{C,U,r_0}\sigma$. Let $F_\alpha = \{a \in U : \mathcal{M}_{C,U,r_0}\sigma(a) > \alpha\}$. For each $a \in F_\alpha$, choose $r_a \in \mathcal{R}_C(a)$ with $r_a < r_0$ and

$$\sigma(B_p(a, r_a)) > \alpha\mu(B_p(a, r_a)).$$

Lemma 5.2.1, applied to $\{B_p(a, r_a) : a \in F_\alpha\}$, gives a countable pairwise disjoint subfamily $B_p(a_i, r_i)$ such that

$$F_\alpha \subset \bigcup_i B_p(a_i, 5r_i).$$

Every selected ball and its fivefold enlargement lie in U'. Consequently,

$$\begin{aligned}\mu^*(F_\alpha) &\le \sum_i \mu(B_p(a_i, 5r_i)) \le C\sum_i \mu(B_p(a_i, r_i)) \\ &\le \frac{C}{\alpha}\sum_i \sigma(B_p(a_i, r_i)) \le \frac{C}{\alpha}\sigma(U').\end{aligned}$$

The second inequality uses $r_i \in \mathcal{R}_C(a_i)$, and the last uses disjointness. □

Vitali relations for doubling balls. We use Vitali relations in the sense of [13, 2.8.16]; see also [2, Definition 4.1]. A covering relation is a family $\mathcal{V} \subset \{(a, S) : a \in S \subset \mathbb{P}^n,\ S \text{ Borel}\}$. It is *fine at a* if $\inf\{\operatorname{diam}_p S : (a, S) \in \mathcal{V}\} = 0$. For a positive Radon measure ν, it is a *ν-Vitali relation* if it is fine everywhere and, whenever $\mathcal{C} \subset \mathcal{V}$ is fine at every point of a set $E \subset \mathbb{P}^n$, the family $\{S : (a, S) \in \mathcal{C},\ a \in E\}$ has an at most countable disjoint subfamily $(S_i)_{i\in I}$ such that $\nu^*(E \setminus \bigcup_{i\in I} S_i) = 0$.

Compactness of $B_p(0, 1)$ permits a choice of $N_0 \ge 2$ such that $B_p(0, 1)$ is covered by N_0 open balls of radius $1/2$, and put $d = \log_2 N_0$. Translation and parabolic dilation give the same covering bound for every ball, so $\mathbb{P}^n$ is geometrically doubling in the sense of [21, Lemma 2.3 and Definition 2.4]. Let ν be a finite positive Radon measure on $\mathbb{P}^n$, and let $D > 6^d$. Define

$$\begin{aligned}G_D(\nu) = \operatorname{spt}\nu \cap \bigcap_{j=1}^{\infty}\bigcup_{k\ge j}\big\{a \in \mathbb{P}^n : \nu(U_p(a, 6^{1-k})) \\ \le D\nu(U_p(a, 6^{-k}))\big\}.\end{aligned} \tag{11.0.2}$$

For each fixed $r > 0$, the function $a \mapsto \nu(U_p(a, r))$ is Borel in a by [14, Proposition 1.5 and Theorem 2.36], applied to $(a, z) \mapsto \mathbf{1}_{\{d_p(a,z)<r\}}$. Thus $G_D(\nu)$ is Borel. Since $\nu(\mathbb{P}^n \setminus \operatorname{spt}\nu) = 0$, the assertion about radii 6^{-k} in [21, Lemma 3.3] gives $\nu(\mathbb{P}^n \setminus G_D(\nu)) = 0$. For $a \in G_D(\nu)$ there are arbitrarily large integers k such that, with $r = 6^{-k}$,

$$\nu(B_p(a, 11r/2)) \le \nu(U_p(a, 6r)) \le D\nu(U_p(a, r)) \le D\nu(B_p(a, 11r/10)). \tag{11.0.3}$$

Writing $5B_p(a,r) = B_p(a,5r)$, closed balls satisfying $0 < \nu(B)$ and $\nu(5B) \le D\nu(B)$ are therefore fine at every $a \in G_D(\nu)$. Indeed, (11.0.3) supplies such balls of radius $11 \cdot 6^{-k}/10$ for arbitrarily large k, and $a \in \operatorname{spt}\nu$ ensures that each has positive mass. Define

$$\begin{aligned}\mathcal{V}_D(\nu) = \{(a, B_p(a,r)) : a \in G_D(\nu),\ r > 0,\ 0 < \nu(B_p(a,r)),\\ \nu(B_p(a,5r)) \le D\nu(B_p(a,r))\}\\ \cup \{(a,\{a\}) : a \notin G_D(\nu)\}.\end{aligned} \tag{11.0.4}$$

The singletons at centers outside $G_D(\nu)$ have zero ν-mass and make the relation fine at those centers. When applying differentiation theorems to $\mathcal{V}_D(\nu)$, we extend each quotient to these singleton pairs by assigning the value zero. This extension does not affect differentiation at $a \in G_D(\nu)$, since $\nu(S) > 0$ for every $(a,S) \in \mathcal{V}_D(\nu)$ with such a center.

Proposition 11.1 (Vitali relation for doubling balls)**.** *For every finite positive Radon measure ν on $\mathbb{P}^n$ and every $D > 6^d$, the relation $\mathcal{V}_D(\nu)$ is a ν-Vitali relation.*

Proof. The relation is fine everywhere by its construction. We verify its Vitali covering property directly; compare the covering criterion in [13, Theorem 2.8.17], as stated in [16, Proposition 3.2]. Suppose $\mathcal{C} \subset \mathcal{V}_D(\nu)$ is fine on E. Apply the selection in the proof of Lemma 5.2.1 to its balls centered in $E \cap G_D(\nu)$ with radii at most one. It gives disjoint balls B_i such that every omitted ball meets a selected ball of at least half its radius. If $a \in (E \cap G_D(\nu)) \setminus \bigcup_i B_i$, fineness at a and closedness of $B_1, \dots, B_J$ give a ball from $\mathcal{C}$ centered at a and disjoint from all of them. Hence $a \in \bigcup_{i>J} 5B_i$, and

$$\nu^*\left((E \cap G_D(\nu)) \setminus \bigcup_i B_i\right) \le \sum_{i>J} \nu(5B_i) \le D \sum_{i>J} \nu(B_i) \longrightarrow 0.$$

Here $J \to \infty$, and the series converges because $\sum_i \nu(B_i) \le \nu(\mathbb{P}^n) < \infty$. If the selection is finite, the same argument shows that it covers $E \cap G_D(\nu)$. The remaining set $E \setminus G_D(\nu)$ is ν-null. Consequently $\mathcal{V}_D(\nu)$ is a ν-Vitali relation. □

Lemma 11.2 (Differentiation along doubling radii)**.** *Let $g \in L^1_{\mathrm{loc}}(\Omega,\mu)$. For μ-almost every $a \in \Omega$ and every $1 \le C < \infty$,*

$$\lim_{\substack{r \downarrow 0 \\ r \in \mathcal{R}_C(a)}} \frac{1}{\mu(B_p(a,r))} \int_{B_p(a,r)} |g(z) - g(a)|\, d\mu(z) = 0.$$

For any countable family of such functions, the exceptional μ-null set may be chosen independently of the function and of C.

Proof. We apply differentiation with respect to a Vitali relation [13, Corollary 2.9.9], in the mean-oscillation formulation of [2, Proposition 4.8]. For the formulation along doubling balls, see also [21, Corollary 3.6 and its proof]. For $m, k \ge 1$, put

$$\begin{aligned}U_m &= \{z \in \Omega : d_p(z,0) < m,\ \operatorname{dist}_p(z, \mathbb{P}^n \setminus \Omega) > m^{-1}\},\\ \mu_m(A) &= \mu(A \cap U_{m+1}) \quad (A \subset \mathbb{P}^n \text{ Borel}), \qquad D_k = 6^d + k,\end{aligned} \tag{11.0.5}$$

where $\operatorname{dist}_p(z,\varnothing) = \infty$. Properness gives compactness of $\overline{U}_m$; the inequalities $m < m+1$ and $m^{-1} > (m+1)^{-1}$ imply $U_m \Subset U_{m+1} \Subset \Omega$. Every $a \in \Omega$ belongs to U_m for all sufficiently large m, so $\Omega = \bigcup_m U_m$.

Let $(g^{(\ell)})_{\ell \ge 1}$ be the given countable family, taking $g^{(\ell)} = g$ for a single function. Define $h_m^{(\ell)}$ to equal $g^{(\ell)}$ on U_{m+1} and zero on $\mathbb{P}^n \setminus U_{m+1}$. Then μ_m is finite and $h_m^{(\ell)} \in L^1(\mathbb{P}^n, \mu_m)$. By Proposition 11.1, $\mathcal{V}_{D_k}(\mu_m)$ is a μ_m-Vitali relation. Applying [2, Proposition 4.8] to this relation and to $h_m^{(\ell)}$ gives a μ-null set $N_{m,k,\ell} \subset U_m$, containing $U_m \setminus G_{D_k}(\mu_m)$, such that for

$a \in U_m \setminus N_{m,k,\ell}$,

$$\lim_{\substack{r\downarrow 0 \\ (a,B_p(a,r))\in\mathcal{V}_{D_k}(\mu_m)}} \frac{\displaystyle\int_{B_p(a,r)} |h_m^{(\ell)}(z) - h_m^{(\ell)}(a)|\, d\mu_m(z)}{\mu_m(B_p(a,r))} = 0. \tag{11.0.6}$$

Set

$$N = \bigcup_{m,k,\ell\ge 1} N_{m,k,\ell}, \qquad \mu^*(N) \le \sum_{m,k,\ell\ge 1} \mu^*(N_{m,k,\ell}) = 0. \tag{11.0.7}$$

For $a \in \Omega \setminus N$ and $1 \le C < \infty$, choose m with $a \in U_m$ and k with $D_k \ge C$. If $r \in \mathcal{R}_C(a)$ is small enough that $B_p(a,5r) \subset U_{m+1}$, then $(a, B_p(a,r)) \in \mathcal{V}_{D_k}(\mu_m)$, and its μ_m-average of $|h_m^{(\ell)} - h_m^{(\ell)}(a)|$ equals its μ-average of $|g^{(\ell)} - g^{(\ell)}(a)|$. Equation (11.0.6) therefore proves the assertion for every ℓ and C. □

Lemma 11.3 (Differentiation of Radon measures)**.** *Let λ be a positive Radon measure on Ω, and write*

$$\lambda = f\mu + \lambda_s, \qquad f \ge 0, \qquad \lambda_s \perp \mu \tag{11.0.8}$$

for its Lebesgue decomposition relative to μ. For μ-almost every $a \in \Omega$ and every $1 \le C < \infty$,

$$\lim_{\substack{r\downarrow 0 \\ r\in\mathcal{R}_C(a)}} \frac{\lambda(B_p(a,r))}{\mu(B_p(a,r))} = f(a) < \infty. \tag{11.0.9}$$

For any countable family of positive Radon measures on Ω, the exceptional μ-null set may be chosen independently of the measure and of C.

Proof. The decomposition (11.0.8) follows from [14, Theorem 3.8], since λ and μ are σ-finite. Moreover, $\int_K f\, d\mu \le \lambda(K) < \infty$ for every compact $K \subset \Omega$, so f is finite μ-almost everywhere. We use Federer's differentiation theorem for measures [13, Theorems 2.9.5 and 2.9.7], in the formulation of [16, Proposition 3.4 and Corollary 3.5(a)].

For a countable family $(\lambda^{(\ell)})_{\ell\ge 1}$, denote the corresponding densities by $f^{(\ell)}$ and, with U_m, μ_m and D_k as in (11.0.5), define

$$\lambda_m^{(\ell)}(A) = \lambda^{(\ell)}(A \cap U_{m+1}) \qquad (A \subset \mathbb{P}^n \text{ Borel}).$$

These measures are finite. Their absolutely continuous parts relative to μ_m have density $f^{(\ell)}$ on U_{m+1}. The auxiliary measure in [16, Proposition 3.4] is

$$\psi_m^{(\ell)}(A) = \inf\bigl\{\lambda_m^{(\ell)}(B) : B \subset \mathbb{P}^n \text{ Borel},\ \mu_m(B \mathbin{\triangle} A) = 0\bigr\} \tag{11.0.10}$$

for Borel sets $A \subset \mathbb{P}^n$. We claim that

$$\psi_m^{(\ell)}(A) = \int_{A\cap U_{m+1}} f^{(\ell)}\, d\mu \qquad \text{for every Borel } A \subset \mathbb{P}^n. \tag{11.0.11}$$

Write $\lambda^{(\ell)} = f^{(\ell)}\mu + \lambda_s^{(\ell)}$, and choose a Borel set $S_m^{(\ell)} \subset U_{m+1}$ such that $\mu_m(S_m^{(\ell)}) = 0$ and $\lambda_s^{(\ell)}(U_{m+1} \setminus S_m^{(\ell)}) = 0$. Every Borel set B with $\mu_m(B \mathbin{\triangle} A) = 0$ satisfies

$$\lambda_m^{(\ell)}(B) \ge \int_{B\cap U_{m+1}} f^{(\ell)}\, d\mu = \int_{A\cap U_{m+1}} f^{(\ell)}\, d\mu.$$

The choice $B = A \setminus S_m^{(\ell)}$ gives equality and is admissible in (11.0.10). This proves (11.0.11); in particular, $d\psi_m^{(\ell)}/d\mu_m = f^{(\ell)}$ μ_m-almost everywhere on U_{m+1}.

By Proposition 11.1, $\mathcal{V}_{D_k}(\mu_m)$ is a μ_m-Vitali relation. Apply [16, Proposition 3.4] with $\kappa_1 = \mu_m$ and $\kappa_2 = \lambda_m^{(\ell)}$. Together with (11.0.11), it gives a μ-null set $E_{m,k,\ell} \subset U_m$, containing $U_m \setminus G_{D_k}(\mu_m)$ and $U_m \cap \{f^{(\ell)} = \infty\}$, such that, for every $a \in U_m \setminus E_{m,k,\ell}$, the ratios $\lambda_m^{(\ell)}(B)/\mu_m(B)$ tend to $f^{(\ell)}(a)$ as $(a,B) \in \mathcal{V}_{D_k}(\mu_m)$ and $\operatorname{diam}_p B \to 0$.

Put $E = \bigcup_{m,k,\ell\geq 1} E_{m,k,\ell}$. Then $\mu^*(E) \leq \sum_{m,k,\ell\geq 1} \mu^*(E_{m,k,\ell}) = 0$. For $a \in \Omega \setminus E$ and $1 \leq C < \infty$, choose m with $a \in U_m$ and k with $D_k \geq C$. Every sufficiently small $r \in \mathcal{R}_C(a)$ satisfies $B_p(a, 5r) \subset U_{m+1}$; $(a, B_p(a,r)) \in \mathcal{V}_{D_k}(\mu_m)$ and

$$\frac{\lambda^{(\ell)}(B_p(a,r))}{\mu(B_p(a,r))} = \frac{\lambda_m^{(\ell)}(B_p(a,r))}{\mu_m(B_p(a,r))} \longrightarrow f^{(\ell)}(a).$$

This proves (11.0.9) and the simultaneous assertion. A single measure is obtained by taking $\lambda^{(\ell)} = \lambda$. □

12. Parabolic Rectifiability by Translation Invariant Tangents

12.1. **Structure of invariant tangents.**

Lemma 12.1.1 (Product representation of an invariant measure)**.** *Let $V = L \times \mathbb{R}_t \in \mathbb{P}G$, choose a unit vector $e_x \in L^\perp$, and put $e = (e_x, 0)$ and $q = n + 1$. Define*

$$F_e : V \times \mathbb{R} \longrightarrow \mathbb{P}^n, \qquad F_e(v,s) = v + se, \qquad \pi_e : \mathbb{P}^n \longrightarrow \mathbb{R}, \qquad \pi_e(x,t) = e_x \cdot x. \tag{12.1.1}$$

If τ is a positive Radon measure on $\mathbb{P}^n$ invariant under translations by V, there is a unique positive Radon measure σ on $\mathbb{R}$ such that

$$\tau = (F_e)_\#(\lambda_V \otimes \sigma) \qquad \text{as Radon measures on } \mathbb{P}^n. \tag{12.1.2}$$

Moreover, for every $z \in \mathbb{R}$ and $r > 0$,

$$\tau(B_p(ze,r)) = \tau(U_p(ze,r)) = b_v(q) \int_{|s-z|<r} \bigl(r^2 - |s-z|^2\bigr)^{q/2}\, d\sigma(s). \tag{12.1.3}$$

Proof. For the product decomposition, compare [10, arXiv version, proof of Lemma 3.1, p. 9]. For $0 \leq f \in C_c(\mathbb{R};\mathbb{R})$, let m_f be the positive Radon measure on V defined by

$$\int_V g\, dm_f = \int_{\mathbb{P}^n} g(P_V w) f(\pi_e(w))\, d\tau(w) \qquad \bigl(g \in C_c(V;\mathbb{R})\bigr). \tag{12.1.4}$$

The function $w \mapsto g(P_V w) f(\pi_e(w))$ has support in the compact set $F_e(\operatorname{supp} g \times \operatorname{supp} f)$, so existence of m_f follows from the Riesz representation theorem [14, Theorem 7.2]. Translation invariance of τ along V implies translation invariance of m_f. Uniqueness of Haar measure [14, Theorem 11.9, p. 344] therefore gives $m_f = c_f \lambda_V$ for some $c_f \geq 0$, taking $c_f = 0$ if $m_f = 0$.

Fix $0 \leq g_0 \in C_c(V;\mathbb{R})$ with $\int_V g_0\, d\lambda_V = 1$. The Riesz representation theorem gives a unique positive Radon measure σ on $\mathbb{R}$ satisfying

$$\int_{\mathbb{R}} f\, d\sigma = \int_{\mathbb{P}^n} g_0(P_V w) f(\pi_e(w))\, d\tau(w) \qquad \bigl(f \in C_c(\mathbb{R};\mathbb{R})\bigr). \tag{12.1.5}$$

For nonnegative f, this integral equals c_f by (12.1.4). Hence, for every $f \in C_c(\mathbb{R};\mathbb{R})$ and $g \in C_c(V;\mathbb{R})$,

$$\int_{\mathbb{P}^n} g(P_V w) f(\pi_e(w))\, d\tau(w) = \left(\int_V g\, d\lambda_V\right)\left(\int_{\mathbb{R}} f\, d\sigma\right). \tag{12.1.6}$$

The product measure $\lambda_V \otimes \sigma$ is Radon [14, Theorem 7.20(c)]. Every $h \in C_c(\mathbb{P}^n;\mathbb{R})$ can be approximated uniformly by finite sums of the functions $w \mapsto g(P_V w) f(\pi_e(w))$ in (12.1.6), with all supports in one compact set [14, Proposition 7.21]. Thus (12.1.6) proves the Radon-measure identity (12.1.2). Taking $g = g_0$ in (12.1.6) also proves uniqueness of σ in (12.1.2).

For $v \in V$ and $s, z \in \mathbb{R}$, (5.1.1) gives

$$d_p(F_e(v,s), ze)^2 = \|v\|_p^2 + |s-z|^2.$$

If $|s-z| < r$, the section of $B_p(ze,r)$ at normal coordinate s is the V-ball of radius $(r^2 - |s-z|^2)^{1/2}$. The open and closed V-balls of radius $\rho > 0$ have the same λ_V-mass $b_v(q)\rho^q$ by (5.6.4) and continuity from below. At $|s-z| = r$, the closed-ball section is a singleton,

of zero λ_V-mass since $\dim V = n \geq 1$. Tonelli's theorem [14, Theorem 2.36], applied to (12.1.2), proves (12.1.3). □

Proposition 12.1.1 (Finite parallel-plane representation)**.** *Let ϕ be a finite positive Radon measure on $\mathbb{P}^n$, $n \geq 1$, and put $q = n+1$. Suppose that*

$$0 < \Theta^q_{p*}(\phi, a) \leq \Theta^{q*}_p(\phi, a) < \infty \qquad \text{for } \phi\text{-almost every } a \in \mathbb{P}^n, \tag{12.1.7}$$

and that, for ϕ-almost every a, there is a plane $V_a \in \mathbb{P}G$ such that every $\tau \in \operatorname{Tan}^q_p(\phi, a)$ is invariant under translations by V_a. Then, for ϕ-almost every a, every $\tau \in \operatorname{Tan}^q_p(\phi, a)$ has the form

$$\tau = \sum_{j=1}^{N} c_j \mathscr{H}^q_p \llcorner (V_a + z_j) \qquad \text{as Radon measures on } \mathbb{P}^n, \tag{12.1.8}$$

where $1 \leq N < \infty$, $c_j > 0$, and $z_1, \ldots, z_N \in V_a^\perp$ are distinct.

Proof. Choose a Borel ϕ-null set N_0 containing the exceptional sets for (12.1.7), the assumed translation invariance, and Lemma 9.1. Fix $a \in \mathbb{P}^n \setminus N_0$, put $V = V_a = L \times \mathbb{R}_t$, and set

$$d_- = \Theta^q_{p*}(\phi, a), \qquad d_+ = \Theta^{q*}_p(\phi, a), \qquad 0 < d_- \leq d_+ < \infty. \tag{12.1.9}$$

Let $\tau \in \operatorname{Tan}^q_p(\phi, a)$, and choose $r_i \downarrow 0$ with

$$r_i^{-q} (T_{a,r_i})_\# \phi \rightharpoonup \tau \qquad \text{in } \mathbb{P}^n. \tag{12.1.10}$$

Choose a unit vector $e_x \in L^\perp$ and put $e = (e_x, 0)$. Lemma 12.1.1 gives a positive Radon measure σ on $\mathbb{R}$ with $\tau = (F_e)_\#(\lambda_V \otimes \sigma)$, where F_e is defined in (12.1.1).

If $\zeta \in \operatorname{Tan}^q_p(\phi, a)$ is generated by $t_i \downarrow 0$, the compact-set inequality for weak convergence [25, Theorem 1.24(1)] gives

$$\zeta(B_p(0,1)) \geq \limsup_{i \to \infty} t_i^{-q} \phi(B_p(a, t_i)) \geq d_-, \tag{12.1.11}$$

by (12.1.9). If $s \in \operatorname{supp} \sigma$, then $se \in \operatorname{supp} \tau$ by (12.1.2). Lemma 9.1 therefore yields $r^{-q}(T_{se,r})_\# \tau \in \operatorname{Tan}^q_p(\phi, a)$ for every $r > 0$. Applying (12.1.11) and then (12.1.3), we obtain

$$d_- \leq r^{-q} \tau(B_p(se, r)) \leq b_v(q) \sigma((s-r, s+r)) \qquad \big(s \in \operatorname{supp} \sigma,\ r > 0\big). \tag{12.1.12}$$

Since σ is finite on bounded intervals, letting $r \downarrow 0$ in (12.1.12) gives

$$\sigma(\{s\}) \geq \frac{d_-}{b_v(q)} \qquad \big(s \in \operatorname{supp} \sigma\big). \tag{12.1.13}$$

For each $R > 0$, the definition of d_+ and the open-set inequality for weak convergence [25, Theorem 1.24(2)], applied to (12.1.10), give

$$\tau(U_p(0, R)) \leq \liminf_{i \to \infty} r_i^{-q} \phi(U_p(a, R r_i)) \leq 2 d_+ R^q. \tag{12.1.14}$$

On $|s| < R/2$, $(R^2 - s^2)^{q/2} \geq (3/4)^{q/2} R^q$. Consequently, (12.1.3) and (12.1.14) imply

$$b_v(q)(3/4)^{q/2} R^q \sigma((-R/2, R/2)) \leq \tau(U_p(0, R)) \leq 2 d_+ R^q.$$

Letting $R \to \infty$, we obtain

$$\sigma(\mathbb{R}) \leq \frac{2 d_+}{b_v(q)(3/4)^{q/2}} < \infty. \tag{12.1.15}$$

Equations (12.1.13) and (12.1.15) show that $\operatorname{supp} \sigma$ is finite. Since $\tau \neq 0$, also $\sigma \neq 0$, and we may write

$$\sigma = \sum_{j=1}^{N} a_j \delta_{s_j}, \qquad a_j = \sigma(\{s_j\}) > 0, \qquad 1 \leq N < \infty,$$

with distinct $s_1, \ldots, s_N \in \mathbb{R}$.

For $1 \leq j \leq N$, define $\iota_j : V \to \mathbb{P}^n$ by $\iota_j(v) = v + s_j e$. Then (12.1.2) gives $\tau = \sum_{j=1}^N a_j (\iota_j)_\# \lambda_V$. The map ι_j preserves d_p, so (5.6.3) yields

$$(\iota_j)_\# \lambda_V = \frac{b_v(q)}{v(q)} \mathscr{H}^q_p \llcorner (V + s_j e).$$

Thus (12.1.8) holds with $z_j = s_j e \in V^\perp$ and $c_j = a_j b_v(q)/v(q) > 0$. □

12.2. **Rectifiability of compact graph pieces.**

Proposition 12.2.1 (Tangents on a parabolic Lipschitz graph)**.** *Let $n \geq 1$, put $q = n + 1$, and let $S \in \mathbb{P}G$. Let $K \subset S$ be compact and let $f : K \to S^\perp$ be Λ-Lipschitz for d_p, where $0 < \Lambda < \infty$. Set*

$$\Gamma_f = \{w + f(w) : w \in K\},$$

as in Definition 5.7.1. Let ψ be a finite positive Radon measure on $\mathbb{P}^n$ with $\psi(\mathbb{P}^n \setminus \Gamma_f) = 0$, and suppose that

$$0 < \Theta^q_{p*}(\psi, a) \leq \Theta^{q*}_p(\psi, a) < \infty \qquad \text{for } \psi\text{-almost every } a \in \mathbb{P}^n. \tag{12.2.1}$$

Assume also that, for ψ-almost every a, every $\tau \in \operatorname{Tan}^q_p(\psi, a)$ admits a representation

$$\tau = \sum_{j=1}^N c_j \mathscr{H}^q_p \llcorner (V + z_j), \tag{12.2.2}$$

where $V \in \mathbb{P}G$, $1 \leq N < \infty$, $c_j > 0$, and the vectors $z_1, \ldots, z_N \in V^\perp$ are distinct. Then, for ψ-almost every a, every $\tau \in \operatorname{Tan}^q_p(\psi, a)$ has the form

$$\tau = c_\tau \mathscr{H}^q_p \llcorner V_\tau \qquad \text{as Radon measures on } \mathbb{P}^n, \tag{12.2.3}$$

for some $c_\tau > 0$ and $V_\tau \in \mathbb{P}G$.

Proof. As in the proof of [27, Theorem 2.34, (2.15), p. 20], we use their Proposition 2.16 to approximate points of the tangent support by rescaled graph points. We include the details needed to exclude multiple parallel planes.

The graph Γ_f is compact, so $\operatorname{supp} \psi \subset \Gamma_f$. Fix $a \in \operatorname{supp} \psi$ where (12.2.1) holds and every geometric tangent has the representation (12.2.2). Let $\tau \in \operatorname{Tan}^q_p(\psi, a)$, and choose $r_i \downarrow 0$ with

$$r_i^{-q} (T_{a,r_i})_\# \psi \rightharpoonup \tau \qquad \text{in } \mathbb{P}^n. \tag{12.2.4}$$

Define $P_{S^\perp} : \mathbb{P}^n \to S^\perp$ by $P_{S^\perp}(z) = z - P_S z$.

For $y, z \in \operatorname{supp} \tau$, [27, Proposition 2.16, p. 15] gives sequences $y_i, z_i \in \operatorname{supp} \psi \subset \Gamma_f$ such that

$$u_i := T_{a,r_i}(y_i) \longrightarrow y, \qquad v_i := T_{a,r_i}(z_i) \longrightarrow z. \tag{12.2.5}$$

Since $u_i - v_i = \delta_{1/r_i}(y_i - z_i)$, the graph inequality (5.7.2) gives

$$\begin{aligned} d_p(P_{S^\perp}(u_i - v_i), 0) &= r_i^{-1} d_p(P_{S^\perp}(y_i - z_i), 0) \\ &\leq \Lambda r_i^{-1} d_p(P_S(y_i - z_i), 0) \\ &= \Lambda d_p(P_S(u_i - v_i), 0). \end{aligned}$$

Passing to the limits in (12.2.5), we obtain

$$d_p(P_{S^\perp}(y - z), 0) \leq \Lambda d_p(P_S(y - z), 0) \qquad (y, z \in \operatorname{supp} \tau). \tag{12.2.6}$$

Thus P_S is injective on $\operatorname{supp} \tau$.

Choose a representation (12.2.2). The coefficients are positive, and (5.6.4) gives

$$\operatorname{supp} \tau = \bigcup_{j=1}^N (V + z_j). \tag{12.2.7}$$

Injectivity of P_S on $V + z_1$ implies that $P_S|_V$ is injective. Both V and S have Euclidean dimension n, so $P_S(V) = S$. Hence $P_S(V + z_j) = S$ for every j. If $N \geq 2$, choose $y \in V + z_1$ and $z \in V + z_2$ with $P_S y = P_S z = 0$. The two affine planes are disjoint, so $y \neq z$, contrary to injectivity on $\operatorname{supp} \tau$. Therefore $N = 1$.

Choose $0 < d < \Theta^q_{p*}(\psi, a)$ using (12.2.1). For every $R > 0$, the compact-set inequality for weak convergence [25, Theorem 1.24(1)], applied to (12.2.4), yields

$$\tau(B_p(0, R/2)) \geq \limsup_{i \to \infty} r_i^{-q} \psi(B_p(a, Rr_i/2)) \geq d(R/2)^q > 0. \tag{12.2.8}$$

Consequently $0 \in \operatorname{supp} \tau = V + z_1$. Since $z_1 \in V^\perp$, we have $z_1 = 0$. This proves (12.2.3) with $V_\tau = V$ and $c_\tau = c_1$. □

Proposition 12.2.2 (Rectifiability of a compact graph piece)**.** *Let ϕ be a finite positive Radon measure on $\mathbb{P}^n$, $n \geq 1$, and put $q = n + 1$. Suppose that*

$$0 < \Theta^q_{p*}(\phi, a) \leq \Theta^{q*}_p(\phi, a) < \infty \qquad \text{for } \phi\text{-almost every } a \in \mathbb{P}^n, \tag{12.2.9}$$

and that, for ϕ-almost every a, there is a plane $V_a \in \mathbb{P}G$ such that every $\tau \in \operatorname{Tan}^q_p(\phi, a)$ is invariant under translations by V_a. Let $S \in \mathbb{P}G$, let $K \subset S$ be compact, and let $f : K \to S^\perp$ be Λ-Lipschitz for d_p, where $0 < \Lambda < \infty$. Set

$$\Gamma_f = \{w + f(w) : w \in K\}.$$

Suppose that $\phi(\Gamma_f) > 0$ and that, for some $0 < c \leq C < \infty$,

$$c\, \mathscr{H}^q_p \llcorner \Gamma_f \leq \phi \llcorner \Gamma_f \leq C\, \mathscr{H}^q_p \llcorner \Gamma_f \qquad \text{as Borel measures on } \mathbb{P}^n. \tag{12.2.10}$$

Then $\mathscr{H}^q_p(\Gamma_f) < \infty$ and Γ_f is countably $\mathbb{P}$-rectifiable. Moreover, for $\mathscr{H}^q_p$-almost every $a \in \Gamma_f$, there is a plane $P_a \in \mathbb{P}G$ such that

$$\operatorname{Tan}_p(\mathscr{H}^q_p \llcorner \Gamma_f, a) = \{b\, \mathscr{H}^q_p \llcorner P_a : b > 0\}. \tag{12.2.11}$$

Proof. Put $\psi = \phi \llcorner \Gamma_f$ and $\nu = \mathscr{H}^q_p \llcorner \Gamma_f$. By (12.2.10),

$$\nu(\mathbb{P}^n) = \nu(\Gamma_f) \leq c^{-1} \phi(\Gamma_f) < \infty.$$

Both ψ and ν are finite Borel measures, hence Radon by [14, Theorem 7.8]. Since ϕ is finite, $\mathbf{1}_{\Gamma_f} \in L^1(\phi)$. Applying [27, Proposition 2.13(ii)–(iii), pp. 14–15] with this function gives, for ψ-almost every a,

$$\begin{aligned} \Theta^q_{p*}(\psi, a) &= \Theta^q_{p*}(\phi, a), \qquad \Theta^{q*}_p(\psi, a) = \Theta^{q*}_p(\phi, a), \\ \operatorname{Tan}^q_p(\psi, a) &= \operatorname{Tan}^q_p(\phi, a). \end{aligned} \tag{12.2.12}$$

Thus ψ satisfies (12.1.7), and every geometric q-tangent of ψ at a is invariant under translations by V_a, for ψ-almost every a. Proposition 12.1.1, followed by Proposition 12.2.1, shows that each such tangent is a positive multiple of $\mathscr{H}^q_p$ on a vertical plane through the origin.

The Radon–Nikodym theorem [14, Theorem 3.8] and (12.2.10) give a Borel function $g : \mathbb{P}^n \to [c, C]$ with $\psi = g\nu$. In particular, $g \in L^1(\nu)$ and ψ and ν have the same null sets. Equations (12.2.9), (12.2.10) and (12.2.12) imply

$$0 < \frac{\Theta^q_{p*}(\psi, a)}{C} \leq \Theta^q_{p*}(\nu, a) \leq \Theta^{q*}_p(\nu, a) \leq \frac{\Theta^{q*}_p(\psi, a)}{c} < \infty \qquad \text{for } \nu\text{-almost every } a. \tag{12.2.13}$$

We may therefore apply [27, Proposition 2.13(iii) and (2.7), pp. 14–15] to ν and g, obtaining

$$\operatorname{Tan}^q_p(\psi, a) = g(a) \operatorname{Tan}^q_p(\nu, a) \qquad \text{for } \nu\text{-almost every } a. \tag{12.2.14}$$

For ν-almost every a, every member of $\operatorname{Tan}^q_p(\nu, a)$ is a positive multiple of $\mathscr{H}^q_p$ on a vertical plane through the origin. If a satisfies (12.2.13) and a sequence $r_i \downarrow 0$ satisfies $r_i^{-q}(T_{a,r_i})_\# \nu \rightharpoonup \lambda$ in $\mathbb{P}^n$, then

$$\lambda(B_p(0, 1)) \geq \limsup_{i \to \infty} r_i^{-q} \nu(B_p(a, r_i)) \geq \Theta^q_{p*}(\nu, a) > 0, \tag{12.2.15}$$

by [25, Theorem 1.24(1)]. Thus these limits are nonzero, as required by our tangent definition.

Apply [27, Theorem 2.34, pp. 18–20] to the compact graph Γ_f, with $h = q$, base plane S, and the family $\mathbb{P}G$ of vertical planes (the stratum $(n-1,1)$ in [27, Definition 2.9, p. 13]). For $n = 1$ this family consists only of the time axis. The Hausdorff measure used in [27, Theorem 2.34] is normalized by $\mathscr{H}^q_{\mathrm{MMP}} = 2^{-q}\mathscr{H}^q_p$ [27, Definition 2.1, p. 12]; its restriction to Γ_f is therefore $2^{-q}\nu$. The flat measures in [27, Definition 2.8, p. 13] are positive Haar multiples on the corresponding planes. Multiplying a measure by a fixed positive constant multiplies each geometric tangent by that constant, so the theorem applies with our Hausdorff normalization. Together with [27, Proposition 2.13(i), p. 14], it gives, for ν-almost every a, a plane $P_a \in \mathbb{P}G$ such that

$$\varnothing \neq \operatorname{Tan}^q_p(\nu, a) \subset \{b\,\mathscr{H}^q_p \llcorner P_a : b > 0\}. \tag{12.2.16}$$

We use the normalization argument in [25, Remark 14.4(3), pp. 187–188]; compare its parabolic application in [26, proof of Lemma 4.8, p. 865]. Fix a where (12.2.13) and (12.2.16) hold. Choose $0 < d_- < \Theta^q_{p*}(\nu, a) \le \Theta^{q*}_p(\nu, a) < d_+ < \infty$. Let $\sigma \in \operatorname{Tan}_p(\nu, a)$, and choose $r_i \downarrow 0$ and $b_i > 0$ such that $b_i (T_{a,r_i})_\# \nu \rightharpoonup \sigma$. Define

$$\nu_i = r_i^{-q} (T_{a,r_i})_\# \nu, \qquad k_i = b_i r_i^q,$$

so $k_i \nu_i \rightharpoonup \sigma$. For each fixed $R > 0$ and all sufficiently large i,

$$\nu_i(B_p(0,1)) \ge d_-, \qquad \nu_i(B_p(0,R)) \le d_+ R^q.$$

Choose $R > 0$ with $\sigma(U_p(0,R)) > 0$. The open- and compact-set inequalities for weak convergence [25, Theorem 1.24(1)–(2)] give

$$0 < \frac{\sigma(U_p(0,R))}{d_+ R^q} \le \liminf_{i\to\infty} k_i \le \limsup_{i\to\infty} k_i \le \frac{\sigma(B_p(0,1))}{d_-} < \infty. \tag{12.2.17}$$

After taking a subsequence, $k_i \to k \in (0,\infty)$ and $\nu_i \rightharpoonup k^{-1}\sigma$. Equation (12.2.16) therefore implies $\sigma \in \{b\,\mathscr{H}^q_p \llcorner P_a : b > 0\}$. Conversely, choose $b_0 \mathscr{H}^q_p \llcorner P_a \in \operatorname{Tan}^q_p(\nu, a)$ using (12.2.16), and let $r_i \downarrow 0$ generate this tangent. For every $b > 0$,

$$\frac{b}{b_0} r_i^{-q} (T_{a,r_i})_\# \nu \rightharpoonup b\,\mathscr{H}^q_p \llcorner P_a.$$

This proves (12.2.11).

Finally, Γ_f is compact and has finite $\mathscr{H}^q_p$-measure. The vertical implication (4)⇒(2) in [26, Theorem 4.12, p. 867], applied to (12.2.11), shows that Γ_f is countably $\mathbb{P}$-rectifiable in the sense of Definition 5.7.2. □

12.3. **Density and thickness estimates.**

Lemma 12.3.1 (Thickness from invariant tangents)**.** *Let η be a positive Radon measure on $\mathbb{P}^n$, $n \ge 1$, and put $q = n+1$. Let $a \in \mathbb{P}^n$ and $V \in \mathbb{P}G$. Suppose that there are constants $0 < c_- \le c_+ < \infty$ and $r_0 > 0$ such that*

$$c_- r^q \le \eta(B_p(a,r)) \le c_+ r^q \qquad (0 < r < r_0), \tag{12.3.1}$$

and that every $\tau \in \operatorname{Tan}^q_p(\eta, a)$ is invariant under translations by V. Then, for every $0 < \beta < 1$,

$$\liminf_{r\downarrow 0} \inf_{z \in (a+V)\cap B_p(a,r)} \frac{\eta(B_p(z,\beta r))}{\eta(B_p(a,r))} \ge \frac{c_-}{c_+} \left(\frac{\beta}{8}\right)^q. \tag{12.3.2}$$

Proof. For Euclidean antecedents of the moving-center argument below, see De Philippis, De Rosa and Ghiraldin [10, journal version, Appendix B, condition (II), pp. 1145–1146] and the earlier proof of Fragalà and Mantegazza [15, author manuscript, proof of Theorem 4.5, condition B, (4.5)–(4.7), pp. 11–12]. For the compactness and nonvanishing steps, compare [25, proof of Theorem 14.3 and Remark 14.4(2), p. 187]. Fix $0 < \beta < 1$ and put $D =$

$(c_-/c_+)(\beta/8)^q$. If (12.3.2) fails, there are $0 < d < D$, radii $r_i \downarrow 0$ with $r_i < r_0$, and points $z_i \in (a+V) \cap B_p(a, r_i)$ such that

$$\eta(B_p(z_i, \beta r_i)) \le d\,\eta(B_p(a, r_i)) \qquad \text{for every } i. \tag{12.3.3}$$

Define

$$\eta_i = r_i^{-q}(T_{a,r_i})_{\#}\eta, \qquad \zeta_i = T_{a,r_i}(z_i). \tag{12.3.4}$$

Each η_i is a positive Radon measure on $\mathbb{P}^n$, since T_{a,r_i} is a homeomorphism. By (5.4.1), $\zeta_i \in V \cap B_p(0,1)$. This set is compact, so, after passing to a subsequence, $\zeta_i \to \zeta \in V \cap B_p(0,1)$.

For each $R > 0$, (12.3.1) gives

$$c_- R^q \le \eta_i(B_p(0,R)) = r_i^{-q}\eta(B_p(a, Rr_i)) \le c_+ R^q \qquad (Rr_i < r_0). \tag{12.3.5}$$

Every compact set $K \subset \mathbb{P}^n$ is contained in some $B_p(0,R)$. Equation (12.3.5) bounds $\eta_i(K)$ for all sufficiently large i, and the remaining finitely many values are finite because η_i is Radon. Hence $\sup_i \eta_i(K) < \infty$ for every compact $K \subset \mathbb{P}^n$. The space $\mathbb{P}^n$ is locally compact and is the union of the compact balls $B_p(0,m)$, $m \ge 1$. The compactness theorem for Radon measures [33, Chapter 1, Theorem 4.16, pp. 37–38] therefore gives a further subsequence and a positive Radon measure τ on $\mathbb{P}^n$ such that $\eta_i \rightharpoonup \tau$ in $\mathbb{P}^n$. For every $s > 0$, the compact-set inequality [25, Theorem 1.24(1), p. 19] and (12.3.5) give

$$\tau(B_p(0,s)) \ge \limsup_{i\to\infty} \eta_i(B_p(0,s)) \ge c_- s^q. \tag{12.3.6}$$

In particular, $\tau \ne 0$, so (9.0.7) gives $\tau \in \operatorname{Tan}_p^q(\eta, a)$. By hypothesis, τ is invariant under translations by V.

Since $\zeta_i \to \zeta$, for all sufficiently large i,

$$B_p(\zeta, \beta/8) \subset U_p(\zeta, \beta/4) \subset B_p(\zeta_i, \beta).$$

The open-set inequality [25, Theorem 1.24(2), p. 19], translation invariance with $\zeta \in V$, and (12.3.6) yield

$$\begin{aligned} \liminf_{i\to\infty} \eta_i(B_p(\zeta_i,\beta)) &\ge \tau(U_p(\zeta,\beta/4)) \\ &\ge \tau(B_p(\zeta,\beta/8)) = \tau(B_p(0,\beta/8)) \\ &\ge c_- \left(\frac{\beta}{8}\right)^q. \end{aligned} \tag{12.3.7}$$

On the other hand, (12.3.3), (12.3.4) and (12.3.5) give, for every i,

$$\begin{aligned} \eta_i(B_p(\zeta_i,\beta)) &= r_i^{-q}\eta(B_p(z_i,\beta r_i)) \\ &\le d\,\eta_i(B_p(0,1)) \le dc_+ < c_- \left(\frac{\beta}{8}\right)^q. \end{aligned}$$

This contradicts (12.3.7). □

Lemma 12.3.2 (A central flat window at the lower density)**.** *Let ϕ and the planes V_a satisfy the hypotheses of Proposition 12.1.1, and put $q = n+1$. Fix any point a at which* (12.1.7) *holds and every $\tau \in \operatorname{Tan}_p^q(\phi, a)$ has the representation* (12.1.8)*. Put*

$$\ell = \Theta_{p*}^q(\phi, a) = \liminf_{r\downarrow 0} \frac{\phi(B_p(a,r))}{r^q}, \qquad W = V_a. \tag{12.3.8}$$

There are radii $R_i \downarrow 0$ and a tangent $\tau \in \operatorname{Tan}_p^q(\phi, a)$ such that

$$\frac{\phi(B_p(a,R_i))}{R_i^q} \longrightarrow \ell, \qquad R_i^{-q}(T_{a,R_i})_{\#}\phi \rightharpoonup \tau \quad \text{in } \mathbb{P}^n, \tag{12.3.9}$$

and

$$\begin{aligned} \tau \llcorner B_p(0,1) &= \frac{\ell}{v(q)}\, \mathscr{H}^q_p \llcorner \big(W \cap B_p(0,1)\big), \\ \tau &\geq \frac{\ell}{v(q)}\, \mathscr{H}^q_p \llcorner W \qquad \text{on } \mathbb{P}^n. \end{aligned} \tag{12.3.10}$$

The identity and inequality in (12.3.10) *are between Radon measures on* $\mathbb{P}^n$*. In particular, these conclusions hold for* ϕ*-almost every* a*.*

Proof. Choose $R_i \downarrow 0$ such that $\phi(B_p(a,R_i))/R_i^q \to \ell$, and define

$$\phi_i = R_i^{-q}(T_{a,R_i})_{\#}\phi.$$

The finite upper density in (12.1.7) gives, for every $R > 0$ and all sufficiently large i,

$$\phi_i(B_p(0,R)) = R_i^{-q}\phi(B_p(a,RR_i)) \leq 2\Theta^{q*}_p(\phi,a)R^q.$$

Each ϕ_i is Radon, so the finitely many remaining indices also have bounded mass on any fixed compact set. Thus $\sup_i \phi_i(K) < \infty$ for every compact $K \subset \mathbb{P}^n$. The space $\mathbb{P}^n$ is locally compact and is the union of the compact balls $B_p(0,m)$, $m \geq 1$. By the compactness theorem [33, Chapter 1, Theorem 4.16, pp. 37–38], a subsequence converges weakly in $\mathbb{P}^n$ to a positive Radon measure τ. For every $\rho > 0$, the compact-set inequality [25, Theorem 1.24(1), p. 19] and (12.3.8) give

$$\tau(B_p(0,\rho)) \geq \limsup_{i\to\infty} R_i^{-q}\phi(B_p(a,\rho R_i)) \geq \ell\rho^q. \tag{12.3.11}$$

Consequently, $\tau \neq 0$ and $0 \in \operatorname{supp}\tau$. By (9.0.7), $\tau \in \operatorname{Tan}^q_p(\phi,a)$, and (12.3.9) holds along this subsequence. For these compactness and nonvanishing steps, compare [25, proof of Theorem 14.3 and Remark 14.4(2), p. 187].

By the choice of a, the measure τ has the finite representation (12.1.8) with $V_a = W$. Its support is the union of the finitely many distinct affine planes in that representation. Since $0 \in \operatorname{supp}\tau$, exactly one of these planes is W. Write the summand supported on W in (12.1.8) as $c_0\, \mathscr{H}^q_p \llcorner W$, where $c_0 > 0$. For all sufficiently small $\rho > 0$, the ball $B_p(0,\rho)$ meets none of the remaining planes. Hence (5.6.4) and (12.3.11) give

$$c_0 v(q)\rho^q = \tau(B_p(0,\rho)) \geq \ell\rho^q, \qquad c_0 v(q) \geq \ell. \tag{12.3.12}$$

The representation (12.1.8) also shows that τ is invariant under translations by W. Equation (12.1.3), with $z = 0$ and $r = 1$, therefore gives $\tau(B_p(0,1)) = \tau(U_p(0,1))$. The open-set inequality [25, Theorem 1.24(2), p. 19] and (12.3.9) imply

$$\tau(B_p(0,1)) = \tau(U_p(0,1)) \leq \liminf_{i\to\infty} \phi_i(U_p(0,1)) \leq \lim_{i\to\infty} \phi_i(B_p(0,1)) = \ell.$$

Together with (12.3.11), this yields

$$\tau(B_p(0,1)) = \ell. \tag{12.3.13}$$

The plane W contributes $c_0 v(q) \geq \ell$ to this mass, by (12.3.12), while all other summands in (12.1.8) are nonnegative. Thus $c_0 v(q) = \ell$, and each remaining summand has zero mass in $B_p(0,1)$. Restricting the representation to $B_p(0,1)$ proves the identity in (12.3.10). Retaining its positive summand on the whole plane W proves the global inequality. □

Lemma 12.3.3 (A compact set with uniform density and thickness bounds)**.** *Let* ϕ *be a nonzero finite positive Radon measure on* $\mathbb{P}^n$*,* $n \geq 1$*, and put* $q = n+1$*. Let* $R \subset \mathbb{P}^n$ *be Borel with* $\phi(\mathbb{P}^n \setminus R) = 0$*, and let* $V : R \to \mathbb{P}G$ *be Borel. Write* $V_x = V(x)$*. Suppose that* (12.1.7) *holds and that every* $\tau \in \operatorname{Tan}^q_p(\phi,x)$ *is invariant under translations by* V_x *for* ϕ*-almost every* $x \in R$*. Fix* $0 < \delta < 1$ *and* $0 < \beta_0 < 1$*.*

There are constants $b, c_+, u_0, d_0, \gamma > 0$*, a Borel set* G*, a compact set* E_0*, and a point* $z \in E_0$ *such that* $E_0 \subset G \subset R$*,* $\phi(E_0) > 0$*,*

$$b \leq \Theta^q_{p*}(\phi,x) < (1+\delta)b \qquad (x \in G), \tag{12.3.14}$$

and

$$(1-\delta)bu^q \le \phi(B_p(x,u)) \le c_+u^q \qquad (x\in G,\ 0<u<u_0). \tag{12.3.15}$$

The constants satisfy

$$2d_0 < \frac{(1-\delta)b}{c_+}\left(\frac{\beta_0}{8}\right)^q, \qquad \frac{\gamma 2^q c_+}{(1-\delta)b} < d_0. \tag{12.3.16}$$

For every $x\in E_0$ and $0<u<u_0$,

$$\inf_{y\in(x+V_x)\cap B_p(x,u)} \frac{\phi(B_p(y,\beta_0u))}{\phi(B_p(x,u))} > 2d_0 \tag{12.3.17}$$

and

$$\phi(B_p(x,u)\setminus G) \le \gamma\phi(B_p(x,u)). \tag{12.3.18}$$

Lemma 12.3.2 applies at every $x\in E_0$, with $a=x$ and $W=V_x$, and

$$\frac{\phi(B_p(z,u)\setminus E_0)}{\phi(B_p(z,u))} \longrightarrow 0 \qquad (u\downarrow 0). \tag{12.3.19}$$

Proof. For the density localization and selection of positive compact sets, compare [27, proof of Theorem 1.3, pp. 40–41]. Remove a Borel ϕ-null set containing the exceptional sets for (12.1.7), the assumed translation invariance, and Proposition 12.1.1. Denote the remaining subset of R by X. Thus $\phi(X)=\phi(\mathbb{P}^n)>0$, and Lemma 12.3.2 applies at every point of X.

For $j\in\mathbb{Z}$ put $b_j=(1+\delta)^j$. For positive integers k,m, define

$$\begin{aligned} G_{jkm} = \big\{x\in X :\ & b_j \le \Theta^q_{p*}(\phi,x) < (1+\delta)b_j, \\ & (1-\delta)b_ju^q \le \phi(B_p(x,u)) \le ku^q \text{ for every } 0<u<1/m\big\}. \end{aligned} \tag{12.3.20}$$

The proof of Lemma 5.3.1 shows that $x\mapsto\Theta^q_{p*}(\phi,x)$ and, for each fixed $u>0$, $x\mapsto\phi(B_p(x,u))$ are Borel on $\mathbb{P}^n$. For each fixed x, continuity from above of closed-ball mass shows that the growth inequalities in (12.3.20) need only be checked at rational u. Thus each G_{jkm} is Borel. Equation (12.1.7), valid at every point of X, implies

$$X = \bigcup_{\substack{j\in\mathbb{Z}\\ k,m\ge1}} G_{jkm}.$$

Choose $j\in\mathbb{Z}$ and positive integers k,m with $\phi(G_{jkm})>0$, and put $G=G_{jkm}$, $b=b_j$, $c_+=k$ and $u_0=1/m$. Then (12.3.14) and (12.3.15) hold.

Put

$$D = \frac{(1-\delta)b}{c_+}\left(\frac{\beta_0}{8}\right)^q, \qquad d_0 = D/4.$$

Lemma 12.3.1, applied at each $x\in G$ with $c_-=(1-\delta)b$, gives

$$\liminf_{u\downarrow0} \inf_{y\in(x+V_x)\cap B_p(x,u)} \frac{\phi(B_p(y,\beta_0u))}{\phi(B_p(x,u))} \ge D. \tag{12.3.21}$$

For each positive integer m, set $r_m=\min\{u_0,1/m\}$ and define

$$\begin{aligned} H_m = \big\{x\in G :\ & \phi(B_p(y,\beta_0u)) \ge 3d_0\phi(B_p(x,u)) \\ & \text{for every } 0<u<r_m \text{ and every } y\in(x+V_x)\cap B_p(x,u)\big\}. \end{aligned} \tag{12.3.22}$$

These sets increase with m, and (12.3.21) gives $\bigcup_m H_m = G$.

We show that each H_m is Borel. Put $A=(\mathbb{Q}^n\times\mathbb{Q})\cap U_p(0,1)$. For $x\in G$, the projection formula in Section 5.5 gives $\|P_{V_x}v\|_p \le \|v\|_p$ for $v\in\mathbb{P}^n$ and $P_{V_x}w=w$ for $w\in V_x$. Since $\overline{A}=B_p(0,1)$, for every $u>0$ the set

$$\{x+\delta_u(P_{V_x}a) : a\in A\}$$

is contained in and dense in $(x + V_x) \cap B_p(x, u)$. For every fixed $s > 0$, the function $y \mapsto \phi(B_p(y, s))$ is upper semicontinuous on $\mathbb{P}^n$ [33, Chapter 1, Remark 3.2, p. 13]. Thus, at fixed u, the non-strict inequality in (12.3.22) need only be checked at these countably many centers.

It also suffices to check rational $u \in (0, r_m)$. Indeed, for any $0 < u < r_m$, choose rational $u_i \downarrow u$ with $u_i < r_m$. Every $y \in (x + V_x) \cap B_p(x, u)$ also belongs to $(x + V_x) \cap B_p(x, u_i)$. Continuity from above gives

$$\phi(B_p(y, \beta_0 u_i)) \longrightarrow \phi(B_p(y, \beta_0 u)), \qquad \phi(B_p(x, u_i)) \longrightarrow \phi(B_p(x, u)),$$

so the inequalities at u_i imply the inequality at u. Consequently,

$$H_m = \bigcap_{\substack{u \in \mathbb{Q} \\ 0<u<r_m}} \bigcap_{a \in A} \big\{ x \in G : \phi(B_p(x + \delta_u(P_{V_x} a), \beta_0 u)) \geq 3 d_0 \phi(B_p(x, u)) \big\}. \tag{12.3.23}$$

For fixed u and a, the map $x \mapsto x + \delta_u(P_{V_x} a)$ is Borel on G, since V is Borel. Hence both ball-mass functions in (12.3.23) are Borel in x, and the countable intersection is Borel. Choose m with $\phi(H_m) > 0$ and put $G_{\mathrm{th}} = H_m$. Decrease u_0 to r_m. Equation (12.3.22) gives

$$\inf_{y \in (x+V_x) \cap B_p(x,u)} \frac{\phi(B_p(y, \beta_0 u))}{\phi(B_p(x, u))} \geq 3 d_0 > 2 d_0 \qquad (x \in G_{\mathrm{th}},\ 0 < u < u_0). \tag{12.3.24}$$

Choose $\gamma > 0$ satisfying the second inequality in (12.3.16). For $x \in G$ and $0 < u < u_0/5$, (12.3.15) gives

$$\frac{\phi(B_p(x, 5u))}{\phi(B_p(x, u))} \leq \frac{5^q c_+}{(1 - \delta) b} =: C. \tag{12.3.25}$$

In particular, all such radii belong to $\mathcal{R}_C(x)$ in (11.0.1), with $\mu = \phi$ and $\Omega = \mathbb{P}^n$. Apply Lemma 11.2 to $1_G \in L^1(\mathbb{P}^n, \phi)$, and let N_G be a Borel ϕ-null set containing its exceptional set. For every $x \in G \setminus N_G$,

$$\begin{aligned} \frac{\phi(B_p(x, u) \setminus G)}{\phi(B_p(x, u))} &= \frac{1}{\phi(B_p(x, u))} \int_{B_p(x,u)} |1_G(w) - 1| \, d\phi(w) \\ &\longrightarrow 0 \qquad (u \downarrow 0). \end{aligned} \tag{12.3.26}$$

For each positive integer m, define

$$\begin{aligned} F_m = \big\{ x \in G \setminus N_G : \phi(B_p(x, u) \setminus G) \leq \gamma \phi(B_p(x, u)) \\ \text{for every } 0 < u < \min\{u_0, 1/m\} \big\}. \end{aligned} \tag{12.3.27}$$

Both closed-ball masses in (12.3.27) are Borel in x at fixed radius and continuous from above in the radius. The inequalities again need only be tested at rational radii, so F_m is Borel. Equation (12.3.26) gives $\bigcup_m F_m = G \setminus N_G$. Choose m such that $\phi(G_{\mathrm{th}} \cap F_m) > 0$. Inner regularity gives a compact set $E_0 \subset G_{\mathrm{th}} \cap F_m$ with $\phi(E_0) > 0$ [14, Proposition 7.5, p. 216]. Decrease u_0 to $\min\{u_0, 1/m\}$. Equations (12.3.24) and (12.3.27) give (12.3.17) and (12.3.18). Since $E_0 \subset X$, Lemma 12.3.2 applies at each point of E_0.

Finally, apply Lemma 11.2 to $1_{E_0} \in L^1(\mathbb{P}^n, \phi)$ and choose $z \in E_0$ outside its exceptional set. Equation (12.3.25) makes every sufficiently small radius admissible at z. Since $1_{E_0}(z) = 1$, the resulting differentiation identity is (12.3.19). □

12.4. **The MMP graph extraction criterion.** Proposition 12.4.1 specializes the graph-extraction result of Merlo, Mourgoglou and Puliatti [27, Proposition 4.12, pp. 30–31] to the square-sum metric and the vertical planes $\mathbb{P}G$, of homogeneous dimension $q = n + 1$. They allow any homogeneous dimension $h \in \{1, \ldots, n + 2\}$, fix the spatial and temporal dimensions of the planes, and work with the class of metrics described in [27, Chapter 4, p. 27]. Their result is a parabolic version of Preiss's quantitative projection lemma [29, Lemma 5.2, pp. 603–605], as explained in [27, pp. 29–30].

Merlo, Mourgoglou and Puliatti adapt Antonelli and Merlo's stopping and projection argument in Carnot groups [3, arXiv v1, Proposition 3.6, pp. 25–26]. Their proof of part (vi) draws on [3, arXiv v1, Proposition 3.10, p. 34].

We retain the compact-set and graph conclusions in parts (ii), (v) and (vi) of their proposition. The stopping-radius bounds, decomposition and projection estimates in parts (i), (iii) and (iv) are omitted from our statement. The good-triple, property-(H) and cylinder conditions are unchanged under this specialization. We impose a smaller upper bound on ϵ to meet the strict inequality used in their positivity argument, and state the measure comparison on every Borel subset of the extracted set. The proof below derives this comparison from their pointwise estimates and converts their Hausdorff normalization to ours.

Definition 12.4.1 (Good triples and property (H))**.** Let $n \geq 1$, let ϕ be a positive Radon measure on $\mathbb{P}^n$, let $z \in \operatorname{supp}\phi$, and put $q = n+1$. Fix $r > 0$ and

$$0 < \epsilon \leq 2^{-3q}5^{-10(q+5)}, \qquad r_1 = (1-\epsilon/q)r, \qquad \mu_{\mathrm{MMP}} = 2^{-7-5q}q^{-3}\epsilon^2. \tag{12.4.1}$$

A triple (x, s, V) with $x \in B_p(z, 2r_1)$, $0 < s \leq 2r$ and $V \in \mathbb{P}G$ is *good* if

$$\phi(B_p(y,t)) \geq (1-\epsilon)\left(\frac{t}{2r}\right)^q \phi(B_p(z,2r)) \tag{12.4.2}$$

for every $y \in B_p(x,2s) \cap (x+V)$ and every $t \in [\mu_{\mathrm{MMP}}s, 2s]$. We say that a compact set $E \subset B_p(z,2r_1)$ and a plane $W \in \mathbb{P}G$ satisfy *property* (H) if

(a)

$$\phi(B_p(z,2r_1) \setminus E) \leq 2^{-q}\mu_{\mathrm{MMP}}^{q+1}\phi(B_p(z,2r_1)); \tag{12.4.3}$$

(b) for every $x \in E$ and every $0 < s \leq 2r - d_p(x,z)$, there is a plane $V \in \mathbb{P}G$ for which (x,s,V) is good;

(c) the triple (z, r, W) is good.

These are the good-triple and property-(H) conditions in [27, (4.8)–(4.9), p. 30], with $h = q$ and vertical planes.

Proposition 12.4.1 (Graph extraction; Merlo–Mourgoglou–Puliatti)**.** *Let $\phi, z, r, \epsilon, r_1, \mu_{\mathrm{MMP}}$ be as in Definition 12.4.1, and suppose that E and W satisfy property (H). There is a compact set $A \subset P_W(B_p(z,r_1))$ such that the compact set*

$$F = E \cap P_W^{-1}(A) \tag{12.4.4}$$

satisfies $P_W(F) = A$, $\mathscr{H}_p^q(F) > 0$, and

$$c_F\,\mathscr{H}_p^q \llcorner F \leq \phi \llcorner F \leq C_F\,\mathscr{H}_p^q \llcorner F \tag{12.4.5}$$

as Radon measures on $\mathbb{P}^n$, for some constants $0 < c_F \leq C_F < \infty$. If, in addition, the cylinder in (5.5.1) *satisfies*

$$\phi\big(C_W(z,r) \cap B_p(z,2r)\big) \leq (1-\epsilon)^{-1}2^{-q}\phi(B_p(z,2r)), \tag{12.4.6}$$

then there is a map $f : W \to W^{\perp}$, Lipschitz for d_p, such that

$$\phi(F \cap \Gamma_f) > 0, \qquad \Gamma_f = \{w + f(w) : w \in W\}. \tag{12.4.7}$$

Proof. Apply [27, Proposition 4.12(ii), (v), (vi), pp. 30–31] with $h = q$ and the vertical stratum $(n-1,1)$ of [27, Definition 2.9, p. 13]. The square-sum metric is translation invariant and homogeneous under δ_r, and its unit ball $\{(x,t) : |x|^2 + |t| \leq 1\}$ is Euclidean convex and invariant under spatial rotations. These are the metric hypotheses imposed in [27, Chapter 4, p. 27]. The projection formula in Section 5.5 gives $d_p(P_V x, P_V y) \leq d_p(x,y)$ for every $V \in \mathbb{P}G$. This verification includes $n = 1$, when the only vertical plane is the time axis.

The published parameter range, with $h = q$, is $0 < \epsilon \le 2 \cdot 5^{-q-5}$. The proof of part (v) uses the stricter inequality $\epsilon < 2^{-3q}5^{-q-5}$ to obtain positive measure [27, p. 37]. Our choice in (12.4.1) satisfies

$$\epsilon \le 2^{-3q}5^{-10(q+5)} < 2^{-3q}5^{-q-5} < 2 \cdot 5^{-q-5}.$$

In the estimate corresponding to [27, (4.30), p. 34], the needed bound is

$$2r_1 + \frac{\epsilon r}{q} = \left(2 - \frac{\epsilon}{q}\right) r < 2r, \tag{12.4.8}$$

which replaces the incorrect intermediate equality printed there. Items (ii) and (v) give the compact set A, the identity $P_W(F) = A$ and positive Hausdorff measure of F. The Hausdorff normalization in [27, Definition 2.1, p. 12] is $\mathscr{H}^q_{\mathrm{MMP}} = 2^{-q}\mathscr{H}^q_p$, so positivity is unchanged.

Part (v) of [27, Proposition 4.12, p. 31] states a comparison between $\phi(F)$ and $\mathscr{H}^q_{\mathrm{MMP}}(F)$. To obtain the restriction inequalities in (12.4.5), we use the pointwise bounds in its proof. Put

$$M = \frac{\phi(B_p(z, 2r))}{(2r)^q} > 0, \qquad s_0 = \min\{r, 2r - 2r_1\} > 0. \tag{12.4.9}$$

For $x \in F$ and $0 < s < s_0$, one has $B_p(x, s) \subset B_p(z, 2r)$, since $E \subset B_p(z, 2r_1)$. Thus [27, (4.47), p. 37] gives the upper bound in

$$(1 - \epsilon)Ms^q \le \phi(B_p(x, s)) \le \mu_{\mathrm{MMP}}^{-q} M s^q \qquad (x \in F,\ 0 < s < s_0). \tag{12.4.10}$$

For the lower bound, property (H)(b) provides a good triple (x, s, V); take $y = x$ and $t = s$ in (12.4.2), using $0 < \mu_{\mathrm{MMP}} < 1$. The metric density comparisons [13, 2.10.17(2) and 2.10.19(3)], applied to the finite measure $\phi \llcorner B_p(z, 2r)$ on each Borel subset of F, now give (12.4.5) with finite positive constants in our Hausdorff normalization. In particular, $\mathscr{H}^q_p(F) < \infty$, since F is compact. The text before [27, (4.48), p. 37] instead states $E \subset B_p(z, r_1)$, and (4.48) changes the coefficient to $\phi(B_p(z, r))$. The bounds (12.4.10) use $E \subset B_p(z, 2r_1)$ and retain $\phi(B_p(z, 2r))$ throughout.

Finally, after translating z to 0, (12.4.6) is precisely [27, (4.10), p. 31]. In the first display of [27, p. 39], multiplying $(1 - \epsilon)^{-5}$ by the factor $(1 - \epsilon)^{-1}$ from (12.4.6) gives the coefficient $\frac{1}{2}(1 - \epsilon)^{-6}$ in place of the printed $\frac{1}{2}(1 - \epsilon)^{-4}$. By (12.4.1), $\epsilon < 1/100$, so $(1 - \epsilon)^6 \ge 1 - 6\epsilon > 47/50$ and

$$\frac{1}{2}(1 - \epsilon)^{-6} < \frac{25}{47} < \frac{27}{50}.$$

Thus the numerical bound used in the proof of part (vi) is preserved. Item (vi) therefore gives a map $f : W \to W^\perp$, Lipschitz for d_p, and (12.4.7). □

12.5. **Positive rectifiable pieces.**

Lemma 12.5.1 (Reference good triples and the cylinder bound)**.** *Let ϕ and the planes V_a satisfy the hypotheses of Lemma 12.3.2, and fix a point z satisfying its pointwise hypotheses. Put $q = n+1$, $\ell = \Theta^q_{p*}(\phi, z)$ and $W = V_z$, and choose R_i and τ as in* (12.3.9) *and* (12.3.10)*, with $a = z$. Fix ϵ in the range* (12.4.1)*, with the corresponding μ_{MMP}, and set*

$$r_i = R_i/2, \qquad r_{1,i} = (1 - \epsilon/q)r_i. \tag{12.5.1}$$

For every sufficiently large i, the triple (z, r_i, W) is good in the sense of Definition 12.4.1, with $r = r_i$ and $r_1 = r_{1,i}$. Moreover,

$$\phi\big(C_W(z, r_i) \cap B_p(z, 2r_i)\big) \le (1 - \epsilon)^{-1}2^{-q}\phi(B_p(z, 2r_i)). \tag{12.5.2}$$

Proof. Define

$$\phi_i = R_i^{-q}(T_{z,R_i})_\# \phi, \qquad m_i = \phi_i(B_p(0, 1)) = \frac{\phi(B_p(z, R_i))}{R_i^q}. \tag{12.5.3}$$

By (12.3.9), $\phi_i \rightharpoonup \tau$ in $\mathbb{P}^n$ and $m_i \to \ell \in (0,\infty)$. In particular, $z \in \operatorname{supp}\phi$. The global inequality in (12.3.10), the plane measure formula (5.6.4), and continuity from below give

$$\tau(U_p(\xi,s)) \geq \frac{\ell}{v(q)}\mathscr{H}^q_p(W \cap U_p(\xi,s)) = \ell s^q \qquad (\xi \in W,\ s>0). \tag{12.5.4}$$

Choose $0<\alpha<1$ such that

$$\frac{(1-\alpha)^q}{(1+\alpha)^{q+2}} > 1-\epsilon. \tag{12.5.5}$$

The parameter set

$$\mathcal{K} = \big(W \cap B_p(0,1)\big) \times [\mu_{\mathrm{MMP}}/2, 1] \tag{12.5.6}$$

is compact. Thus there are $(\xi_j,\sigma_j) \in \mathcal{K}$, $j=1,\ldots,N$, such that every $(\xi,\sigma)\in\mathcal{K}$ satisfies

$$d_p(\xi,\xi_j) + |\sigma-\sigma_j| < \alpha\sigma_j \tag{12.5.7}$$

for some j. Set $U_j = U_p(\xi_j,(1-\alpha)\sigma_j)$. Whenever (12.5.7) holds, the triangle inequality gives

$$U_j \subset B_p(\xi,\sigma), \qquad \sigma_j \geq \frac{\sigma}{1+\alpha}. \tag{12.5.8}$$

Apply the open-set inequality [25, Theorem 1.24(2), p. 19] to the finitely many fixed balls U_j. Equation (12.5.4) and $m_i \to \ell$ give, for all sufficiently large i,

$$\phi_i(U_j) > \frac{\ell(1-\alpha)^q\sigma_j^q}{1+\alpha} \quad (j=1,\ldots,N), \qquad 0<m_i<(1+\alpha)\ell. \tag{12.5.9}$$

For each $(\xi,\sigma)\in\mathcal{K}$, choose j satisfying (12.5.7). Equations (12.5.3), (12.5.8), (12.5.9) and (12.5.5) yield

$$\begin{aligned}\frac{\phi(B_p(z+\delta_{R_i}\xi,R_i\sigma))}{\phi(B_p(z,R_i))} &= \frac{\phi_i(B_p(\xi,\sigma))}{m_i} \geq \frac{\phi_i(U_j)}{m_i}\\ &> \frac{(1-\alpha)^q}{(1+\alpha)^2}\sigma_j^q\\ &\geq \frac{(1-\alpha)^q}{(1+\alpha)^{q+2}}\sigma^q > (1-\epsilon)\sigma^q.\end{aligned} \tag{12.5.10}$$

There is therefore an i_0 such that (12.5.10) holds for every $i\geq i_0$ and every $(\xi,\sigma)\in\mathcal{K}$. Since $\delta_{R_i}W = W$, the substitutions $y = z+\delta_{R_i}\xi$ and $t = R_i\sigma$ range over every $y \in (z+W)\cap B_p(z,2r_i)$ and $t\in[\mu_{\mathrm{MMP}}r_i, 2r_i]$. Thus (12.5.10) proves (12.4.2) for (z,r_i,W).

For the cylinder estimate, define the compact set

$$F_0 = B_p(0,1)\cap C_W(0,1/2). \tag{12.5.11}$$

The unit-ball identity in (12.3.10) and (5.6.4) give

$$\tau(F_0) = \frac{\ell}{v(q)}\mathscr{H}^q_p(W\cap B_p(0,1/2)) = \ell 2^{-q}. \tag{12.5.12}$$

The projection formula in Section 5.5 gives $P_W\delta_s = \delta_s P_W$ for every $s>0$. Using (5.5.1) and $2r_i = R_i$, we obtain

$$T_{z,R_i}\big(C_W(z,r_i)\cap B_p(z,2r_i)\big) = F_0. \tag{12.5.13}$$

The compact-set inequality [25, Theorem 1.24(1), p. 19], (12.5.12) and $m_i\to\ell>0$ therefore imply

$$\begin{aligned}\limsup_{i\to\infty}\frac{\phi(C_W(z,r_i)\cap B_p(z,2r_i))}{\phi(B_p(z,2r_i))} &= \limsup_{i\to\infty}\frac{\phi_i(F_0)}{m_i}\\ &\leq \frac{\tau(F_0)}{\ell} = 2^{-q} < (1-\epsilon)^{-1}2^{-q}.\end{aligned} \tag{12.5.14}$$

This proves (12.5.2) for every sufficiently large i. □

Lemma 12.5.2 (A positive compact graph piece)**.** *Let ϕ, R and V satisfy the hypotheses of Lemma 12.3.3, and put $q = n + 1$. There are a plane $W \in \mathbb{P}G$, a compact set $K \subset W$, a map $f : K \to W^\perp$, Lipschitz for d_p, and constants $0 < c \le C < \infty$ such that the compact graph*

$$\Gamma = \{w + f(w) : w \in K\} \tag{12.5.15}$$

is contained in R, has $\phi(\Gamma) > 0$, and satisfies

$$c\,\mathscr{H}^q_p \llcorner \Gamma \le \phi \llcorner \Gamma \le C\,\mathscr{H}^q_p \llcorner \Gamma \qquad \text{as Radon measures on } \mathbb{P}^n. \tag{12.5.16}$$

Proof. Fix ϵ in the range (12.4.1), with the corresponding μ_{MMP}. Choose $0 < \rho < 1$ with $(1-\rho)^q > 1 - \epsilon$, and then $0 < \delta < 1$ such that

$$(1-\delta)(1-\rho)^q > (1-\epsilon)(1+\delta)^2. \tag{12.5.17}$$

Choose $0 < \beta_0 < 1$ satisfying

$$2\beta_0 \le \rho\mu_{\mathrm{MMP}}. \tag{12.5.18}$$

Apply Lemma 12.3.3 with these δ and β_0, obtaining $b, c_+, u_0, d_0, \gamma, G, E_0, z$. Put $\ell = \Theta^q_{p*}(\phi, z)$ and $W = V_z$. That lemma permits the application of Lemma 12.3.2 at z; choose its sequence $R_i \downarrow 0$, and set $r_i = R_i/2$ and $r_{1,i} = (1 - \epsilon/q)r_i$ as in (12.5.1). Lemma 12.5.1 gives the good triple (z, r_i, W) and the cylinder bound (12.5.2) for every sufficiently large i.

Since $z \in G$, (12.3.14) gives $\ell < (1+\delta)b$. The convergence in (12.3.9), with $a = z$, therefore gives, for every sufficiently large i,

$$\phi(B_p(z, R_i)) \le (1+\delta)^2 b R_i^q, \qquad 4R_i < u_0. \tag{12.5.19}$$

Define the compact sets

$$E_i = E_0 \cap B_p(z, 2r_{1,i}). \tag{12.5.20}$$

Since $2r_{1,i} \downarrow 0$, (12.3.19) gives, for every sufficiently large i,

$$\phi(B_p(z, 2r_{1,i}) \setminus E_i) \le 2^{-q}\mu^{q+1}_{\mathrm{MMP}}\phi(B_p(z, 2r_{1,i})). \tag{12.5.21}$$

Fix i so that (12.5.19), (12.5.21) and the conclusions of Lemma 12.5.1 hold.

Let $x \in E_i$ and $0 < s \le R_i - d_p(x, z)$. Then $4s \le 4R_i < u_0$ by (12.5.19). For every $y \in (x + V_x) \cap B_p(x, 2s)$, (12.3.17), applied with $u = 2s$, gives

$$\phi(B_p(y, 2\beta_0 s)) > 2d_0\phi(B_p(x, 2s)). \tag{12.5.22}$$

We claim that $B_p(y, 2\beta_0 s)$ meets G. Otherwise, since $\beta_0 < 1$, this ball would be contained in $B_p(x, 4s) \setminus G$. Equations (12.3.18), (12.3.15) and (12.3.16) would then give

$$\begin{aligned}\phi(B_p(y, 2\beta_0 s)) &\le \gamma\phi(B_p(x, 4s)) \le \gamma c_+ (4s)^q \\ &\le \frac{\gamma 2^q c_+}{(1-\delta)b}\phi(B_p(x, 2s)) < d_0\phi(B_p(x, 2s)),\end{aligned}$$

contradicting (12.5.22).

Choose $w \in G \cap B_p(y, 2\beta_0 s)$. For every $t \in [\mu_{\mathrm{MMP}}s, 2s]$, (12.5.18) gives

$$\begin{aligned}d_p(w, y) &\le 2\beta_0 s \le \rho\mu_{\mathrm{MMP}} s \le \rho t, \\ B_p(w, (1-\rho)t) &\subset B_p(y, t).\end{aligned} \tag{12.5.23}$$

Since $w \in G$ and $0 < (1-\rho)t < 2s < u_0$, (12.3.15), (12.5.23), (12.5.17) and (12.5.19) imply

$$\begin{aligned}\phi(B_p(y, t)) &\ge \phi(B_p(w, (1-\rho)t)) \ge (1-\delta)b(1-\rho)^q t^q \\ &> (1-\epsilon)(1+\delta)^2 b t^q \\ &\ge (1-\epsilon)\left(\frac{t}{R_i}\right)^q \phi(B_p(z, R_i)).\end{aligned} \tag{12.5.24}$$

Since $R_i = 2r_i$, this proves (12.4.2) for (x, s, V_x), for every $x \in E_i$ and every $0 < s \le 2r_i - d_p(x, z)$. Thus clause (b) of Definition 12.4.1 holds. Equation (12.5.21) gives clause (a), and Lemma 12.5.1 gives clause (c) and the additional cylinder bound (12.4.6).

Apply Proposition 12.4.1 with $E = E_i$ and $r = r_i$. It gives a compact set $F \subset E_i$ satisfying (12.4.5), and a map $g : W \to W^\perp$, Lipschitz for d_p, such that

$$\phi(F \cap \Gamma_g) > 0, \qquad \Gamma_g = \{w + g(w) : w \in W\}. \tag{12.5.25}$$

Set

$$\Gamma = F \cap \Gamma_g, \qquad K = P_W(\Gamma), \qquad f = g|_K. \tag{12.5.26}$$

The graph Γ_g is closed because g is continuous, so Γ is compact; its projection K is compact as well. The map $f : K \to W^\perp$ is Lipschitz for d_p and satisfies (12.5.15). Moreover, $\Gamma \subset E_i \subset E_0 \subset R$ and $\phi(\Gamma) > 0$ by (12.5.25). Restricting (12.4.5) to Γ gives (12.5.16) with $c = c_F$ and $C = C_F$. □

Proposition 12.5.1 (Positive rectifiable pieces in Borel sets)**.** *Let ϕ be a finite positive Radon measure on $\mathbb{P}^n$, $n \ge 1$, and put $q = n + 1$. Suppose that* (12.1.7) *holds. Let $X \subset \mathbb{P}^n$ be Borel with $\phi(\mathbb{P}^n \setminus X) = 0$, and let $V : X \to \mathbb{P}G$ be Borel. Write $V_a = V(a)$, and assume that every $\tau \in \mathrm{Tan}^q_p(\phi, a)$ is invariant under translations by V_a for ϕ-almost every $a \in X$ (Definition 4.2.1).*

For every Borel set $R \subset \mathbb{P}^n$ with $\phi(R) > 0$, there are a plane $W \in \mathbb{P}G$, a compact set $K \subset W$, a map $f : K \to W^\perp$, Lipschitz for d_p, and constants $0 < c \le C < \infty$ such that

$$\Gamma = \{w + f(w) : w \in K\} \subset R, \qquad \phi(\Gamma) > 0, \tag{12.5.27}$$

and

$$c\,\mathscr{H}^q_p \llcorner \Gamma \le \phi \llcorner \Gamma \le C\,\mathscr{H}^q_p \llcorner \Gamma \qquad \text{as Radon measures on } \mathbb{P}^n. \tag{12.5.28}$$

The set Γ is compact and countably $\mathbb{P}$-rectifiable, with $0 < \mathscr{H}^q_p(\Gamma) < \infty$.

Proof. Fix a Borel set $R \subset \mathbb{P}^n$ with $\phi(R) > 0$, and define

$$\psi = \phi \llcorner R, \qquad \psi(A) = \phi(A \cap R) \quad (A \subset \mathbb{P}^n \text{ Borel}). \tag{12.5.29}$$

This is a nonzero finite Borel measure, hence a Radon measure on $\mathbb{P}^n$ by [14, Theorem 7.8, p. 217]. The density hypothesis (12.1.7) allows us to apply [27, Proposition 2.13(ii)–(iii), pp. 14–15] with $h = q$ and $\mathbf{1}_R \in L^1(\phi)$. For ψ-almost every $a \in R \cap X$, this gives

$$\begin{aligned} \Theta^q_{p*}(\psi, a) &= \Theta^q_{p*}(\phi, a), \quad \Theta^{q*}_p(\psi, a) = \Theta^{q*}_p(\phi, a), \\ \mathrm{Tan}^q_p(\psi, a) &= \mathrm{Tan}^q_p(\phi, a). \end{aligned} \tag{12.5.30}$$

Since $\psi(\mathbb{P}^n \setminus (R \cap X)) = 0$, (12.5.30) and the hypotheses on ϕ imply that, for ψ-almost every $a \in R \cap X$, the lower and upper raw q-densities of ψ at a are positive and finite, and every $\tau \in \mathrm{Tan}^q_p(\psi, a)$ is invariant under translations by V_a. The restriction $V|_{R\cap X}$ is Borel. By Lemma 12.5.2, applied to ψ, $R \cap X$ and $V|_{R\cap X}$, there are a plane $W \in \mathbb{P}G$, a compact set $K \subset W$, a map $f : K \to W^\perp$, Lipschitz for d_p, and constants $0 < c \le C < \infty$ such that the compact graph $\Gamma = \{w + f(w) : w \in K\}$ lies in $R \cap X$, has $\psi(\Gamma) > 0$, and satisfies (12.5.16) with ϕ replaced by ψ. Since $\Gamma \subset R$, the definition (12.5.29) gives

$$\psi \llcorner \Gamma = \phi \llcorner \Gamma \qquad \text{as Radon measures on } \mathbb{P}^n. \tag{12.5.31}$$

This proves (12.5.27) and (12.5.28). In particular,

$$0 < \frac{\phi(\Gamma)}{C} \le \mathscr{H}^q_p(\Gamma) \le \frac{\phi(\Gamma)}{c} < \infty.$$

The parent measure ϕ satisfies the density and translation-invariance hypotheses of Proposition 12.2.2. The compact graph (12.5.27) and comparison (12.5.28) supply its remaining hypotheses, with $S = W$. That proposition therefore shows that Γ is countably $\mathbb{P}$-rectifiable. □

12.6. **Countable coverings and weighted representations.**

Proposition 12.6.1 (A countable rectifiable carrier)**.** *Let ϕ, X and V satisfy the hypotheses of Proposition 12.5.1, and put $q = n + 1$. There is a possibly empty, finite or countable family of pairwise disjoint compact vertical parabolic Lipschitz graphs $\Gamma_j \subset X$, $j \in J$ (Definition 5.7.1), such that each Γ_j is countably $\mathbb{P}$-rectifiable, has $\phi(\Gamma_j) > 0$ and $0 < \mathscr{H}^q_p(\Gamma_j) < \infty$, and satisfies*

$$c_j \, \mathscr{H}^q_p \llcorner \Gamma_j \le \phi \llcorner \Gamma_j \le C_j \, \mathscr{H}^q_p \llcorner \Gamma_j \qquad \text{as Radon measures on } \mathbb{P}^n \tag{12.6.1}$$

for some $0 < c_j \le C_j < \infty$. The set

$$K = \bigcup_{j \in J} \Gamma_j \subset X \tag{12.6.2}$$

is Borel and countably $\mathbb{P}$-rectifiable, and

$$\phi(\mathbb{P}^n \setminus K) = 0, \qquad \mathscr{H}^q_p \llcorner K \text{ is } \sigma\text{-finite.} \tag{12.6.3}$$

Proof. For the passage from positive graph pieces to a countable covering, compare [27, proof of Theorem 1.3, p. 40]. For each Borel set $R \subset X$, let $\mathcal{F}(R)$ be the family of compact vertical parabolic Lipschitz graphs $\Gamma \subset R$ that are countably $\mathbb{P}$-rectifiable, have $\phi(\Gamma) > 0$, and satisfy (12.5.28) for some $0 < c \le C < \infty$. Each such graph has $0 < \mathscr{H}^q_p(\Gamma) < \infty$ by that comparison and finiteness of ϕ. Proposition 12.5.1 shows that $\mathcal{F}(R)$ is nonempty whenever $\phi(R) > 0$.

Set $R_0 = X$. Suppose that $\Gamma_1, \dots, \Gamma_{j-1}$ have been chosen and that $R_{j-1} = X \setminus \bigcup_{i<j} \Gamma_i$. If $\phi(R_{j-1}) = 0$, stop the construction. Otherwise, define

$$s_j = \sup\{\phi(\Gamma) : \Gamma \in \mathcal{F}(R_{j-1})\}. \tag{12.6.4}$$

The remainder R_{j-1} is Borel, and $0 < s_j \le \phi(R_{j-1}) < \infty$. Choose $\Gamma_j \in \mathcal{F}(R_{j-1})$ and set R_j so that

$$\phi(\Gamma_j) > \frac{s_j}{2}, \qquad R_j = R_{j-1} \setminus \Gamma_j. \tag{12.6.5}$$

This defines a finite or countable family of pairwise disjoint compact graphs; the family is empty if $\phi(X) = 0$. Let J be its index set, and define K by (12.6.2).

The set K is Borel because every Γ_j is compact. If the construction stops, then $\phi(X \setminus K) = 0$. If it continues indefinitely and $\phi(X \setminus K) > 0$, Proposition 12.5.1 supplies a graph $\Gamma \in \mathcal{F}(X \setminus K)$. Since $X \setminus K \subset R_{j-1}$ for every $j \ge 1$, (12.6.4) and (12.6.5) imply

$$\phi(\Gamma_j) > \frac{s_j}{2} \ge \frac{\phi(\Gamma)}{2} \qquad (j \ge 1).$$

The disjointness of the selected graphs would then give

$$\phi(X) \ge \sum_{j=1}^{\infty} \phi(\Gamma_j) = \infty,$$

contrary to finiteness of ϕ. Thus $\phi(X \setminus K) = 0$ in both cases. Since $\phi(\mathbb{P}^n \setminus X) = 0$, the first assertion in (12.6.3) follows.

Each Γ_j has finite $\mathscr{H}^q_p$-measure, so $\mathscr{H}^q_p \llcorner K$ is σ-finite. To verify countable $\mathbb{P}$-rectifiability, fix $0 < \Lambda < \infty$. For every $j \in J$, Definition 5.7.2 provides vertical parabolic (q, Λ)-Lipschitz graphs $G_{j,i}(\Lambda)$, $i \ge 1$, such that

$$\mathscr{H}^q_p\left(\Gamma_j \setminus \bigcup_{i=1}^{\infty} G_{j,i}(\Lambda)\right) = 0.$$

Consequently,

$$\mathscr{H}^q_p\left(K \setminus \bigcup_{j \in J} \bigcup_{i=1}^{\infty} G_{j,i}(\Lambda)\right) \le \sum_{j \in J} \mathscr{H}^q_p\left(\Gamma_j \setminus \bigcup_{i=1}^{\infty} G_{j,i}(\Lambda)\right) = 0.$$

The combined graph family is countable. Since $\Lambda > 0$ was arbitrary, K is countably $\mathbb{P}$-rectifiable. □

Proposition 12.6.2 (Weighted rectifiable representation)**.** *Let ϕ and K be as in Proposition 12.6.1. There is a Borel function $\theta : K \to (0,\infty)$ such that*

$$\phi = \theta\, \mathscr{H}^q_p \llcorner K \qquad \text{as Radon measures on } \mathbb{P}^n. \tag{12.6.6}$$

Proof. Define the Borel measure ν on $\mathbb{P}^n$ by

$$\nu = \mathscr{H}^q_p \llcorner K, \qquad \nu(A) = \mathscr{H}^q_p(A \cap K) \quad (A \subset \mathbb{P}^n \text{ Borel}). \tag{12.6.7}$$

It is σ-finite by (12.6.3). Let Γ_j, c_j and C_j, $j \in J$, be the pieces and comparison constants in Proposition 12.6.1. For every Borel set $A \subset \mathbb{P}^n$ and every $j \in J$, (12.6.1) gives

$$\phi(A \cap \Gamma_j) = 0 \quad \Longleftrightarrow \quad \nu(A \cap \Gamma_j) = 0.$$

Both measures are concentrated on the countable union $K = \bigcup_{j\in J} \Gamma_j$, by (12.6.3) and (12.6.7). Consequently $\phi \ll \nu$ and $\nu \ll \phi$.

Since ϕ is finite and ν is σ-finite, the Radon–Nikodym theorem [14, Theorem 3.8, pp. 90–91], applied on the Borel σ-algebra of $\mathbb{P}^n$, gives a Borel function $g : \mathbb{P}^n \to [0,\infty]$ such that

$$\phi(A) = \int_A g \, d\nu \qquad (A \subset \mathbb{P}^n \text{ Borel}). \tag{12.6.8}$$

For every $j \in J$ and every Borel set $A \subset \Gamma_j$, (12.6.1) and (12.6.8) imply

$$c_j \nu(A) \le \int_A g \, d\nu \le C_j \nu(A).$$

It follows that

$$c_j \le g(a) \le C_j \qquad \text{for } \nu\text{-almost every } a \in \Gamma_j. \tag{12.6.9}$$

Since J is countable, the Borel set

$$N = \{a \in K : g(a) = 0 \text{ or } g(a) = \infty\}$$

is ν-null by (12.6.9). Define

$$\theta : K \longrightarrow (0,\infty), \qquad \theta(a) = \begin{cases} g(a), & a \in K \setminus N, \\ 1, & a \in N. \end{cases} \tag{12.6.10}$$

This function is Borel in a, and (12.6.7) and (12.6.8) give

$$\phi(A) = \int_{A\cap K} \theta \, d\mathscr{H}^q_p \qquad (A \subset \mathbb{P}^n \text{ Borel}).$$

Thus (12.6.6) holds; the weighted measure is Radon because it equals ϕ. □

Remark. Merlo, Mourgoglou and Puliatti [27, Theorem 1.3, p. 5] prove rectifiability under positive finite lower and upper density bounds and the assumption that every geometric tangent is flat. Proposition 12.6.3 retains these density bounds but assumes that, at almost every point a, every geometric tangent is invariant under translations by a common vertical plane V_a (Definition 4.2.1), with $a \mapsto V_a$ Borel. Such invariance alone does not imply flatness: for $q = n+1$ and $z \in V_a^{\perp} \setminus \{0\}$, the measure

$$\mathscr{H}^q_p \llcorner V_a + \mathscr{H}^q_p \llcorner (V_a + z)$$

is invariant under translations by V_a but is not flat. The plane V_a is common to all tangents at a, whereas [27, Theorem 1.3] allows the supporting plane to depend on the tangent measure.

Proposition 12.1.1 gives finite parallel-plane representations, and Proposition 12.2.1 reduces them to a single plane on Lipschitz graph pieces. Proposition 12.2.2 then applies [27,

Theorem 2.34, pp. 18–20] and [26, Theorem 4.12, p. 867] to obtain rectifiability of those pieces.

Proposition 12.6.3 (Rectifiability on open sets)**.** *Let $\Omega \subset \mathbb{P}^n$ be open, $n \geq 1$, put $q = n+1$, and let η be a positive Radon measure on Ω. Suppose that*

$$0 < \Theta^q_{p*}(\eta, a) \leq \Theta^{q*}_p(\eta, a) < \infty \qquad \text{for } \eta\text{-almost every } a \in \Omega. \tag{12.6.11}$$

Let $X \subset \Omega$ be Borel with $\eta(\Omega \setminus X) = 0$, and let $V : X \to \mathbb{P}G$ be Borel. Write $V_a = V(a)$, and assume that every $\tau \in \operatorname{Tan}^q_p(\eta, a)$ is invariant under translations by V_a for η-almost every $a \in X$ (Definition 4.2.1). Then there are a Borel countably $\mathbb{P}$-rectifiable set $K \subset X$ and a Borel function $\theta : K \to (0, \infty)$ such that $\mathscr{H}^q_p \llcorner K$ is σ-finite, $\eta(\Omega \setminus K) = 0$, and

$$\eta = \theta\, \mathscr{H}^q_p \llcorner K \qquad \text{as Radon measures on } \Omega. \tag{12.6.12}$$

Proof. Choose a countable cover $\Omega = \bigcup_{k \geq 1} U_k$ by open sets $U_k \Subset \Omega$, and define

$$\phi_k(A) = \eta(A \cap U_k) \qquad (A \subset \mathbb{P}^n \text{ Borel}). \tag{12.6.13}$$

Since

$$\phi_k(\mathbb{P}^n) = \eta(U_k) \leq \eta(\overline{U_k}) < \infty,$$

each ϕ_k is a finite positive Borel measure on $\mathbb{P}^n$, hence Radon by [14, Theorem 7.8, p. 217].

Fix $k \geq 1$ and $a \in U_k$. Choose $\rho > 0$ with $B_p(a, \rho) \subset U_k$. For $0 < r < \rho$, the definition (12.6.13) gives $\phi_k(B_p(a, r)) = \eta(B_p(a, r))$. For $f \in C_c(\mathbb{P}^n; \mathbb{R})$, choose $R > 0$ with $\operatorname{supp} f \subset B_p(0, R)$. By (5.4.1), $T^{-1}_{a,r}(\operatorname{supp} f) \subset B_p(a, rR) \subset U_k$ whenever $0 < r < \rho/R$. Consequently,

$$r^{-q} \int_{\mathbb{P}^n} f(T_{a,r}(z))\, d\phi_k(z) = r^{-q} \int_\Omega f(T_{a,r}(z))\, d\eta(z) \qquad (0 < r < \rho/R). \tag{12.6.14}$$

The density definition (5.3.1), the tangent definition (9.0.7) and the convergence convention (4.2.3) therefore imply

$$\begin{aligned} \Theta^q_{p*}(\phi_k, a) = \Theta^q_{p*}(\eta, a), \quad \Theta^{q*}_p(\phi_k, a) = \Theta^{q*}_p(\eta, a), \\ \operatorname{Tan}^q_p(\phi_k, a) = \operatorname{Tan}^q_p(\eta, a) \qquad\qquad (a \in U_k). \end{aligned} \tag{12.6.15}$$

Set $X_k = X \cap U_k$. This is Borel, $\phi_k(\mathbb{P}^n \setminus X_k) = 0$, and $V|_{X_k}$ is Borel. Equations (12.6.11) and (12.6.15), together with the assumed translation invariance for η, verify the hypotheses of Proposition 12.6.1 for ϕ_k, X_k and $V|_{X_k}$. By Propositions 12.6.1 and 12.6.2, there are a Borel countably $\mathbb{P}$-rectifiable set $K_k \subset X_k$ and a Borel function $\theta_k : K_k \to (0, \infty)$ such that $\mathscr{H}^q_p \llcorner K_k$ is σ-finite and

$$\phi_k(\mathbb{P}^n \setminus K_k) = 0, \qquad \phi_k = \theta_k\, \mathscr{H}^q_p \llcorner K_k \quad \text{as Radon measures on } \mathbb{P}^n. \tag{12.6.16}$$

Define

$$K = \bigcup_{k=1}^\infty K_k, \qquad D_k = K_k \setminus \bigcup_{i<k} K_i. \tag{12.6.17}$$

The sets D_k are pairwise disjoint and Borel, with union $K \subset X$. For every k, (12.6.13) and (12.6.16) give $\eta(U_k \setminus K_k) = 0$. Since the sets U_k cover Ω,

$$\eta(\Omega \setminus K) \leq \sum_{k=1}^\infty \eta(U_k \setminus K_k) = 0. \tag{12.6.18}$$

The measure $\mathscr{H}^q_p \llcorner K$ is σ-finite because each $\mathscr{H}^q_p \llcorner K_k$ is σ-finite. For each $\Lambda > 0$, the countable families of vertical parabolic (q, Λ)-Lipschitz graphs covering the sets K_k modulo $\mathscr{H}^q_p$-null sets combine into a countable family covering K modulo an $\mathscr{H}^q_p$-null set. Thus K is countably $\mathbb{P}$-rectifiable by Definition 5.7.2.

Define $\theta : K \to (0, \infty)$ by

$$\theta(a) = \theta_k(a) \qquad (a \in D_k). \tag{12.6.19}$$

The countable Borel partition $\{D_k\}$ and Borelness of each θ_k show that θ is Borel in a. For every Borel set $A \subset \Omega$, the inclusion $D_k \subset K_k \subset U_k$ and (12.6.16) give

$$\eta(A \cap D_k) = \phi_k(A \cap D_k) = \int_{A \cap D_k} \theta_k \, d\mathscr{H}_p^q = \int_{A \cap D_k} \theta \, d\mathscr{H}_p^q.$$

Summing over k and using (12.6.18), we obtain

$$\eta(A) = \sum_{k=1}^{\infty} \eta(A \cap D_k) = \int_{A \cap K} \theta \, d\mathscr{H}_p^q \qquad (A \subset \Omega \text{ Borel}).$$

This proves (12.6.12); the weighted measure is Radon on Ω because it equals η. □

13. Proof of Theorem 1

13.1. **Proof of Theorem 1(1).**

Proof. Apply Lemma 5.3.1 to the positive Radon measure $\mu = \|W\|$ with $q = n + 1$. Its positive lower-density locus is

$$E_+ = \left\{ a \in \Omega : \liminf_{r \downarrow 0} \frac{\mu(B_p(a, r))}{(2r)^q} > 0 \right\}.$$

The lemma gives that E_+ is Borel and $\mathscr{H}_p^q \llcorner E_+$ is σ-finite, as required. □

13.2. **Proof of Theorem 1(2).**

Proof. Write $\mu = \|W\|$, $q = n + 1$, and $\mu_+ = \mu \llcorner E_+$. Suppose there is a Borel set $A \subset \Omega$ with $\mathscr{H}_p^q(A) = 0$ and $\mu_+(A) > 0$. Put $S = A \cap E_+$. By Theorem 1(1), S is Borel, and

$$\mathscr{H}_p^q(S) = 0, \qquad \mu(S) = \mu_+(A) > 0.$$

Apply Lemma 11.2 to $\mathbf{1}_S \in L^1_{\mathrm{loc}}(\Omega, \mu)$. Since $\mathbf{1}_S(a) = 1$ for $a \in S$, it gives, at μ-almost every $a \in S$,

$$\lim_{\substack{r \downarrow 0 \\ r \in \mathcal{R}_C(a)}} \frac{\mu(B_p(a, r) \setminus S)}{\mu(B_p(a, r))} = 0 \qquad \text{for every } 1 \le C < \infty. \tag{13.2.1}$$

The set $\mathcal{R}_C(a)$ is defined in (11.0.1). Let $N \subset E_+$ be a Borel set with $\mu(N) = 0$ containing the intersections of E_+ with the exceptional sets for (13.2.1), Lemmas 9.4.1 and 9.2.1, and the half-time hypothesis $\mathrm{HT}(\mu, \cdot)$ in Theorem 1. Since $S \subset E_+$,

$$\mu(S \setminus N) = \mu(S) > 0.$$

Fix $a \in S \setminus N$.

We shall apply Lemma 9.4.1 at a to obtain a nonzero mass-normalized limit. Choose positive sequences $R_i \downarrow 0$ and $\varepsilon_i \downarrow 0$ such that $B_p(a, R_i) \Subset \Omega$ and $\varepsilon_i \le 1$ for every i. Since $a \in S \subset E_+$, the hypothesis (5.3.3) holds. Apply Lemma 5.3.2 at a to (R_i) and (ε_i) to obtain radii s_i. After omitting finitely many indices and passing to a subsequence, relabel R_i, ε_i, s_i together so that s_i decreases strictly to zero. Put $U = U_p(0, 1)$, and let μ_i and W_i be the mass-normalized rescalings in (9.0.4).

Lemma 9.4.1 gives indices $i_j \uparrow \infty$ and a positive Radon measure τ on U. Replace $(R_i, \varepsilon_i, s_i, \mu_i, W_i)$ by $(R_{i_j}, \varepsilon_{i_j}, s_{i_j}, \mu_{i_j}, W_{i_j})$ and index all five sequences by i. Then

$$\mu_i \rightharpoonup \tau \quad \text{in } U, \qquad W_i \rightharpoonup \tau \otimes \nu_a \quad \text{in } U \times \mathbb{P}G. \tag{13.2.2}$$

Moreover, (9.4.3) gives

$$\tau(B_p(0, t)) \ge 5^{-q} t^q \qquad \text{for every } 0 < t < 1. \tag{13.2.3}$$

In particular, $\tau \ne 0$.

To apply (13.2.1) along (s_i), we verify that $s_i \in \mathcal{R}_{2\cdot 5^q}(a)$ for every i (see (11.0.1)). Lemma 5.3.2 gives $B_p(a, 5s_i) \Subset \Omega$ and $0 < \mu(B_p(a, s_i)) < \infty$. By (5.3.4) and $\varepsilon_i \le 1$,

$$\mu(B_p(a, 5s_i)) \le 5^q(1+\varepsilon_i)\mu(B_p(a, s_i)) \le 2 \cdot 5^q \mu(B_p(a, s_i)).$$

Thus $s_i \in \mathcal{R}_{2\cdot 5^q}(a)$. Since $s_i \downarrow 0$, (13.2.1) with $C = 2 \cdot 5^q$ yields

$$\lim_{i\to\infty} \frac{\mu(B_p(a, s_i) \setminus S)}{\mu(B_p(a, s_i))} = 0. \tag{13.2.4}$$

Equations (8.3.1) and (5.5.2) give

$$M_{\tau\otimes\nu_a} = A(\nu_a)\tau \qquad \text{as Radon measures on } U.$$

Since $a \notin N$, Lemma 9.2.1, applied to (13.2.2), yields

$$\int_U A(\nu_a) : D_x X \, d\tau = 0 \qquad \big(X \in C^1_c(U; \mathbb{R}^n)\big). \tag{13.2.5}$$

The choice of N also ensures $\mathrm{HT}(\mu, a)$. Lemma 9.2.2, applied to the same selected radii and (13.2.2), gives

$$\int_{U_p(0,\kappa)} \partial_t \phi \, d\tau = 0 \qquad \big(\phi \in C^1_c(U_p(0,\kappa); \mathbb{R})\big). \tag{13.2.6}$$

The lower bound (13.2.3) and equations (13.2.5) and (13.2.6) satisfy the hypotheses of Lemma 9.4.2 with $\nu = \nu_a$. Thus there is a plane $V_a = L_a \times \mathbb{R}_t \in \mathbb{P}G$ such that

$$\nu_a = \delta_{V_a}, \qquad W_i \rightharpoonup \tau \otimes \delta_{V_a} \quad \text{in } U \times \mathbb{P}G. \tag{13.2.7}$$

We adapt the projection argument in [10, arXiv version, proof of Lemma 3.4, pp. 12–14]. Put $V = V_a$, choose $0 < r < \kappa$, and take $\chi \in C^\infty_c(U_p(0,\kappa); \mathbb{R})$ with $0 \le \chi \le 1$ and $\chi = 1$ on $B_p(0, r)$. Extend χ by zero to $\mathbb{P}^n$. Set $K = \operatorname{supp} \chi$ and $K_V = P_V(K)$.

Since $B_p(a, 5s_i) \Subset \Omega$, the rescaled domains $T_{a,s_i}(\Omega)$ contain $U' = U_p(0, 2)$, which contains $B_p(0, 1)$. Proposition 9.1 gives locally bounded spatial first chronovariation for W_i, with representing measures $m_{\mathrm{chr},i}$ from (9.0.10). Since $a \notin N$, (9.2.2) and (9.0.4) give

$$\mu_i(K) \le \mu_i(B_p(0, 1)) = 1, \qquad |m_{\mathrm{chr},i}|(K) \longrightarrow 0. \tag{13.2.8}$$

Choose $\zeta \in C^\infty_c(U; \mathbb{R})$ with $0 \le \zeta \le 1$ and $\zeta = 1$ on K. Equation (13.2.7) yields

$$\begin{aligned} &\int_{K\times\mathbb{P}G} \|P_{L'} - P_{L_a}\|_{\mathrm{op}} \, dW_i(z, L' \times \mathbb{R}_t) \\ &\quad \le \int_{U\times\mathbb{P}G} \zeta(z) \|P_{L'} - P_{L_a}\|_{\mathrm{op}} \, dW_i(z, L' \times \mathbb{R}_t) \longrightarrow 0. \end{aligned} \tag{13.2.9}$$

Together with $\mathrm{HT}(\mu, a)$ and $s_i \in \mathcal{R}_{2\cdot 5^q}(a)$, the bounds (13.2.8) and (13.2.9) verify the hypotheses of Proposition 10.1 for the restrictions of W_i to $U' \times \mathbb{P}G$. After passing to a further subsequence, there is a nonnegative $\gamma \in L^1_{\mathrm{loc}}(V, \lambda_V)$ such that

$$u_i = (P_V)_\#(\chi\mu_i), \qquad |u_i - \gamma\lambda_V|(K_V) \longrightarrow 0. \tag{13.2.10}$$

The choice of χ and (5.3.5) give

$$u_i(K_V) = \int \chi \, d\mu_i \ge \mu_i(B_p(0, r)) \ge \frac{r^q}{5^q(1+\varepsilon_i)}.$$

Thus (13.2.10) implies

$$(\gamma\lambda_V)(K_V) \ge 5^{-q} r^q > 0. \tag{13.2.11}$$

For every i, put $S_i = T_{a,s_i}(S)$ and define the positive Radon measure on V

$$u_i^S = (P_V)_\#\big(\chi(\mu_i \llcorner S_i)\big). \tag{13.2.12}$$

By (9.0.4) and (13.2.4),

$$
\begin{aligned}
|u_i - u_i^S|(V) &= \int_{T_{a,s_i}(\Omega)\setminus S_i} \chi \, d\mu_i \\
&\leq \frac{\mu(B_p(a,s_i)\setminus S)}{\mu(B_p(a,s_i))} \longrightarrow 0.
\end{aligned} \tag{13.2.13}
$$

Equations (13.2.10) and (13.2.13) therefore give

$$
|u_i^S - \gamma\lambda_V|(K_V) \longrightarrow 0. \tag{13.2.14}
$$

By (5.4.1) and the projection formula in Section 5.5, $P_V \circ T_{a,s_i}$ is s_i^{-1}-Lipschitz for d_p. The definition (5.2.1) gives

$$
\mathscr{H}^q_p(P_V(S_i)) \leq s_i^{-q}\mathscr{H}^q_p(S) = 0.
$$

By Borel regularity of Hausdorff measure [33, Chapter 1, (2.6), p. 10] and (5.6.3), there is a Borel set $E_i \subset V$ with $P_V(S_i) \subset E_i$ and $\lambda_V(E_i) = 0$. Hence $u_i^S(V \setminus E_i) = 0$ and $u_i^S \perp \gamma\lambda_V$. The total variation (4.2.4) and (13.2.11) give

$$
|u_i^S - \gamma\lambda_V|(K_V) = u_i^S(K_V) + (\gamma\lambda_V)(K_V) \geq 5^{-q}r^q > 0,
$$

contrary to (13.2.14). Therefore $\mu_+(A) = 0$ for every Borel set $A \subset \Omega$ with $\mathscr{H}^q_p(A) = 0$. By Borel regularity of $\mathscr{H}^q_p$, this gives $\mu_+ \ll \mathscr{H}^q_p$. □

13.3. **Proof of Theorem 1(3) and (4).**

Proof. Put $\mu = \|W\|$ and $q = n+1$, and let E_∞ be as in (5.3.2). Lemma 5.3.1 gives $\mathscr{H}^q_p(E_\infty) = 0$. Theorem 1(2) therefore gives $\mu_+(E_\infty) = 0$. The inclusion $\{a \in \Omega : \Theta^q_{p*}(\mu, a) = \infty\} \subset E_\infty$ proves (3). For every $a \in E_+ \setminus E_\infty$, the definition of E_+ in Lemma 5.3.1 and (5.3.1) give

$$
0 < \Theta^q_{p*}(\mu, a) \leq \Theta^{q*}_p(\mu, a) < \infty.
$$

Since $\mu_+ = \mu \llcorner E_+$ and $\mu_+(E_\infty) = 0$, this proves (4). □

13.4. **Proof of Theorem 1(5).**

Proof of the plane-field assertion. Write $W = \mu \otimes \nu_a$, where $\mu = \|W\|$, and put $q = n+1$, $\mu_+ = \mu \llcorner E_+$ and $W_+ = W \llcorner (E_+ \times \mathbb{P}G)$. Let $N \subset E_+$ be a Borel set with $\mu(N) = 0$ containing the intersections of E_+ with the exceptional sets for Lemmas 9.4.1 and 9.2.1, and the half-time hypothesis $\mathrm{HT}(\mu, \cdot)$ in Theorem 1. Fix $a \in E_+ \setminus N$.

Choose positive sequences $R_i \downarrow 0$ and $\varepsilon_i \downarrow 0$ with $B_p(a, R_i) \Subset \Omega$, and select radii s_i by Lemma 5.3.2. Pass to a subsequence so that s_i decreases strictly to zero, and let μ_i, W_i be the rescalings in (9.0.4). Put $U = U_p(0,1)$. Lemma 9.4.1 gives, after passing to a further common subsequence, a positive Radon measure τ on U such that

$$
\mu_i \rightharpoonup \tau \quad \text{in } U, \qquad W_i \rightharpoonup \tau \otimes \nu_a \quad \text{in } U \times \mathbb{P}G, \tag{13.4.1}
$$

and

$$
\tau(B_p(0,t)) \geq 5^{-q}t^q \qquad (0 < t < 1). \tag{13.4.2}
$$

The moment definition (8.3.1) gives $M_{\tau\otimes\nu_a} = A(\nu_a)\tau$. Since $a \notin N$, Lemmas 9.2.1 and 9.2.2, applied to (13.4.1), yield

$$
\begin{aligned}
&\int_U A(\nu_a) : D_x X \, d\tau = 0 \quad \big(X \in C^1_c(U; \mathbb{R}^n)\big), \\
&\int_{U_p(0,\kappa)} \partial_t \phi \, d\tau = 0 \quad \big(\phi \in C^1_c(U_p(0,\kappa); \mathbb{R})\big).
\end{aligned} \tag{13.4.3}
$$

By (13.4.2) and (13.4.3), Lemma 9.4.2 implies that ν_a is a Dirac measure. This holds for every $a \in E_+ \setminus N$.

Each entry of $A(\nu_a)$ is μ-measurable in a. Apply [14, Proposition 2.12] to these entries to choose a Borel map $B : \Omega \to \mathbb{R}^{n\times n}$ such that $B(a) = A(\nu_a)$ for μ-almost every $a \in \Omega$. The map

$$J : \mathbb{P}G \longrightarrow \mathbb{R}^{n\times n}, \qquad J(L \times \mathbb{R}_t) = P_L,$$

has compact image $\mathcal{Q} = J(\mathbb{P}G)$ and continuous inverse $J^{-1} : \mathcal{Q} \to \mathbb{P}G$ by the projection metric in Section 5.5. Fix $V_0 \in \mathbb{P}G$ and define

$$V_a = \begin{cases} J^{-1}(B(a)), & B(a) \in \mathcal{Q}, \\ V_0, & B(a) \notin \mathcal{Q}, \end{cases} \qquad a \in \Omega. \tag{13.4.4}$$

This field is Borel. If $\nu_a = \delta_P$ and $B(a) = A(\nu_a)$, then $B(a) = J(P)$ and $V_a = P$. Consequently,

$$\nu_a = \delta_{V_a} \qquad \text{for } \mu_+\text{-almost every } a \in \Omega. \tag{13.4.5}$$

For every nonnegative Borel function F on $\Omega \times \mathbb{P}G$, (6.1.2) and (13.4.5) give

$$\begin{aligned} \int_{\Omega\times\mathbb{P}G} F\, dW_+ &= \int_{E_+} \left(\int_{\mathbb{P}G} F(a,P)\, d\nu_a(P) \right) d\mu(a) \\ &= \int_\Omega F(a, V_a)\, d\mu_+(a). \end{aligned}$$

By (6.2.2), this proves

$$W_+ = \mu_+ \otimes \delta_{V_a} = (\mathrm{Gr}(V))_{\#}\mu_+ \qquad \text{as Radon measures on } \Omega \times \mathbb{P}G. \tag{13.4.6}$$

□

Proof of the bundle tangent assertion. Let V_a be the Borel plane field in (13.4.4). Choose a Borel set $N \subset E_+$ with $\mu(N) = 0$ containing the intersections of E_+ with the exceptional sets for Lemma 9.3.2, Theorem 1(4) and (13.4.5). Fix $a \in E_+ \setminus N$. The density bounds in Theorem 1(4) and the identity $\nu_a = \delta_{V_a}$ in (13.4.5) verify the hypotheses of Lemma 9.3.2, with $V = V_a$. Hence every $\Upsilon \in \mathrm{Tan}^q_p(W, a)$ equals $\tau \otimes \delta_{V_a}$ for some $\tau \in \mathrm{Tan}^q_p(\mu, a)$. This proves Theorem 1(5)(b). □

Proof of the translation-invariance assertion. Enlarge the Borel μ-null set $N \subset E_+$ in the preceding proof to contain the intersections of E_+ with the exceptional sets for Lemma 9.2.3 and $\mathrm{HT}(\mu, \cdot)$ in Theorem 1. Fix $a \in E_+ \setminus N$. Theorem 1(4), (13.4.5) and $\mathrm{HT}(\mu, a)$ verify (9.2.11) with $V = V_a$. Lemma 9.2.3 therefore shows that every $\tau \in \mathrm{Tan}^q_p(\mu, a)$ is invariant under translations by V_a. This proves the translation-invariance assertion in Theorem 1(5)(a). □

Restriction and recentering. Enlarge the Borel μ-null set $N \subset E_+$ in the preceding proof to contain the intersection of E_+ with the exceptional set for Lemma 11.2 applied to 1_{E_+}. Fix $a \in E_+ \setminus N$. By Theorem 1(4), choose $0 < c_- \le c_+ < \infty$ and $r_0 > 0$ with $B_p(a, r_0) \Subset \Omega$ so that (9.3.9) holds. For $0 < r < r_0/5$,

$$\frac{\mu(B_p(a, 5r))}{\mu(B_p(a, r))} \le 5^q \frac{c_+}{c_-} =: C,$$

so $r \in \mathcal{R}_C(a)$ by (11.0.1). Lemma 11.2 therefore gives

$$\frac{\mu(B_p(a,r) \setminus E_+)}{\mu(B_p(a,r))} = \frac{1}{\mu(B_p(a,r))} \int_{B_p(a,r)} |1_{E_+}(z) - 1|\, d\mu(z) \longrightarrow 0 \qquad (r \downarrow 0). \tag{13.4.7}$$

For each fixed $R > 0$, (9.3.9) and (13.4.7) imply

$$r^{-q}\mu(B_p(a, Rr) \setminus E_+) \le c_+ R^q \frac{\mu(B_p(a, Rr) \setminus E_+)}{\mu(B_p(a, Rr))} \longrightarrow 0. \tag{13.4.8}$$

Taking $R = 1$ in (13.4.8) gives

$$\Theta^q_{p*}(\mu_+, a) = \Theta^q_{p*}(\mu, a), \qquad \Theta^{q*}_p(\mu_+, a) = \Theta^{q*}_p(\mu, a). \tag{13.4.9}$$

Write

$$(\mu_+)_{a,r} = r^{-q}(T_{a,r})_{\#}\mu_+ \qquad \text{on } T_{a,r}(\Omega).$$

For $f \in C_c(\mathbb{P}^n; \mathbb{R})$, choose $R > 0$ with $\operatorname{supp} f \subset B_p(0, R)$. For all sufficiently small r,

$$\left| \int_{T_{a,r}(\Omega)} f \, d\mu_{a,r} - \int_{T_{a,r}(\Omega)} f \, d(\mu_+)_{a,r} \right| \le \|f\|_\infty \, r^{-q} \mu(B_p(a, Rr) \setminus E_+) \longrightarrow 0 \tag{13.4.10}$$

by (13.4.8). Thus the two rescalings have the same limits along the same sequences, and

$$\operatorname{Tan}^q_p(\mu_+, a) = \operatorname{Tan}^q_p(\mu, a). \tag{13.4.11}$$

For these restriction identities, compare [27, Proposition 2.13(ii)–(iii) and (2.7), pp. 14–15].

Choose a countable cover $\Omega = \bigcup_{k \ge 1} O_k$ by open sets $O_k \Subset \Omega$, and define

$$\phi_k(A) = \mu_+(A \cap O_k) \qquad (A \subset \mathbb{P}^n \text{ Borel}). \tag{13.4.12}$$

Since $\phi_k(\mathbb{P}^n) = \mu_+(O_k) \le \mu(\overline{O_k}) < \infty$, each ϕ_k is a finite positive Radon measure on $\mathbb{P}^n$ by [14, Theorem 7.8, p. 217]. For $a \in O_k$, the measures ϕ_k and μ_+ agree in a neighborhood of a. Their densities at a are therefore equal, and their rescalings agree on each compact subset of $\mathbb{P}^n$ for all sufficiently small radii. In particular,

$$\operatorname{Tan}^q_p(\phi_k, a) = \operatorname{Tan}^q_p(\mu_+, a) \qquad (a \in O_k). \tag{13.4.13}$$

Theorem 1(4) and (13.4.9) show that ϕ_k satisfies (9.0.12). For each k, choose a Borel ϕ_k-null set $N_k \subset O_k \cap E_+$ containing the exceptional points of Lemma 9.1 in $O_k \cap E_+$. Since $\mu(N_k) = \phi_k(N_k) = 0$, enlarge N by $\bigcup_{k \ge 1} N_k$.

Fix $a \in E_+ \setminus N$, choose k with $a \in O_k$, and let $\tau \in \operatorname{Tan}^q_p(\mu, a)$. Equations (13.4.11) and (13.4.13) give $\tau \in \operatorname{Tan}^q_p(\phi_k, a)$. Lemma 9.1, followed by the same two identities, yields

$$s^{-q}(T_{z,s})_{\#}\tau \in \operatorname{Tan}^q_p(\mu, a) \qquad \big(z \in \operatorname{supp}\tau, \ s > 0\big). \tag{13.4.14}$$

□

Proof of the finite parallel-plane assertion. Let ϕ_k be the finite Radon measures in (13.4.12). By Theorem 1(4), (13.4.9), and the local equality of ϕ_k and μ_+ on O_k,

$$0 < \Theta^q_{p*}(\phi_k, a) \le \Theta^{q*}_p(\phi_k, a) < \infty \qquad \text{for } \phi_k\text{-almost every } a \in \mathbb{P}^n.$$

Equations (13.4.13) and (13.4.11) identify every geometric tangent of ϕ_k with a geometric tangent of μ at ϕ_k-almost every $a \in O_k$. Lemma 9.2.3, with (13.4.5) and the half-time hypothesis of Theorem 1, shows that all these tangents are invariant under translations by V_a. Thus Proposition 12.1.1 applies to each ϕ_k with the parent plane field V_a.

For each k, let $M_k \subset O_k \cap E_+$ be a Borel ϕ_k-null set containing the exceptional points of that proposition in $O_k \cap E_+$. Since $\mu(M_k) = \phi_k(M_k) = 0$, enlarge the Borel μ-null set N from the preceding proof by $\bigcup_{k \ge 1} M_k$. Fix $a \in E_+ \setminus N$, choose k with $a \in O_k$, and let $\tau \in \operatorname{Tan}^q_p(\mu, a)$. By (13.4.11) and (13.4.13), $\tau \in \operatorname{Tan}^q_p(\phi_k, a)$. Proposition 12.1.1 gives

$$\tau = \sum_{j=1}^{J} c_j \mathscr{H}^q_p \llcorner (V_a + z_j),$$

where $1 \le J < \infty$, $c_j > 0$, and the $z_j \in V_a^\perp$ are distinct. Together with the translation-invariance assertion already proved, this establishes Theorem 1(5)(a) and completes the proof of Theorem 1. □

14. Proof of Theorem 2

14.1. **Proof of Theorem 2(1): mass representation.**

Proof. Write $\mu = \|W\|$, $q = n+1$ and $\mu_+ = \mu \llcorner E_+$. By Theorem 1(1), E_+ is Borel. For every compact $C \subset \Omega$, $\mu_+(C) \le \mu(C) < \infty$, so μ_+ is a positive Radon measure on Ω by [14, Theorem 7.8, p. 217]. Moreover, $\mu_+(\Omega \setminus E_+) = 0$.

Theorem 1(4) and the density identities (13.4.9) imply

$$0 < \Theta^q_{p*}(\mu_+, a) \le \Theta^{q*}_p(\mu_+, a) < \infty \qquad \text{for } \mu_+\text{-almost every } a \in \Omega. \tag{14.1.1}$$

Let $V : \Omega \to \mathbb{P}G$, $V(a) = V_a$, be the Borel field defined in (13.4.4). By (13.4.11) and Theorem 1(5)(a), every $\tau \in \operatorname{Tan}^q_p(\mu_+, a)$ is invariant under translations by V_a for μ_+-almost every $a \in E_+$ (Definition 4.2.1).

By Proposition 12.6.3, applied with $\eta = \mu_+$, $X = E_+$ and the Borel field $V|_{E_+}$, there are a Borel countably $\mathbb{P}$-rectifiable set $K \subset E_+$ and a Borel function $\theta : K \to (0, \infty)$ such that $\mathscr{H}^q_p \llcorner K$ is σ-finite, $\mu_+(\Omega \setminus K) = 0$, and

$$\mu_+ = \theta\, \mathscr{H}^q_p \llcorner K \qquad \text{as Radon measures on } \Omega. \tag{14.1.2}$$

□

14.2. **Proof of Theorem 2(1): varifold representation.**

Proof. Let K and θ be as in (14.1.2), and let $V : \Omega \to \mathbb{P}G$, $V(a) = V_a$, be the Borel field in (13.4.4). Since $0 < \theta < \infty$ on K, the measures μ_+ and $\mathscr{H}^q_p \llcorner K$ have the same null sets. Choose the pieces K_j in Lemma 5.7.1, and put $\sigma_j = \mathscr{H}^q_p \llcorner K_j$, a finite Radon measure on $\mathbb{P}^n$. Lemma 9.1.1 gives the density bounds (9.1.1) for σ_j. Equation (9.1.2) and Definition 9.1.1 imply

$$\varnothing \ne \operatorname{Tan}^q_p(\sigma_j, a) \subset \{c\, \mathscr{H}^q_p \llcorner \operatorname{recTan}(K, a) : c > 0\} \quad \text{for } \sigma_j\text{-almost every } a \in K_j. \tag{14.2.1}$$

Let $O_k \Subset \Omega$ and ϕ_k be the finite localizations in (13.4.12). The measures ϕ_k and μ_+ agree near each point of O_k, so (14.1.1) gives positive finite lower and upper raw densities for ϕ_k at ϕ_k-almost every point. For $j, k \ge 1$, define the Borel function $\rho_{jk} : \mathbb{P}^n \to [0, \infty)$ by

$$\rho_{jk}(z) = \begin{cases} \theta(z), & z \in K_j \cap O_k, \\ 0, & z \notin K_j \cap O_k. \end{cases} \tag{14.2.2}$$

Equation (14.1.2) gives

$$\rho_{jk}\sigma_j = \phi_k \llcorner K_j, \qquad \int_{\mathbb{P}^n} \rho_{jk}\, d\sigma_j = \mu_+(K_j \cap O_k) \le \phi_k(\mathbb{P}^n) < \infty. \tag{14.2.3}$$

Thus $\rho_{jk} \in L^1(\sigma_j)$ and $\mathbf{1}_{K_j} \in L^1(\phi_k)$. Apply [27, Proposition 2.13(iii) and (2.7), pp. 14–15] to σ_j with weight ρ_{jk} and to ϕ_k with weight $\mathbf{1}_{K_j}$. Both applications use the same raw r^{-q} normalization as (9.0.7). Since $\rho_{jk}(a) = \theta(a) > 0$ on $K_j \cap O_k$, equations (14.2.3) and (13.4.13) give

$$\operatorname{Tan}^q_p(\mu_+, a) = \theta(a) \operatorname{Tan}^q_p(\sigma_j, a) \qquad \text{for } \mu_+\text{-almost every } a \in K_j \cap O_k. \tag{14.2.4}$$

Choose a Borel μ_+-null set $N \subset K$ containing $K \setminus \bigcup_j K_j$, the exceptional sets in K_j for (14.2.1) and in $K_j \cap O_k$ for (14.2.4), for all j, k, and the exceptional points in K of (13.4.11) and Theorem 1(5)(a). Fix $a \in K \setminus N$ and choose j, k with $a \in K_j \cap O_k$. By (14.2.1) and (14.2.4), there are $c > 0$ and a nonzero tangent

$$\tau = c\, \mathscr{H}^q_p \llcorner P \in \operatorname{Tan}^q_p(\mu_+, a), \qquad P = \operatorname{recTan}(K, a).$$

Equation (13.4.11) and Theorem 1(5)(a) show that τ is invariant under translations by V_a (Definition 4.2.1). Its support is P by (5.6.4). For every $v \in V_a$, translation of the support therefore gives

$$P + v = \operatorname{supp}\big((z \mapsto z + v)_\# \tau\big) = \operatorname{supp} \tau = P.$$

Since $0 \in P$, this implies $v \in P$, so $V_a \subset P$. Both planes belong to $\mathbb{P}G$ and have Euclidean dimension n. Hence

$$V_a = \operatorname{recTan}(K, a) \qquad \text{for } \mu_+\text{-almost every } a \in K. \tag{14.2.5}$$

The same equality holds $\mathscr{H}^q_p$-almost everywhere on K, so $V|_K$ is a Borel representative of $\operatorname{recTan}(K, \cdot)$. In (6.2.1) we use its global Borel extension V, with base measure $\mu_+ = \theta \mathscr{H}^q_p \llcorner K$. For $F \in C_c(\Omega \times \mathbb{P}G; \mathbb{R})$, equations (13.4.6), (14.1.2) and (14.2.5) give

$$\begin{aligned} \int_{\Omega \times \mathbb{P}G} F \, dW_+ &= \int_K F(a, V_a)\theta(a) \, d\mathscr{H}^q_p(a) \\ &= \int_K F(a, \operatorname{recTan}(K, a))\theta(a) \, d\mathscr{H}^q_p(a). \end{aligned}$$

By (6.2.2), this proves

$$W_+ = (\theta \mathscr{H}^q_p \llcorner K) \otimes \delta_{\operatorname{recTan}(K,a)} \qquad \text{as Radon measures on } \Omega \times \mathbb{P}G. \tag{14.2.6}$$

This completes Theorem 2(1) and proves the plane identification in Theorem 2(2). □

14.3. **Proof of Theorem 2(2).**

Proof. Let K_j and $\sigma_j = \mathscr{H}^q_p \llcorner K_j$ be the pieces and measures used in Section 14.2. By Lemma 5.7.1, each K_j is Borel, has finite $\mathscr{H}^q_p$-measure and admits vertical graph covers with arbitrarily small Lipschitz constants. The vertical case of [26, Theorem 4.9 and its proof, p. 866; Lemmas 2.3–2.4, pp. 858–859] gives density $2^{-q}v(q)$ with denominator $(2r)^q$, where $v(q)$ is the plane constant in (5.6.4). Thus, in the normalization (5.3.1),

$$\lim_{r \downarrow 0} \frac{\sigma_j(B_p(a, r))}{r^q} = v(q) \qquad \text{for } \sigma_j\text{-almost every } a \in K_j. \tag{14.3.1}$$

For each $j \geq 1$, define the Borel function $f_j : \Omega \to [0, \infty)$ by

$$f_j(z) = \begin{cases} \theta(z), & z \in K_j, \\ 0, & z \in \Omega \setminus K_j. \end{cases} \tag{14.3.2}$$

The mass representation (14.1.2) gives the Lebesgue decomposition

$$\mu_+ = f_j \sigma_j + \mu_+ \llcorner (\Omega \setminus K_j) \qquad \text{as Radon measures on } \Omega. \tag{14.3.3}$$

Indeed, $\mu_+ \llcorner K_j = f_j \sigma_j$, and the second summand is singular to σ_j because σ_j is carried by K_j. For every compact $C \subset \Omega$,

$$\int_C f_j \, d\sigma_j = \mu_+(C \cap K_j) \leq \mu_+(C) < \infty,$$

so $f_j \in L^1_{\mathrm{loc}}(\Omega, \sigma_j)$.

Apply Lemma 11.3 on Ω with reference measure σ_j and $\lambda = \mu_+$, for every j. Since $\mu_+ \llcorner K_j = \theta \sigma_j$ and $0 < \theta < \infty$ on K, each σ_j-null exceptional set in K_j is μ_+-null. Choose a Borel μ_+-null set $N \subset K$ containing $K \setminus \bigcup_j K_j$, all exceptional points in K_j for (14.3.1) and these applications of Lemma 11.3, for every j, and the exceptional points in K for (13.4.9) and (14.2.5). Fix $a \in K \setminus N$ and choose j with $a \in K_j$. By (14.3.1),

$$\frac{\sigma_j(B_p(a, 5r))}{\sigma_j(B_p(a, r))} = 5^q \frac{\sigma_j(B_p(a, 5r))/(5r)^q}{\sigma_j(B_p(a, r))/r^q} \longrightarrow 5^q.$$

The denominators are positive and $B_p(a, 5r) \Subset \Omega$ for all sufficiently small r. Hence every such radius satisfies (11.0.1) for the reference measure σ_j with $C = 2 \cdot 5^q$. Lemma 11.3 and (14.3.3) therefore give

$$\lim_{r \downarrow 0} \frac{\mu_+(B_p(a, r))}{\sigma_j(B_p(a, r))} = f_j(a) = \theta(a). \tag{14.3.4}$$

Multiplying (14.3.4) by (14.3.1) yields

$$\lim_{r\downarrow 0} \frac{\mu_+(B_p(a,r))}{r^q} = \theta(a)v(q).$$

Equation (13.4.9) now implies

$$\Theta^q_p(\mu, a) = \Theta^q_p(\mu_+, a) = v(q)\theta(a) \qquad \text{for } \mu_+\text{-almost every } a \in K. \tag{14.3.5}$$

Together with (14.2.5), this proves Theorem 2(2). □

14.4. **Proof of Theorem 2(3).**

Proof. Enlarge the Borel μ_+-null set $N \subset K$ chosen in Section 14.3 to include the exceptional points in K for (13.4.8), (13.4.10), (13.4.11) and Theorem 1(5)(b). Also include the exceptional points in K_j for (14.2.1) and in $K_j \cap O_k$ for (14.2.4), for every j, k. These piecewise exceptional sets are μ_+-null by (14.1.2). Fix $a \in K \setminus N$ and choose j, k with $a \in K_j \cap O_k$. Equations (14.2.1), (14.2.4), (14.2.5) and (13.4.11) give

$$\varnothing \neq \operatorname{Tan}^q_p(\mu, a) = \operatorname{Tan}^q_p(\mu_+, a) \subset \{c\,\mathscr{H}^q_p \llcorner V_a : c > 0\}. \tag{14.4.1}$$

Put $\Omega_r = T_{a,r}(\Omega)$ and use the rescalings (9.0.6). For the restrictions, these are

$$(\mu_+)_{a,r} = r^{-q}(T_{a,r})_\# \mu_+, \qquad (W_+)_{a,r} = r^{-q}(\widehat{T}_{a,r})_\# W_+, \tag{14.4.2}$$

on Ω_r and $\Omega_r \times \mathbb{P}G$, respectively. Every fixed compact subset of $\mathbb{P}^n$ lies in Ω_r for all sufficiently small r, and all limits below use (4.2.3). For each $R > 0$, (14.3.5) gives

$$\mu_{a,r}(B_p(0,R)) = \frac{\mu(B_p(a,Rr))}{r^q} \longrightarrow \theta(a)v(q)R^q. \tag{14.4.3}$$

Let $r_i \downarrow 0$. Equation (14.4.3) bounds the masses on every compact set for all sufficiently large i. For each integer $m \geq 1$, discard finitely many indices so that $B_p(0,m) \Subset \Omega_{r_i}$ and apply [33, Chapter 1, Theorem 4.16, pp. 37–38] on $U_p(0,m)$. Choose nested subsequences as m increases. Their limits agree on overlaps, and a diagonal subsequence gives $\mu_{a,r_i} \rightharpoonup \tau$ in $\mathbb{P}^n$. By [25, Theorem 1.24(1), p. 19],

$$\tau(B_p(0,1)) \geq \limsup_i \mu_{a,r_i}(B_p(0,1)) = \theta(a)v(q) > 0.$$

Thus τ is a nonzero tangent in (9.0.7); by (14.4.1), there is $c > 0$ such that $\tau = c\,\mathscr{H}^q_p \llcorner V_a$. For this compactness and nonvanishing argument, compare [25, proof of Theorem 14.3 and Remark 14.4(2), p. 187].

The ball formula (5.6.4) implies

$$\tau(\partial B_p(0,1)) = cv(q) - \lim_{s\uparrow 1} cv(q)s^q = 0.$$

The open-set and compact-set inequalities in [25, Theorem 1.24(1)–(2), p. 19] give

$$\tau(U_p(0,1)) \leq \liminf_i \mu_{a,r_i}(B_p(0,1)) \leq \limsup_i \mu_{a,r_i}(B_p(0,1)) \leq \tau(B_p(0,1)).$$

The first and last terms agree. Hence (14.4.3) yields

$$cv(q) = \tau(B_p(0,1)) = \lim_i \mu_{a,r_i}(B_p(0,1)) = \theta(a)v(q). \tag{14.4.4}$$

Since $v(q) > 0$, $c = \theta(a)$. Every sequence of radii tending to zero therefore has a subsequence whose mass rescalings converge to $\theta(a)\mathscr{H}^q_p \llcorner V_a$. If convergence failed for some $f \in C_c(\mathbb{P}^n; \mathbb{R})$, a sequence of radii would keep its test integrals a fixed positive distance from $\theta(a)\int_{V_a} f\, d\mathscr{H}^q_p$, contradicting this subsequence conclusion. Equation (13.4.10) transfers the limit to $(\mu_+)_{a,r}$. Hence

$$\mu_{a,r},\ (\mu_+)_{a,r} \rightharpoonup \theta(a)\mathscr{H}^q_p \llcorner V_a \qquad \text{in } \mathbb{P}^n \text{ as } r \downarrow 0. \tag{14.4.5}$$

For the varifold rescalings, (9.0.2) gives

$$W_{a,r}(B_p(0,R) \times \mathbb{P}G) = \mu_{a,r}(B_p(0,R)).$$

By the compactness of $\mathbb{P}G$ in Section 5.5 and (14.4.3), [33, Chapter 1, Theorem 4.16, pp. 37–38] applies on $U_p(0,m)\times\mathbb{P}G$ after discarding finitely many indices for each m. The same nested-subsequence construction gives compatible local limits and a convergent subsequence of W_{a,r_i} along every sequence $r_i\downarrow 0$. Let Υ be such a limit. For $f\in C_c(\mathbb{P}^n;\mathbb{R})$, the function $(y,P)\mapsto f(y)$ has compact support in $\mathbb{P}^n\times\mathbb{P}G$, so

$$\int_{\mathbb{P}^n\times\mathbb{P}G} f(y)\,d\Upsilon(y,P)=\lim_i\int_{\Omega_{r_i}} f\,d\mu_{a,r_i}=\theta(a)\int_{V_a} f\,d\mathscr{H}^q_p$$

by (14.4.5). Thus $\|\Upsilon\|=\theta(a)\mathscr{H}^q_p\llcorner V_a$ as Radon measures on $\mathbb{P}^n$, and $\Upsilon\neq 0$. By (9.0.9) and Theorem 1(5)(b),

$$\Upsilon=\theta(a)(\mathscr{H}^q_p\llcorner V_a)\otimes\delta_{V_a}.$$

Every subsequential limit has this form. The preceding test-function argument, now with $F\in C_c(\mathbb{P}^n\times\mathbb{P}G;\mathbb{R})$, proves convergence of $W_{a,r}$ as $r\downarrow 0$.

For $F\in C_c(\mathbb{P}^n\times\mathbb{P}G;\mathbb{R})$, choose $R>0$ with $\operatorname{supp}F\subset B_p(0,R)\times\mathbb{P}G$. Since $W_+\le W$, (14.4.2) and (13.4.8) imply

$$\left|\int_{\Omega_r\times\mathbb{P}G} F\,dW_{a,r}-\int_{\Omega_r\times\mathbb{P}G} F\,d(W_+)_{a,r}\right|\le\|F\|_\infty\,r^{-q}\mu(B_p(a,Rr)\setminus E_+)\longrightarrow 0. \tag{14.4.6}$$

It follows that

$$W_{a,r},\ (W_+)_{a,r}\rightharpoonup\theta(a)(\mathscr{H}^q_p\llcorner V_a)\otimes\delta_{V_a}\qquad\text{in }\mathbb{P}^n\times\mathbb{P}G\text{ as }r\downarrow 0. \tag{14.4.7}$$

The limits (14.4.5) and (14.4.7) hold at every $a\in K\setminus N$, where the conclusions of Theorem 2(2) also hold. This proves (3) and completes Theorem 2. □

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(Joshua L. Flynn) Department of Mathematics, Massachusetts Institute of Technology
Email address: `jlflynn@mit.edu`

(Ignacio Tejeda) Department of Mathematics, University of Pittsburgh
Email address: `ijt24@pitt.edu`